



Ergonomics Design Best Practice

**Form Factor Research:
Smart Phone Form Factor Predictive Model**

02-07-2011

Design & Technology Research & Strategic Marketing Research
Samsung Telecommunications of America

with HumanCentric

CONFIDENTIAL

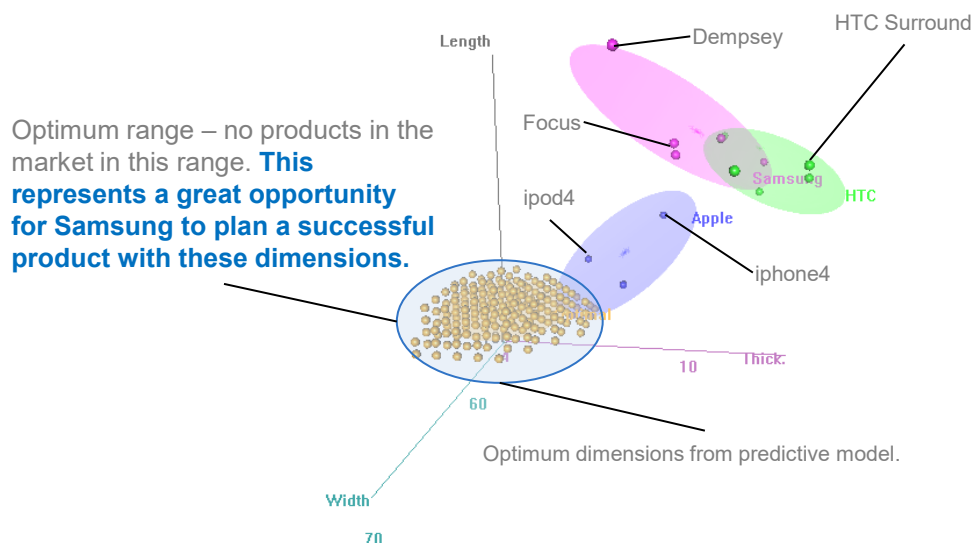
Fast Facts & Recommendations

Fast Facts

- Nineteen mockups of different length, width and thickness were evaluated by 120 smartphone owners or intenders. Other aspects of the industrial design were held constant.
- A predictive model of ideal and acceptable form factor model is developed.
- The model predicts that users will rate relatively short (**up to 108mm**), wide (**61-66mm**) and thin (**4-6mm**) devices as *optimum* in form factor; users will rate devices **up to 127mm** in length, **up to 73mm** in width and **4 to 13mm** in thickness as *acceptable*.
- Comparing Apple, HTC and Samsung devices in the market, only Apple devices come closer to the optimum range (but still not in the optimum range).
- With increasing display size, increase of length of devices beyond the acceptable range is tolerated by some users.*

Insights/Recommendations

- From the research results and their interpretations, there is an *unfulfilled* product gap of a relatively short, wide and thin device with a large display (at or over 4.4" but *not* too large to reach with one-hand use) in the *optimum* range as indicated in the results. If Samsung can deliver a smartphone product with optimum form factor and a large good quality display, this can become a winner (with competitive user experience design, product performance and marketing support).
- Users prefer reachability of touch display to bigger screen.



*After they rated and ranked each mockup in their form factor preference, a web page overlay (cnn.com and mobile cnn.com) was presented with each model where respondents ranked their preference.

Table of Contents

[Executive Summary](#)

[Results – Regression](#)

[Method](#)

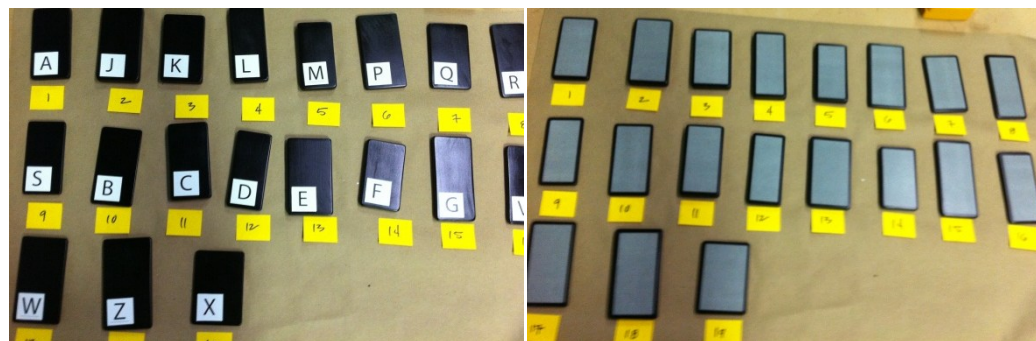
[Results – User Ratings of Models](#)

[Appendix A: Statistical Output](#)

[Appendix B: Model Ratings, Comments](#)

[Appendix C: Additional Data](#)

[Appendix D: Device Dimensions](#)





Executive Summary

Study Overview

Goal: Explore User Preferences for Size

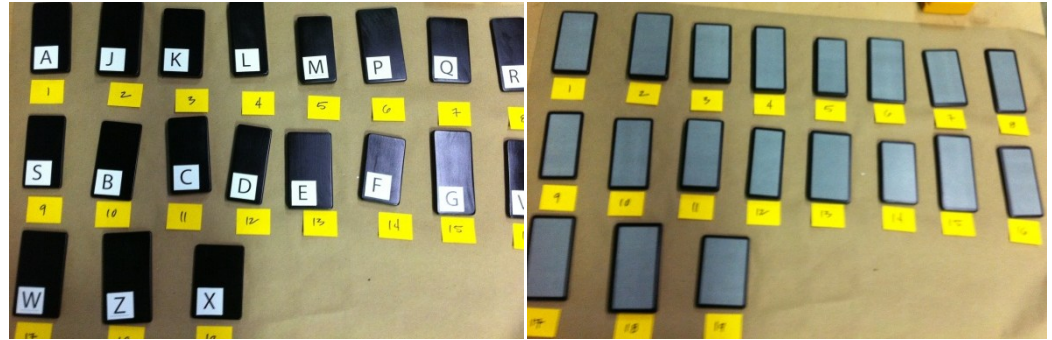
- What is the optimal range for each dimension?
- How do the dimensions affect each other?
 - Example: does the thickness of the device change how wide users want it to be?

Approach

- Create 19 different physical models
- Models represent a variety of sizes
 - Different width, length and thickness
 - 15 models represented strictly manipulated combinations of width, length and thickness
 - Short, thick, wide • Long, thick, wide
 - Short, thick, narrow • Long, thick, narrow
 - Short, thin, wide • Long, thin, wide
 - Short, thin, narrow • Long, thin, narrow
 - One model that was average on all dimensions
 - Six models that were average on two dimensions and extreme on the third: short, long, wide, narrow, thick or thin
- 4 devices represented specific model dimensions of interest.

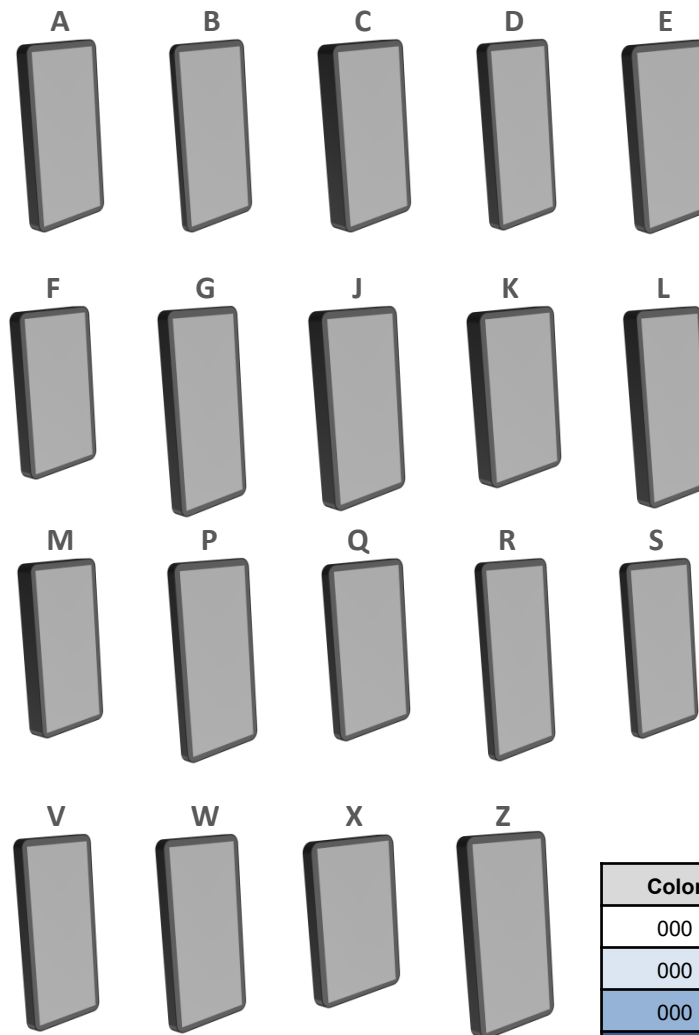
Study Format

- Participants spent 2 hours with the 19 device models
- For each model, participants rated
 - Where would you carry the device?
 - How comfortable is it to hold in your hand?
 - What do you think of the width, length, thickness?
 - Overall, do you like or dislike the size/shape?
- Which are your three favorite models?
- Which are your three least favorite models?
- The session concluded with a group discussion of why they liked and disliked the different models



Summary of Mockup Models

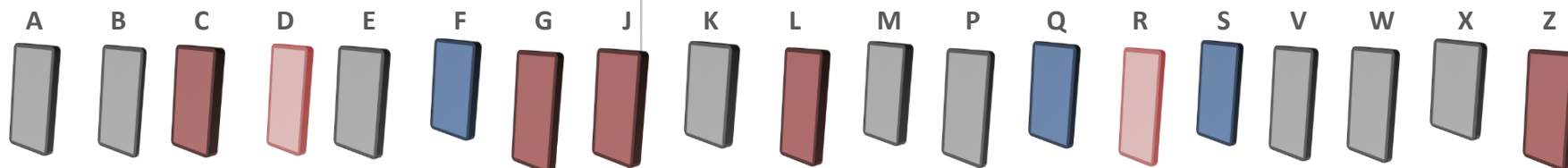
Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal
A	119	61	9	124mm (4.9")
J	127	66	12	132mm (5.2")
K	112	66	12	119mm (4.7")
L	127	57	12	129mm (5.1")
M	112	57	12	114mm (4.5")
P	127	66	6	132mm (5.2")
Q	112	66	6	119mm (4.7")
R	127	57	6	130mm (5.1")
S	112	57	6	114mm (4.5")
B	119	61	4	124mm (4.9")
C	119	61	14	124mm (4.9")
D	119	54	9	122mm (4.8")
E	119	69	9	127mm (5.0")
F	107	61	9	112mm (4.4")
G	132	61	9	135mm (5.3")
V	123	60	8	127mm (5.0")
W	123	66	8	127mm (5.0")
Z	129	69	9	135mm (5.3")
X	105	73	8	117mm (4.6")



Color	Meaning
000	Minimum
000	
000	Average
000	
000	Maximum
000	Other

User Ratings of Mockups Tested

Letter	Overall	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)	Length	Width	Thickness
A	+ Like	119	61	9	124mm (4.9")	118	+ Big but acc.	+ Just right	+ Just right
J	- Really dislike	127	66	12	132mm (5.2")	155	- Too big	+ Just right	+ Just right
K	+ Like	112	66	12	119mm (4.7")	142	+ Just right	+ Just right	- Too big
L	- Really dislike	127	57	12	129mm (5.1")	140	- Too big	+ Small but acc.	- Too big
M	+ Like	112	57	12	114mm (4.5")	129	+ Just right	+ Just right	- Too big
P	+ Like	127	66	6	132mm (5.2")	101	+ Big but acc.	+ Just right	+ Just right
Q	+ Really like	112	66	6	119mm (4.7")	95	+ Just right	+ Just right	+ Just right
R	- Dislike	127	57	6	130mm (5.1")	94	+ Big but acc.	+ Small but acc.	+ Small but acc.
S	+ Really Like	112	57	6	114mm (4.5")	88	+ Just right	+ Just right	+ Just right
B	+ Like	119	61	4	124mm (4.9")	79	+ Big but acc.	+ Just right	+ S but acc.
C	- Really dislike	119	61	14	124mm (4.9")	158	+ Big but acc.	+ Just right	- Too big
D	- Dislike	119	54	9	122mm (4.8")	110	+ Big but acc.	+ Small but acc.	+ Just right
E	+ Like	119	69	9	127mm (5.0")	127	+ Big but acc.	+ Big but acc.	+ Just right
F	+ Really like	107	61	9	112mm (4.4")	111	+ Just Right	+ Just Right	+ Just right
G	- Really dislike	132	61	9	135mm (5.3")	126	- Too big	+ Just right	+ Just right
V	+ Like	123	60	8	127mm (5.0")	107	+ Big but acc.	+ Just right	+ Just right
W	+ Like	123	66	8	127mm (5.0")	113	+ Big but acc.	+ Just right	+ Just right
Z	- Really dislike	129	69	9	135mm (5.3")	133	- Too big	+ Big but acc.	+ Just right
X	+ Like	105	73	8	117mm (4.6")	109	+ Just right	+ Big but acc.	+ Just right



Recommended Dimensions

	Width	Length	Thickness
Range of Interest*	54-69mm	107-132mm	4-14mm
Range that users “really like”	61-66mm	min-108mm	min-6mm
Range that users “like”	full	min-127mm	min-13mm

Optimum (mm)

		107	111	115	119	123	127			4	5	6	7	8	9	10	11	12	13			107	111	115	119	123	127		
mum)	4	63	63	64	64	65		4	54	107	107	107	107	107	107	107	107		54	54	6.5	6.5	7					54	
	5	63	63	63	64	64	65	5	56	107	107	107	107	107	107	107	107	107	56	56	6	6.5	6.5	7				56	
	6	62	63	63	64	64	64	6	58	107	107	107	107	107	107	107	107	107	58	58	6	6	6.5	7	7			58	
	7	62	62	63	63	64	64	7	60	107	107	107	107	107	107	107	107	107	60	60	5.5	6	6	6.5	7			60	
	8	62	62	63	63	63	64	8	62	107	107	107	107	107	107	107	107	107	62	62	5	5.5	6	6	6.5	7		62	
	9	61	62	62	63	63		9	64	107	107	107	107	107	107	107	107	108	64	64	5	5	5.5	6	6	6.5		64	
	10	61	62	62	62	63		10	66	107	107	107	107	107	107	107	108	66	66	4.5	5	5	5.5	5.5	6	6		66	
	11	61	61	62	62			11	68	107	107	107	107	107	108	108	109	68	68	4.5	4.5	5	5.5	5.5				68	
Thickness	12	61	61	61	62			12																					
	13	60	61					13	Width																				

Acceptable Range (mm)

Thickness	Length						Thickness											Length						
	107	111	115	119	123	127	4	5	6	7	8	9	10	11	12	13	107	111	115	119	123	127		
4	54-69	54-69	55-69	57-69	59-69		54	114	115	116	117	116	116	114	110		54	4-12	4-11	5-9			54	
5	54-69	54-69	54-69	56-69	58-69	63-67	56	118	120	120	121	121	120	119	116	112	56	4-12	4-12	4-11	5-10		56	
6	54-69	54-69	54-69	56-69	58-69	62-67	58	122	123	123	124	124	123	122	120	116	58	4-13	4-13	4-12	4-11	5-9	58	
7	54-69	54-69	54-69	55-69	58-69	62-67	60	124	125	125	126	126	125	124	122	118	60	4-13	4-13	4-13	4-12	4-11	60	
8	54-69	54-69	54-69	55-69	59-69	62-67	62	125	126	127	127	127	126	125	122	119	62	4-13	4-13	4-13	4-12	4-11	6-8	62
9	54-69	54-69	54-69	56-69	58-68		64	126	127	127	127	127	126	125	122	118	64	4-13	4-13	4-12	4-12	4-11	5-8	64
10	54-69	54-69	55-69	56-68	59-66		66	126	127	127	127	127	125	123	120	113	66	4-12	4-12	4-12	4-11	4-10	5-8	66
11	54-68	55-68	56-68	56-66			68	125	126	126	126	125	123	121	115		68	4-11	4-11	4-11	4-10	4-9		68
12	55-66	56-66	58-65	61-63																				
13	60-62	60-61																						

Represents a length range of 107-125mm. The shortest device tested was 107mm, and we can only draw conclusions within the range tested in the study, so the minimum for all combinations of width/thickness is 107.

The thinnest mockup tested was 4mm, and we can only draw conclusions within the range tested in the study, so the minimum for most combinations of width/length is 4mm.

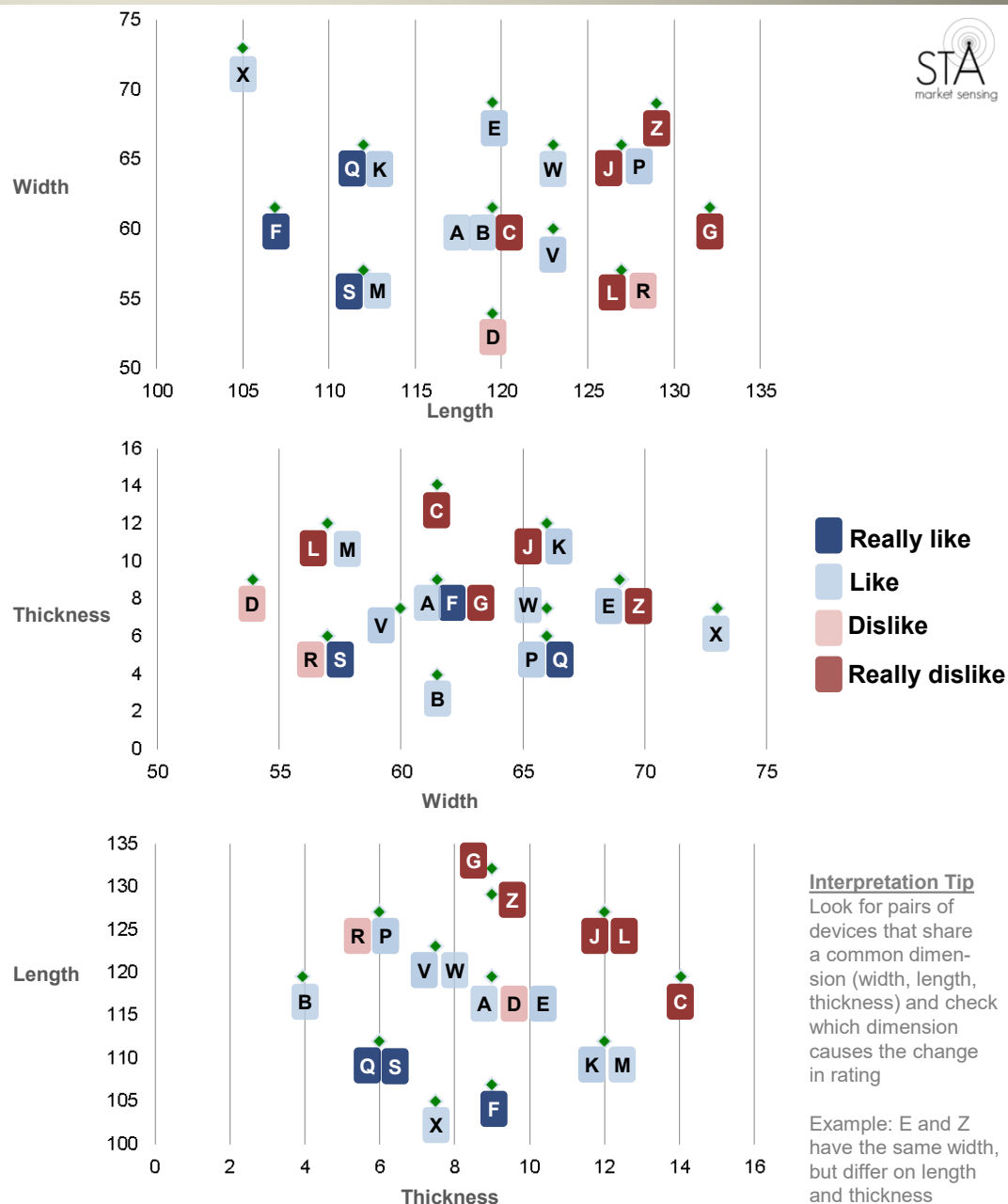
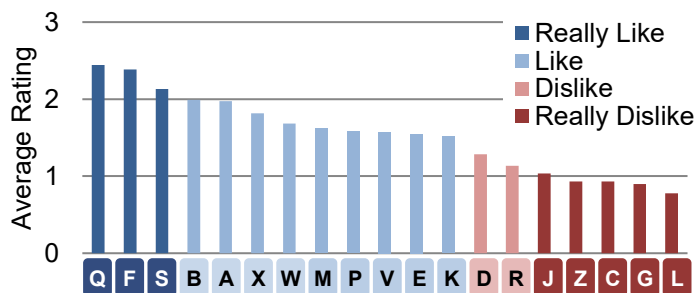
User Ratings of Mockups Tested

Description of the Graphs

- Each graph at the right shows the devices by two of their three physical dimensions
- The most and least favorite devices are highlighted with color

Discussion

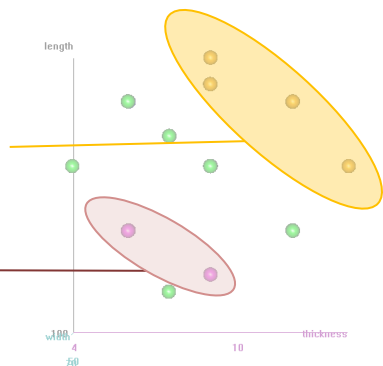
- Length is a good predictor of how much users will like a device
 - Short devices are liked
 - Mid-length devices can be liked or disliked
 - Long devices are disliked or neutral
- Thickness is a good predictor of liking a device
 - Thicker devices are more disliked
- Width alone is not a good predictor
- There is an interaction among all three dimensions that is complex, but when looking for a single predictor, the combination of length and thickness is the best predictor of liking a device
 - Long, narrow, thick is least preferred
 - Long, wide, thick is somewhat disliked
 - Short/thin devices are liked



Mockup Dimensions & Ratings

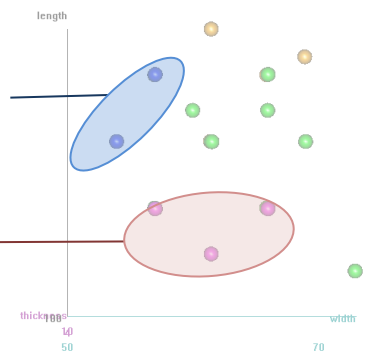
The mockups that were "really disliked" are longer than 127mm or at 14mm thickness.

The mockups that were "really liked" are shorter than 119mm

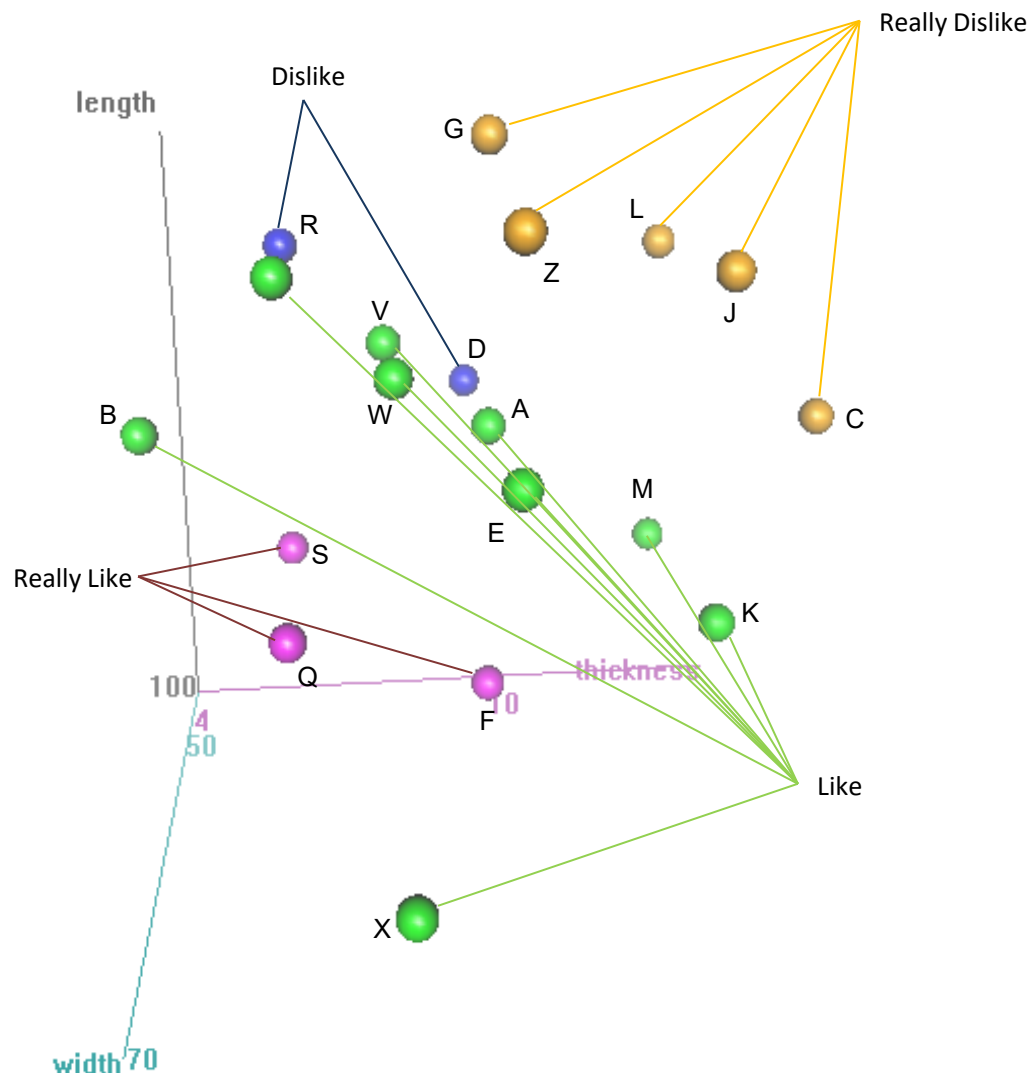
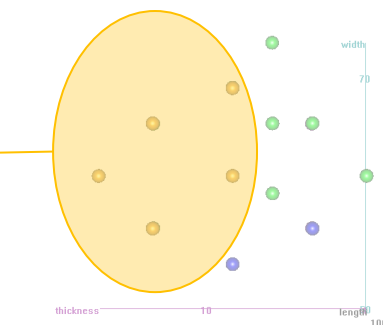


The mockups that are narrow and relatively long were disliked

The really liked mockups are relatively short but wide.



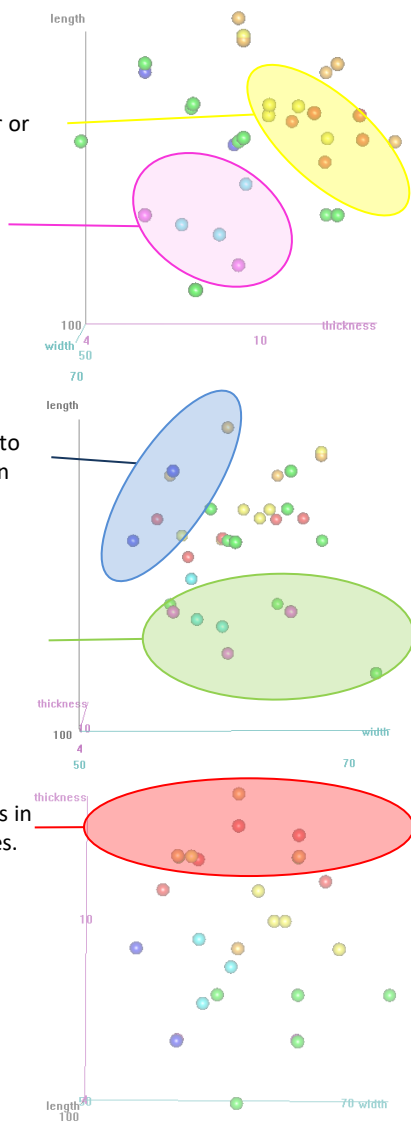
Thick mockups were disliked



Mockup vs. Apple, HTC, Samsung Dimensions

Many Samsung and HTC products are either longer or thicker.

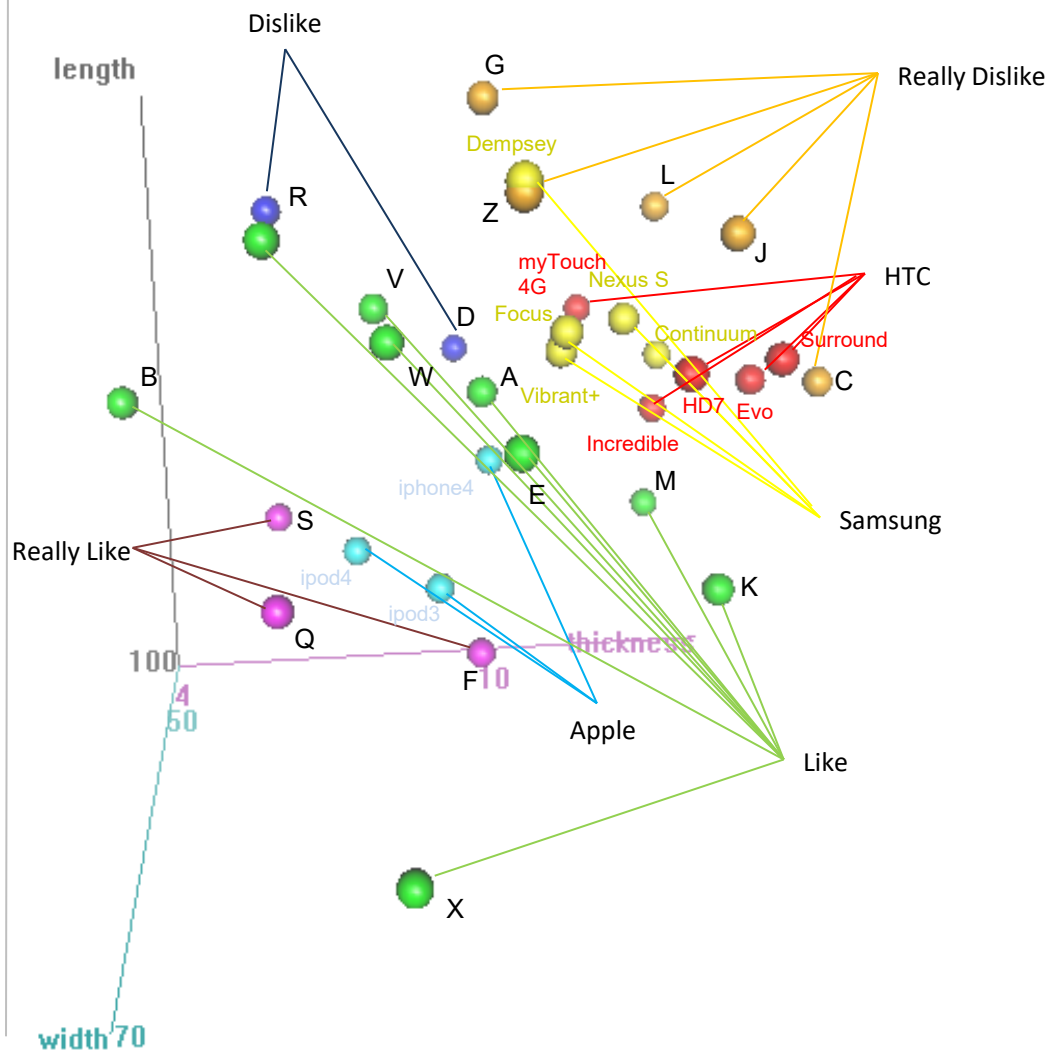
Apple products are closest to the “Really Like” mockup dimensions



HTC MyTouch 4G and Samsung Continuum fall into the “relative” disproportion group (length/width)

Respondents have a large tolerance for width.

HTC have “thicker” devices in the representative samples.



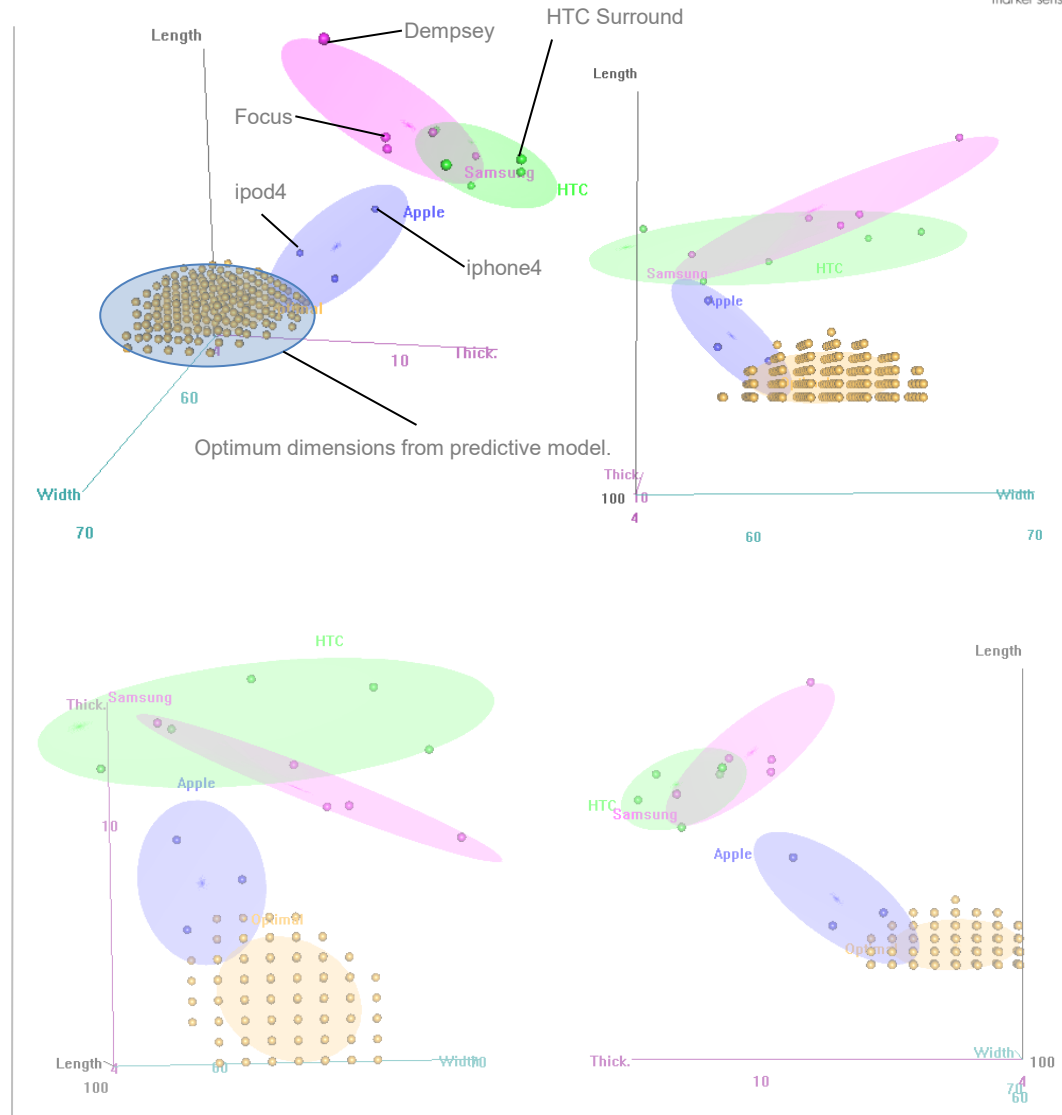
Optimum Dimensions vs. Apple, HTC and Samsung Devices

Description of the Chart

- The optimum dimensions are represented by orange dots.
- The dimensions of representative Apple, HTC and Samsung products are represented in blue, green and magenta shadings.

Discussion

- Apple products come closer to the optimum dimensions (iPod Touch 4G and 3G and iPhone 4 are represented)
- HTC products (Incredible, Evo, MyTouch 4G, HD7 and Surround) are further away from the optimum dimensions than Apple products
- Samsung products (Focus, Dempsey, Nexus S, Continuum and Vibrant+) are further away from the optimum dimensions than Apple products.
- Samsung products also trend longer in length while HTC products are generally thicker.
- Thinner devices are desired by the consumers and will be a good strategic direction to pursue.
- Length of the devices should be kept shorter. There is a design gap at this point in the market for a device shorter than Apple products but thinner and with a wider form factor. The product with this range of form factor can be a winner especially if a larger screen can be used (e.g., without physical buttons on the front chassis).



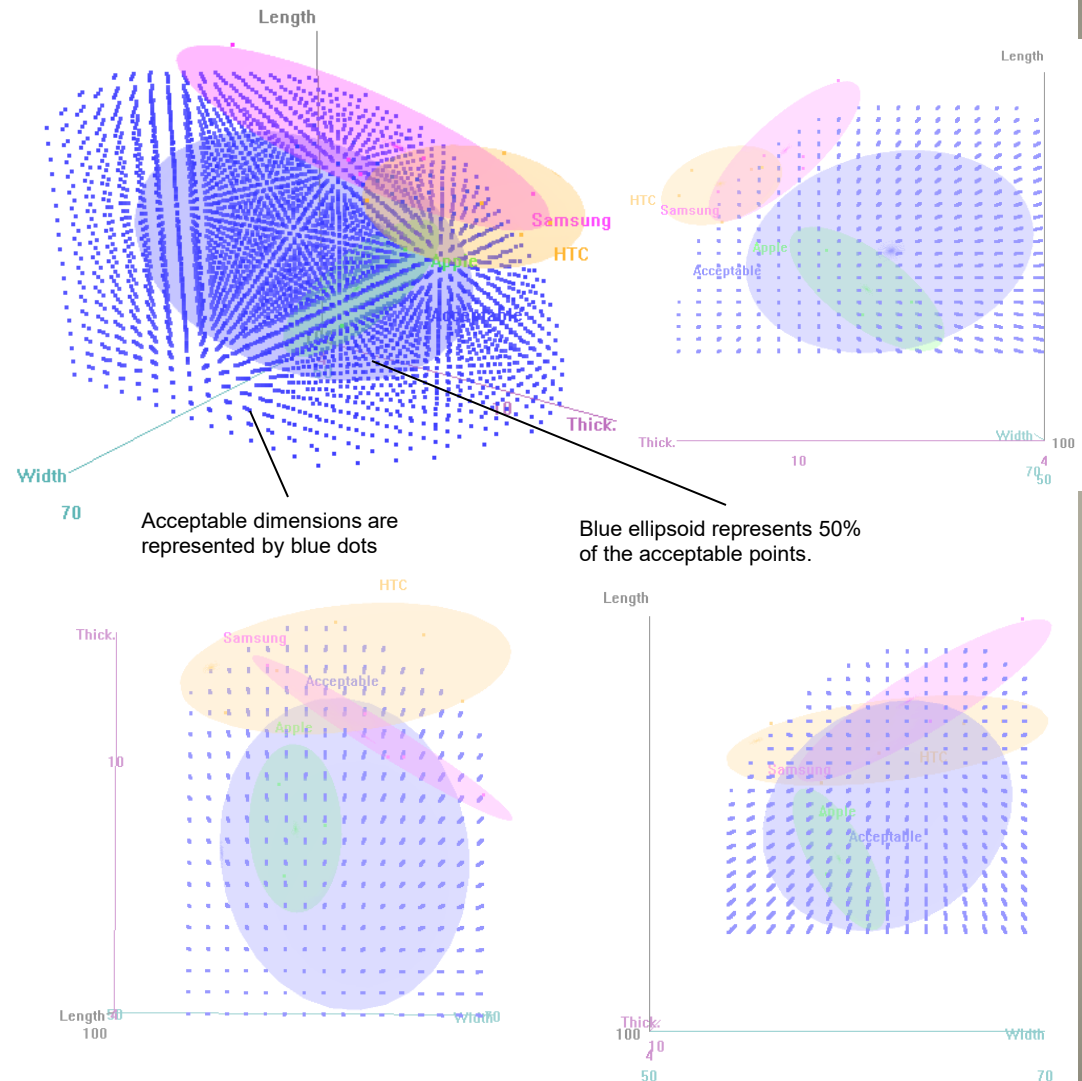
Acceptable Dimensions vs. Apple, HTC and Samsung Devices

Description of the Chart

- The acceptable dimensions are represented by blue dots.
- The dimensions of representative Apple, HTC and Samsung products are represented in green, yellow and magenta shadings.

Discussion

- All Apple products are within acceptable dimensions (iPod Touch 4G and 3G and iPhone 4 are represented)
- Four of the five HTC products (Evo, MyTouch 4G, HD7 and Surround) are outside of acceptable dimensions. Only HTC Incredible is barely within the acceptable range.
- Two of the five Samsung (Dempsey and Continuum) are outside of acceptable dimensions. Samsung Focus, Vibrant+ and Nexus S are within acceptable range.



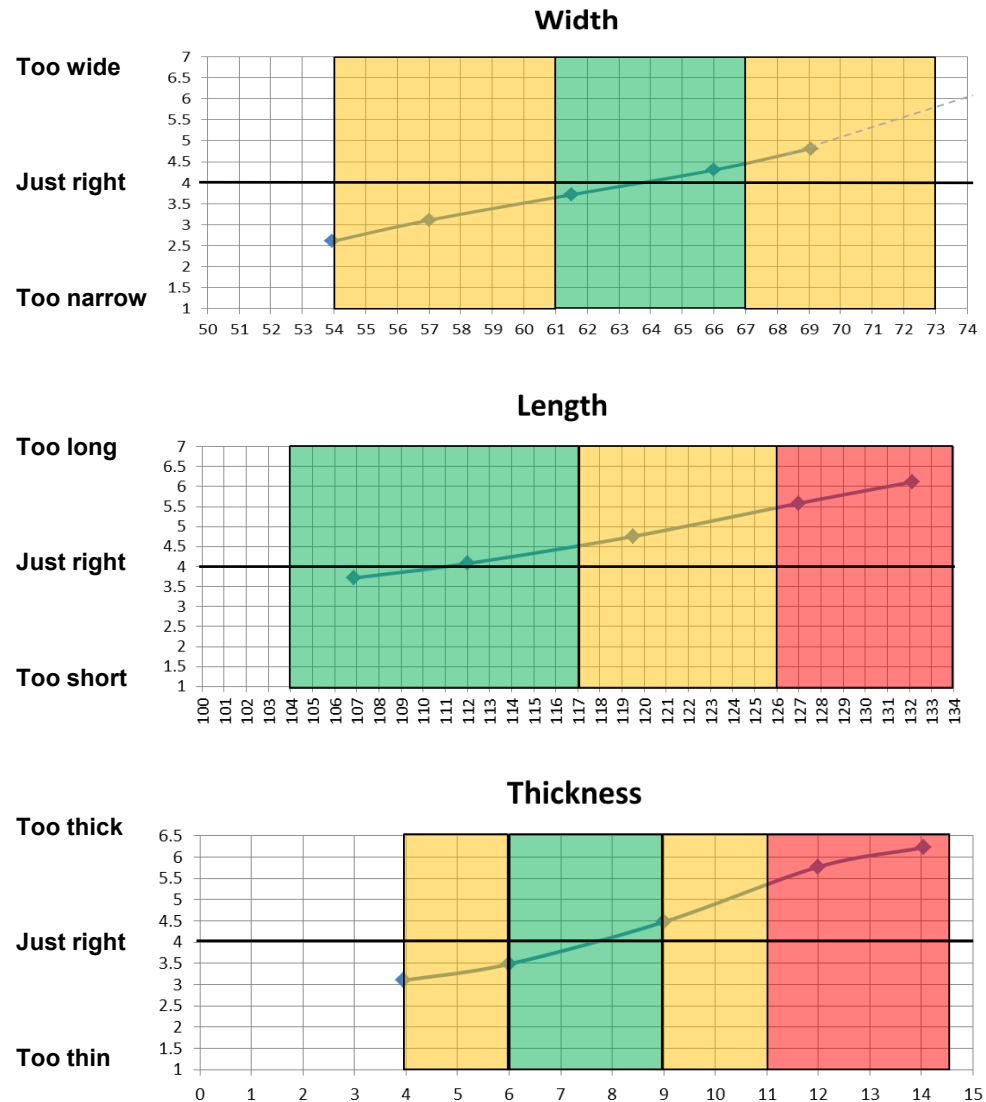
Recommended Dimensions Based on Ratings by Participants

Description of Graphs

- Each point represents a size tested in the study
- The graphs at the right represent average user ratings of each dimension
 - 7 = far too much (wide/long/thick)
 - 6 = too much
 - 5 = too much but acceptable
 - 4 = just right
 - 3 = too little but acceptable
 - 2 = too little
 - 1 = far too little (narrow/short/thin)
- The graphs are based only on the 15 devices where we consistently varied the size (excluding VWXZ)

Overview of results

- These ratings are for individual dimensions
 - In reality, they affect/depend on each other.
- These ratings are likely to be based more on the physicality of the device than screen size.
- Width
 - 61-67 optimal
 - 54-73 acceptable
 - The range tested were all acceptable
- Length
 - 104-117 optimal
 - ???-126 acceptable
 - No devices tested were too short
 - No devices tested were short but acceptable
- Thickness
 - 6-9 optimal
 - 4-11 acceptable
 - No devices tested were too thin





Results – Regression

Relationships between width, length, thickness, and the range of dimensions the regression predicts users will like

This section discusses the 15 devices that draw their dimensions from the central composite design (excluding WXVZ)

Recommended Dimensions

	Width	Length	Thickness
Range of Interest*	54-69mm	107-132mm	4-14mm
Range that users "really like"	61-66mm	min-108mm	min-6mm
Range that users "like"	full	min-127mm	min-13mm

Optimum (mm)

	107	111	115	119	123	127	
4	63	63	64	64	65		4
5	63	63	63	64	64	65	5
6	62	63	63	64	64	64	6
7	62	62	63	63	64	64	7
8	62	62	63	63	63	64	8
9	61	62	62	63	63		9
10	61	62	62	62	63		10
11	61	61	62	62			11
12	61	61	61	62			12
Thickness 13	60	61					13

	4	5	6	7	8	9	10	11	12	13		107	111	115	119	123	127	
54	107	107	107	107	107	107	107	107	107		54	6.5	6.5	7				54
56	107	107	107	107	107	107	107	107	107	107	56	6	6.5	6.5	7			56
58	107	107	107	107	107	107	107	107	107	107	58	6	6	6.5	7	7		58
60	107	107	107	107	107	107	107	107	107	107	60	5.5	6	6	6.5	7		60
62	107	107	107	107	107	107	107	107	107	107	62	5	5.5	6	6	6.5	7	62
64	107	107	107	107	107	107	107	107	107	108	64	5	5	5.5	6	6	6.5	64
66	107	107	107	107	107	107	107	108	109		66	4.5	5	5	5.5	6	6	66
Width 68	107	107	107	107	107	108	108	109			68	4.5	4.5	5	5.5	5.5		68

Acceptable Range (mm)

	107	111	115	119	123	127	
4	54-69	54-69	55-69	57-69	59-69		4
5	54-69	54-69	54-69	56-69	58-69	63-67	5
6	54-69	54-69	54-69	56-69	58-69	62-67	6
7	54-69	54-69	54-69	55-69	58-69	62-67	7
8	54-69	54-69	54-69	55-69	59-69	62-67	8
9	54-69	54-69	54-69	56-69	58-68		9
10	54-69	54-69	55-69	56-68	59-66		10
11	54-68	55-68	56-68	56-66			11
12	55-66	56-66	58-65	61-63			12
Thickness 13	60-62	60-61					13

	4	5	6	7	8	9	10	11	12	13		107	111	115	119	123	127	
54	114	115	116	117	116	116	114	110			54	4-12	4-11	5-9				54
56	118	120	120	121	121	120	119	116	112		56	4-12	4-12	4-11	5-10			56
58	122	123	123	124	124	123	122	120	116		58	4-13	4-13	4-12	4-11	5-9		58
60	124	125	125	126	126	125	124	122	118	111	60	4-13	4-13	4-13	4-12	4-11		60
62	125	126	127	127	127	126	125	122	119	110	62	4-13	4-13	4-13	4-12	4-11	6-8	62
64	126	127	127	127	127	126	125	122	118		64	4-13	4-13	4-12	4-12	4-11	5-8	64
66	126	127	127	127	127	125	123	120	113		66	4-12	4-12	4-12	4-11	4-10	5-8	66
Width 68	125	126	126	126	125	123	121	115			68	4-11	4-11	4-11	4-10	4-9		68

Represents a length range of 107-125mm. The shortest device tested was 107mm, and we can only draw conclusions within the range tested in the study, so the minimum for all combinations of width/thickness is 107.

The thinnest device tested was 4mm, and we can only draw conclusions within the range tested in the study, so the minimum for most combinations of width/length is 4mm.

Regression Results

Using linear regression, we built an equation that models how the physical size of the device affects how much users will like the device

Understanding the Equation

- Liking is on a range of 1 to 4
 - 1 = Really dislike
 - 2 = Dislike
 - 2.5 would represent neutral
 - 3 = Like
 - 4 = Really Like
- Because width, length and thickness have such different ranges of millimeters, you can't compare the coefficients in the equation directly to compare how much changes in each dimension affect liking
- The second table uses values that are standardized based on the amount of change in each dimensions from smallest to largest tested in the current study

Discussion

- Length and thickness have the highest impact
- Width also has a high impact
- All 9 terms, including interactions, are in the model
 - This means that understanding the relationship between the variables will be difficult because the relationship is complex

Optimum

- The optimum point for this equation within the range tested is
 - width = 62.6mm,
 - length = 107mm,
 - thickness = 5.1mm
 - predicted value = 3.6 out of 4

$$\begin{aligned} \text{Liking} = & 3.96 - 0.70 \text{ Length} - 0.29 \text{ Length}^2 \\ & - 0.50 \text{ Thickness} - 0.46 \text{ Thickness}^2 \\ & + 0.17 \text{ Width} - 0.51 \text{ Width}^2 \\ & - 0.21 \text{ Width} * \text{Thickness} \\ & + 0.19 \text{ Length} * \text{Thickness} \\ & + 0.18 \text{ Width} * \text{Thickness} \end{aligned}$$

These coefficients assumed centered and standardized the variables were used

Relative contribution to liking*		Interpretation
Length	– .45	Shorter is preferred
Length ²	– .10	
Thickness	– .32	Thinner is preferred
Thickness ²	– .16	
Width	.11	Moderate width is preferred (as opposed to narrow or wide)
Width ²	– .18	
Width * Thickness	– .06	There is a tradeoff between width and thickness Wide/thin or narrow/thick is preferred
Length * Thickness	.05	Length and thickness should be proportional Long/thick or short/thin is preferred
Width * Length	.05	Width and length should be proportional Long/wide or short/narrow is preferred

* Standardized equation coefficient

Optimal Thickness

Description of the Table

- We predicted how much users will like every combination of width, length and thickness
- The values in the table are the optimal thickness for each combination of width x length
- The four zones represent how much users like/dislike a particular combination of width/length/thickness
 - Deviations in thickness from the values in the table decrease the predicted "like" rating
 - Really dislike (predicted rating $1 \leq 1.5$)
 - Dislike (< 2.5)
 - Like (< 3.5)
 - Really Like (≤ 4)

Thickness	
Range of Interest	4-14mm
Range that users "really like"	min-6mm
Range that users "like"	min-13mm

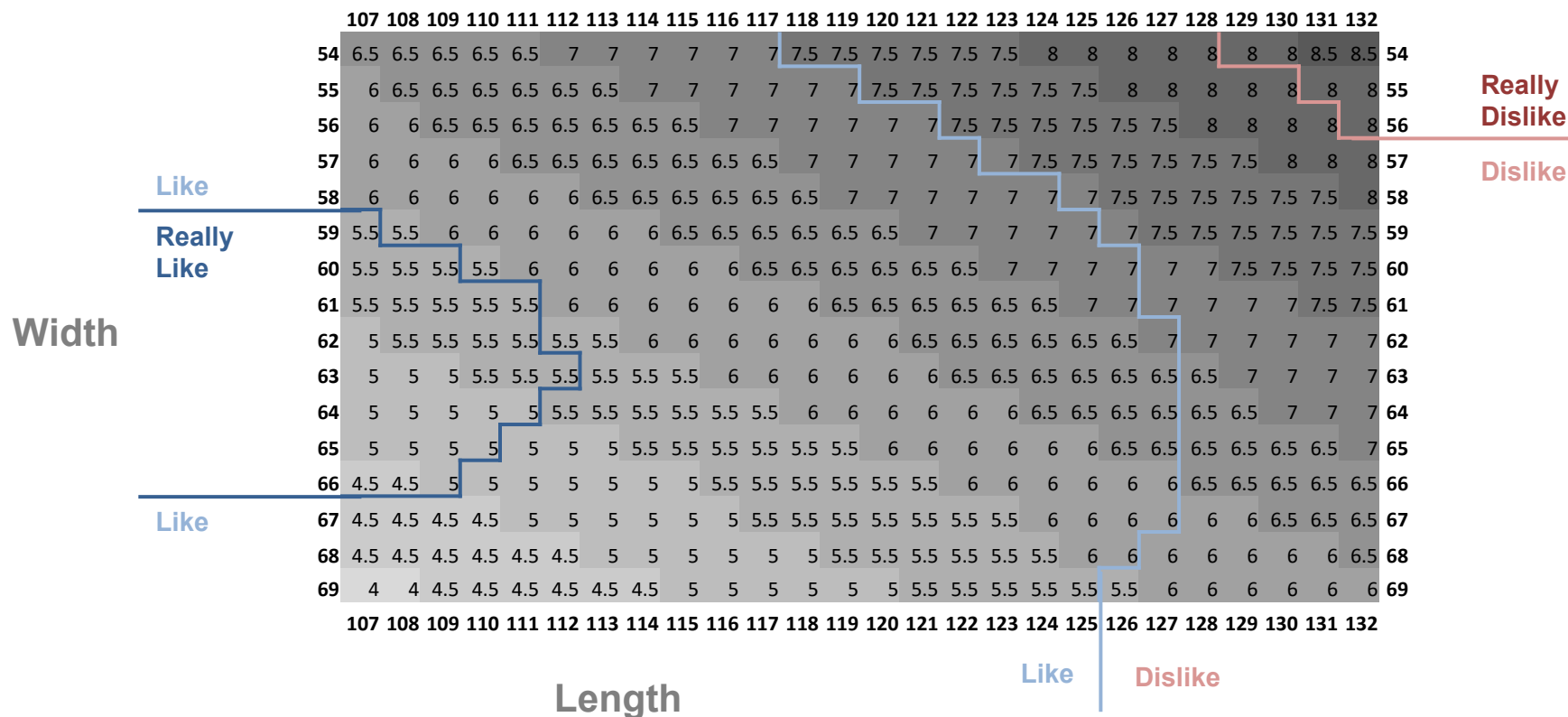
		107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132				
Width	54	6.5	6.5	6.5	6.5	6.5	7	7	7	7	7	7	7.5	7.5	7.5	7.5	7.5	7.5	8	8	8	8	8	8	8	8.5	8.5	54	Really Dislike		
	55	6	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	7	7	7	7.5	7.5	7.5	7.5	7.5	7.5	8	8	8	8	8	8	8	8		55	
	56	6	6	6.5	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	7	7	7	7.5	7.5	7.5	7.5	7.5	7.5	8	8	8	8	8	8		56	
	57	6	6	6	6	6.5	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	7	7	7	7.5	7.5	7.5	7.5	7.5	7.5	7.5	8	8	8		57	Dislike
	58	6	6	6	6	6	6	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	7	7	7	7	7.5	7.5	7.5	7.5	7.5	7.5	7.5	8	58		
	59	5.5	5.5	6	6	6	6	6	6	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	7	7	7	7.5	7.5	7.5	7.5	7.5	7.5	7.5	59		
	60	5.5	5.5	5.5	5.5	6	6	6	6	6	6.5	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	7	7	7	7	7.5	7.5	7.5	7.5	60		
	61	5.5	5.5	5.5	5.5	5.5	6	6	6	6	6	6	6	6	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	7	7	7	7.5	7.5	61		
	62	5	5.5	5.5	5.5	5.5	5.5	5.5	6	6	6	6	6	6	6	6	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	7	7	7	62		
	63	5	5	5	5.5	5.5	5.5	5.5	5.5	5.5	6	6	6	6	6	6	6	6.5	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	7	63		
	64	5	5	5	5	5	5.5	5.5	5.5	5.5	5.5	5.5	5.5	6	6	6	6	6	6.5	6.5	6.5	6.5	6.5	6.5	6.5	7	7	7	64		
	65	5	5	5	5	5	5	5	5.5	5.5	5.5	5.5	5.5	5.5	5.5	6	6	6	6	6	6	6.5	6.5	6.5	6.5	6.5	6.5	7	65		
	66	4.5	4.5	5	5	5	5	5	5	5	5	5.5	5.5	5.5	5.5	5.5	5.5	6	6	6	6	6	6	6	6.5	6.5	6.5	6.5	66	Like	
	67	4.5	4.5	4.5	4.5	5	5	5	5	5	5	5.5	5.5	5.5	5.5	5.5	5.5	5.5	6	6	6	6	6	6	6	6	6.5	6.5	6.5		67
	68	4.5	4.5	4.5	4.5	4.5	4.5	5	5	5	5	5	5	5	5.5	5.5	5.5	5.5	5.5	5.5	6	6	6	6	6	6	6	6	6.5		68
	69	4	4	4.5	4.5	4.5	4.5	4.5	4.5	5	5	5	5	5	5	5	5.5	5.5	5.5	5.5	5.5	5.5	6	6	6	6	6	6	6		69
		107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132				

Optimal Thickness

Discussion

- For width/length combinations that users like, optimum thickness ranges from 4mm to 7mm
 - Optimum thickness increases with length
 - Optimum thickness decreases with width
- For width/length combinations that users really like, optimum thickness ranges from 4.5-5.5mm

Thickness	
Range of Interest	4-14mm
Range that users "really like"	min-6mm
Range that users "like"	min-13mm



Thickness Range

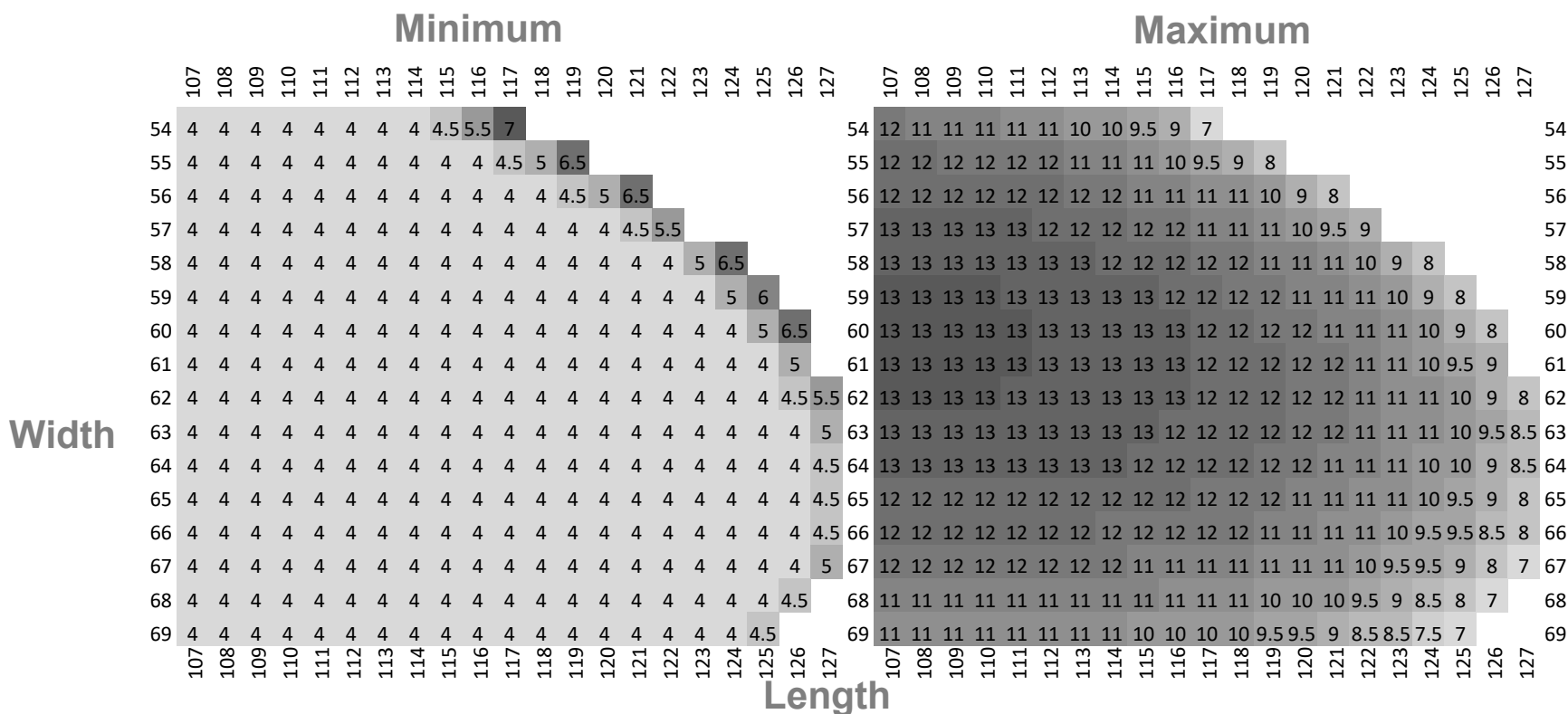
Description of the Table

- We predicted how much users will like every combination of width, length and thickness
- Values in the table represent the range of thickness that the model predicts will be "liked" (≥ 2.5 on a scale of 1, 2, 3, or 4)

Discussion

- The maximum liked width is 127mm (we tested up to 132mm)
- The range of liked thickness is largest for width ≈ 64 , and decreases as the width increases or decreases
- The range is largest for the shorter lengths and decreases as the device gets longer, especially for very narrow or wide devices
- $>13\text{mm}$ is not predicted to be acceptable for any combination of width/length
- The range of liked thickness is small at the edges of the colored areas below

Thickness	
Range of Interest	4-14mm
Range that users "really like"	min-6mm
Range that users "like"	min-13mm



Optimal Length

Description of the Table

- We predicted how much users will like every combination of width, length and thickness
- The values in the table are the optimal length for each combination of width x thickness
- The four zones represent how much users like/dislike a particular combination of width/length/thickness
 - Deviations in length from the values in the table decrease the predicted "like" rating
 - Really dislike (predicted rating 1 <= 1.5)
 - Dislike (< 2.5)
 - Like (< 3.5)
 - Really Like (<= 4) – this category was empty

Range of Interest

Length

107-132mm

Range that users "really like"

min-108mm

Range that users "like"

min-127mm

Width

Like

Really
Like

Like

	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10	10.5	11	11.5	12	12.5	13	13.5	14	
54	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	54
55	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	55
56	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	56
57	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	57
58	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	58
59	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	59
60	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	108	60
61	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	108	108	61
62	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	108	108	109	62
63	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	108	108	109	109	63
64	107	107	107	107	107	107	107	107	107	107	107	107	107	107	107	108	108	108	109	109	110	64
65	107	107	107	107	107	107	107	107	107	107	107	107	107	107	108	108	109	109	109	110	110	65
66	107	107	107	107	107	107	107	107	107	107	107	107	107	108	108	109	109	109	110	110	111	66
67	107	107	107	107	107	107	107	107	107	107	107	108	108	108	109	109	110	110	110	111	111	67
68	107	107	107	107	107	107	107	107	107	107	108	108	108	109	109	110	110	110	111	111	112	68
69	107	107	107	107	107	107	107	107	107	108	108	109	109	109	110	110	111	111	111	112	112	69
	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10	10.5	11	11.5	12	12.5	13	13.5	14	

Thickness

Like

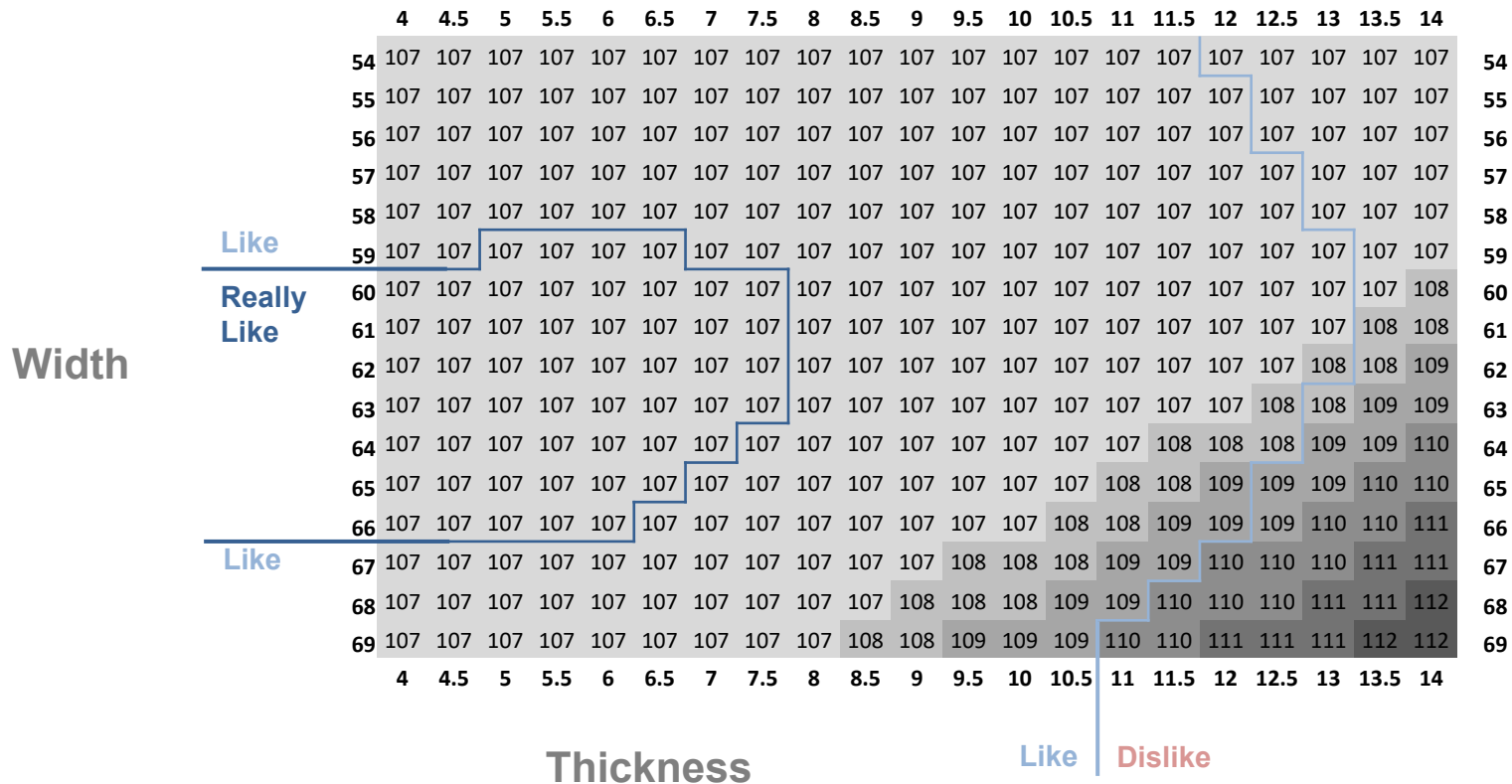
Dislike

Optimal Length

Discussion

- For width/thickness combinations that users like, optimum length ranges from 107-109mm
 - Optimum length increases with thickness
 - Optimum length increases with width
- For width/thickness combinations that users really like, optimum length is 107mm
- The actual optimum length is likely to be shorter, outside of the bounds of sizes tested in this study
 - This could explain why X tested relatively well despite its wide form, it was the shortest device in the study at 105mm. It was excluded from this analysis because its dimensions were not drawn from the central composite design used to pick dimensions for the study

Length	
Range of Interest	107-132mm
Range that users “really like”	min-108mm
Range that users “like”	min-127mm



Length Range

Description of the Table

- We predicted how much users will like every combination of width, length and thickness
- **The minimum length was 107**, the minimum tested, for all combinations of width x thickness
- Values in the table represent the maximum length that the model predicts will be "liked" (≥ 2.5 on a scale of 1, 2, 3, or 4)

Range of Interest

Length

107-132mm

Range that users “really like”

min-108mm

Range that users “like”

min-127mm

Discussion

- **The maximum length is 127mm**
- The maximum length is centered 4.5-8.5mm thickness and 62-67mm wide, and drops in all directions from there
- For thick devices, the model predicts that users will not like very narrow or very wide devices

[illegible]

Optimal Width

Description of the Table

- We predicted how much users will like every combination of width, length and thickness
- The values in the table are the optimal length for each combination of width x thickness
- The four zones represent how much users like/dislike a particular combination of width/length/thickness
 - Deviations in length from the values in the table decrease the predicted “like” rating
 - Really dislike (predicted rating $1 \leq 1.5$)
 - Dislike (< 2.5)
 - Like (< 3.5)
 - Really Like (≤ 4) *this category was empty*

Range of Interest

Range that users “really like”

Range that users “like”

Width

54-69mm

61-66mm

full

		107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132				
Thickness	Really Like	4	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	65	65	65	65	65	65	65	65	65	66	4		
		4.5	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	65	65	65	65	65	65	65	65	65	4.5		
		5	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	65	65	65	65	65	65	65	5		
		5.5	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	65	65	65	65	65	65	5.5		
		6	62	62	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	65	65	65	65	65	6		
		6.5	62	62	62	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	65	65	65	6.5	
		7	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	65	7		
	Like	7.5	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	7.5		
		8	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	8		
		8.5	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	8.5			
		9	61	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	9		
		9.5	61	61	62	62	62	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	64	64	9.5		
		10	61	61	61	61	62	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	64	64	10	
		10.5	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	64	10.5		
Dislike	11	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	11		
	11.5	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	11.5		
	12	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	12			
	12.5	60	60	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	12.5			
	13	60	60	60	61	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	13			
	13.5	60	60	60	60	61	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	13.5			
	14	60	60	60	60	60	60	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	63	14			
		107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132				
																										Really Dislike					
																										Dislike					

Optimal Width

Discussion

- For thickness/length combinations that users like, optimum width ranges from 60mm to 65mm
 - Optimum width increases with length
 - Optimum width decreases with thickness
- For thickness/length combinations that users really like, optimum width ranges from 62-63mm

	Width
Range of Interest	54-69mm
Range that users “really like”	61-66mm
Range that users “like”	full

		107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132			
Thickness	Really Like	4	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	65	65	65	65	65	65	65	65	65	66	4		
		4.5	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	65	65	65	65	65	65	65	65	65	4.5	
		5	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	65	65	65	65	65	65	65	5	
		5.5	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	65	65	65	65	65	65	5.5	
	6	62	62	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	65	65	65	65	65	6		
	6.5	62	62	62	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	64	65	65	6.5		
	7	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	65	65	7		
	7.5	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	64	64	64	7.5		
	Like	8	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	8	
		8.5	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	64	8.5	
	Dislike	Like	9	61	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	63	64	64	64	64	64	9
			9.5	61	61	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	63	64	64	64	9.5	
			10	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	64	64	10	
			10.5	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	64	10.5	
11		61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	63	11		
11.5		61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	63	63	63	63	63	63	63	63	11.5		
12		61	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	63	63	63	12		
12.5		60	60	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	63	63	12.5		
Really Dislike		13	60	60	60	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	63	13	
		13.5	60	60	60	60	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	62	63	63	13.5	
	14	60	60	60	60	60	60	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	62	62	62	62	63	14		
		107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132			
		Length																				Dislike	Really Dislike							

Width Range

Description of the Table

- We predicted how much users will like every combination of width, length and thickness
- **The minimum length was 107**, the minimum tested, for all combinations of width x thickness
- Values in the table represent the maximum length that the model predicts will be “liked” (≥ 2.5 on a scale of 1 to 5)

Discussion

- **The maximum length is 127mm**
- The maximum length is centered 4.5-8.5mm thickness and 62-67mm wide, and drops in all directions from there
- For thick devices, the model predicts that users will not like very narrow or very wide devices

	Width
Range of Interest	54-69mm
Range that users “really like”	61-66mm
Range that users “like”	full

Minimum

[illegible]

Maximum

	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127		
4	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	67	4		
4.5	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	66	4.5	
5	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	67	5	
5.5	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	67	5.5	
6	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	67	6	
6.5	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	67	6.5	
7	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	67	7	
7.5	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	67	66	7.5
8	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	68	67	66	8
8.5	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	67	66	64	8.5
9	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	68	67	67	65	9	
9.5	69	69	69	69	69	69	69	69	69	69	69	69	69	69	69	68	68	67	67	66	63	9.5	
10	69	69	69	69	69	69	69	69	69	69	69	69	69	68	68	68	67	66	65	64	10		
10.5	69	69	69	69	69	69	69	69	69	68	68	68	68	67	67	67	66	65	63	10.5			
11	68	68	68	68	68	68	68	68	68	67	67	67	66	66	65	64	11	11					
11.5	67	67	67	67	67	67	67	67	66	66	66	66	65	64	63	11.5	11.5						
12	66	66	66	66	66	66	66	65	65	65	64	64	63	12	12								
12.5	64	64	64	64	64	64	64	64	63	62	12.5	12.5											
13	62	62	62	62	61	13	13																

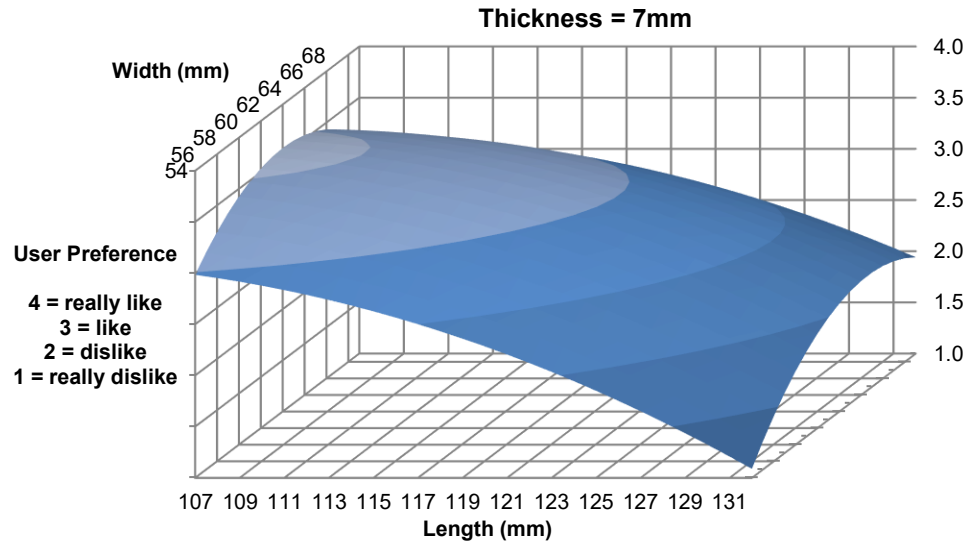
Thickness

Length

Width, Length at Thickness = 7mm

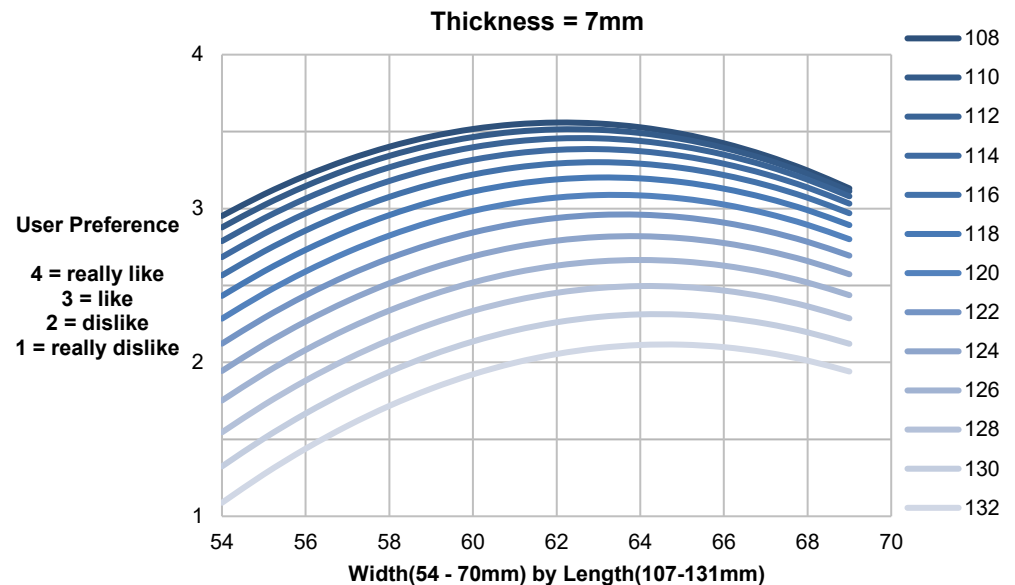
Description of the Graph

- Graph shows how much users like various combinations of width, length at Thickness = 7mm
- The color bands correspond to ranges of "like"
 - Lighter blues are liked more
- The two graphs show the same data
 - Top graph is a surface graph
 - Bottom graph shows a line graph for each length



Discussion

- Like value decreases with the length of the devices
 - The drop in liking for length is much more than for width
- The width has a peak, and liking drops as the width gets either longer or shorter
 - At 107mm length, the peak width is ~62mm
 - At 131mm length, the peak width is ~64mm
- The best devices are
 - Short
 - Moderate width, centered around 63mm
- Within the range tested, at 7mm thick
 - Users like all widths
 - Users like length $\leq$ ~127mm at moderate widths
 - Users like length $\leq$ ~117mm at all widths



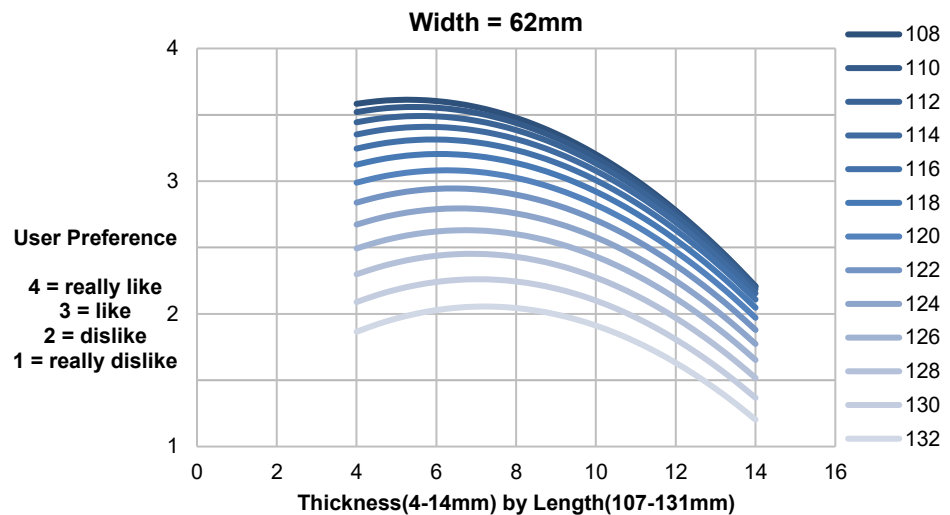
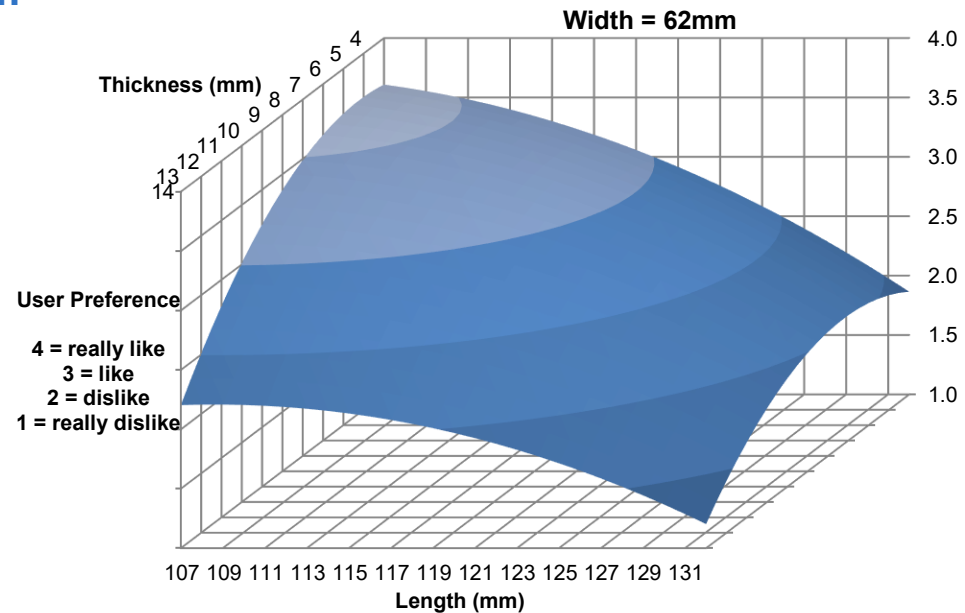
Length, Thickness at Width = 62mm

Description of the Graph

- Graph shows how much users like various combinations of thickness, length at width = 62mm
- The color bands correspond to ranges of "like"
 - Lighter blues are liked more
- The two graphs show the same data
 - Top graph is a surface graph
 - Bottom graph shows a line graph for each length

Discussion

- Like value decreases with the length of the devices
- Like value decreases with the thickness of the device
 - For longer lengths, the peak is a little higher than the minimum length, peaking at ~7mm
- The best devices are
 - Thin
 - Short
- Within the range tested, at 62mm wide
 - Users like thickness ≤ 11 mm
 - Users like length $\leq \sim 127$ mm



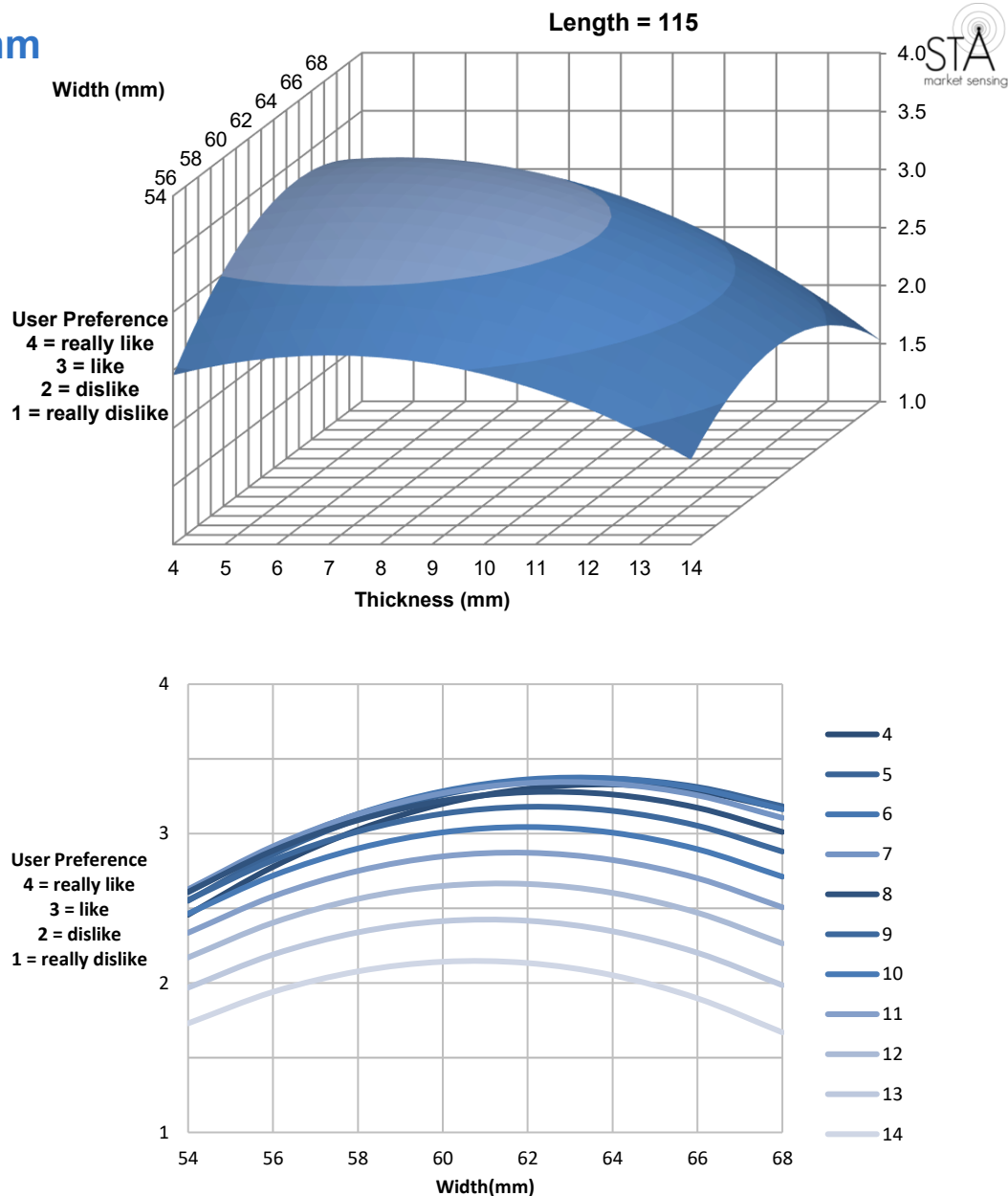
Width, Thickness at Length = 115mm

Description of the Graph

- Graph shows how much users like various combinations of width, thickness at length = 115mm
- The color bands correspond to ranges of “like”
 - Lighter blues are liked more
- The two graphs show the same data
 - Top graph is a surface graph
 - Bottom graph shows a line graph for each thickness

Discussion

- Like value decreases as the width drops away from the optimum
 - Optimum is ~63mm for thin devices
 - Optimum is about 61mm for thicker devices
- Like value decreases as the thickness drops away from the optimum
 - Optimum is ~9mm for narrow devices
 - Optimum is ~6mm for wider devices
- The best devices are
 - Somewhat thin, centered around ~7-8mm
 - Moderate width, centered around 63mm
- Within the range tested, at 115mm long
 - Users like all widths and thicknesses
 - Users dislike wide/thin and narrow/thick devices





Method

Study Format and Materials

Study Overview

Goal: Explore User Preferences for Size

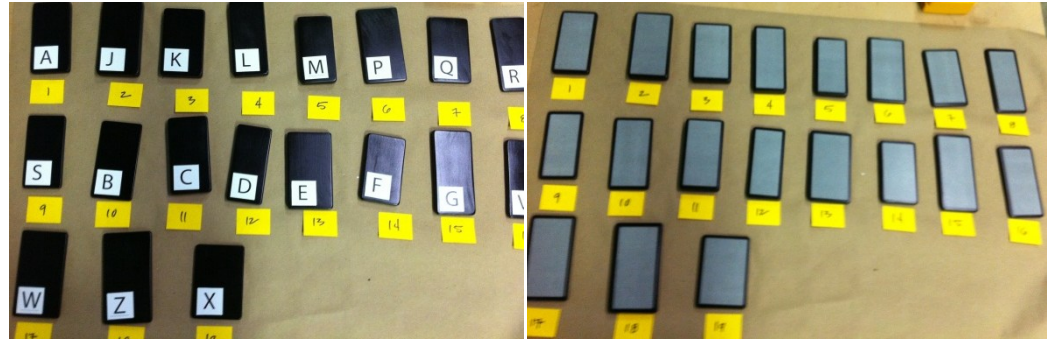
- What is the optimal range for each dimension?
- How do the dimensions affect each other?
 - Example: does the thickness of the device change how wide users want it to be?

Approach

- Create 19 different physical models
- Models represent a variety of sizes
 - Different width, length and thickness
 - 15 models represented strictly manipulated combinations of width, length and thickness
 - Short, thick, wide • Long, thick, wide
 - Short, thick, narrow • Long, thick, narrow
 - Short, thin, wide • Long, thin, wide
 - Short, thin, narrow • Long, thin, narrow
 - One model that was average on all dimensions
 - Six models that were average on two dimensions and extreme on the third: short, long, wide, narrow, thick or thin
- 4 devices represented specific model dimensions of interest

Study Format

- Participants spent 2 hours with the 19 device models
- For each model, participants rated
 - Where would you carry the device?
 - How comfortable is it to hold in your hand?
 - What do you think of the width, length, thickness?
 - Overall, do you like or dislike the size/shape?
- Which are your three favorite models?
- Which are your three least favorite models?
- The session concluded with a group discussion of why they liked and disliked the different models



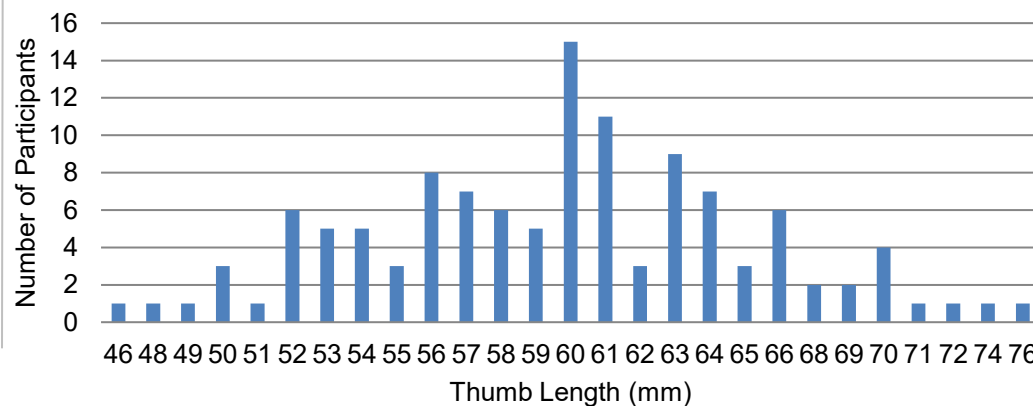
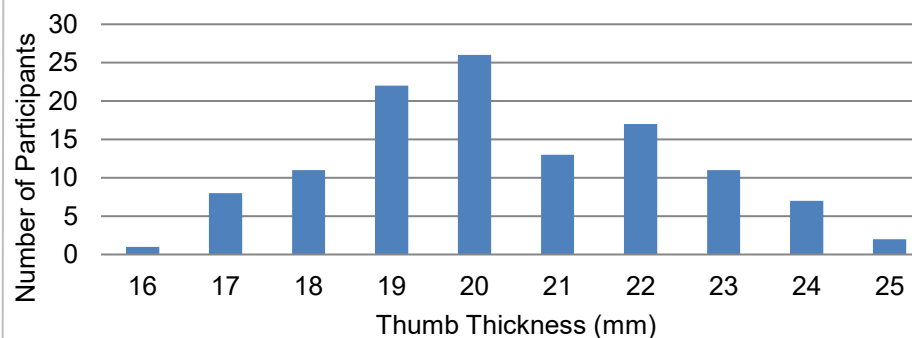
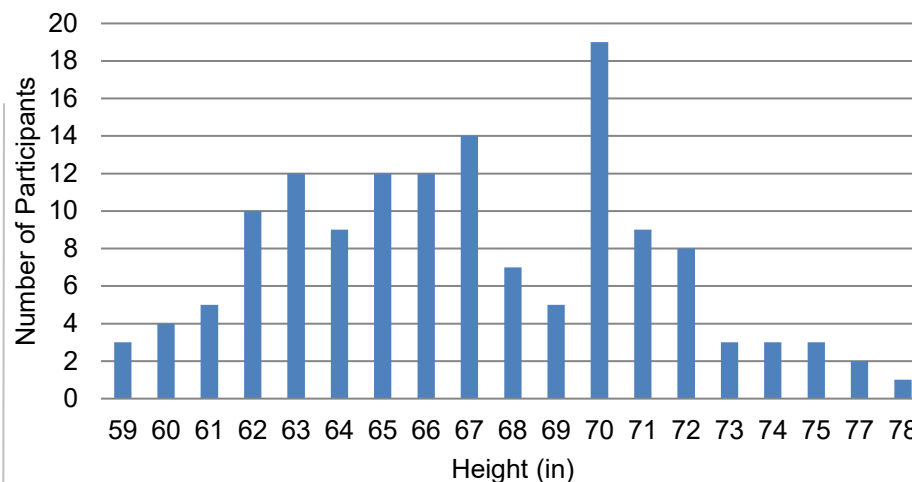
Participant Demographics

Participant Screening Criteria

- 120 participants
- Users that own
 - a touchscreen only phone
 - a touchscreen phone with a physical keyboard, or
 - intend to purchase a touchscreen phone in the next year

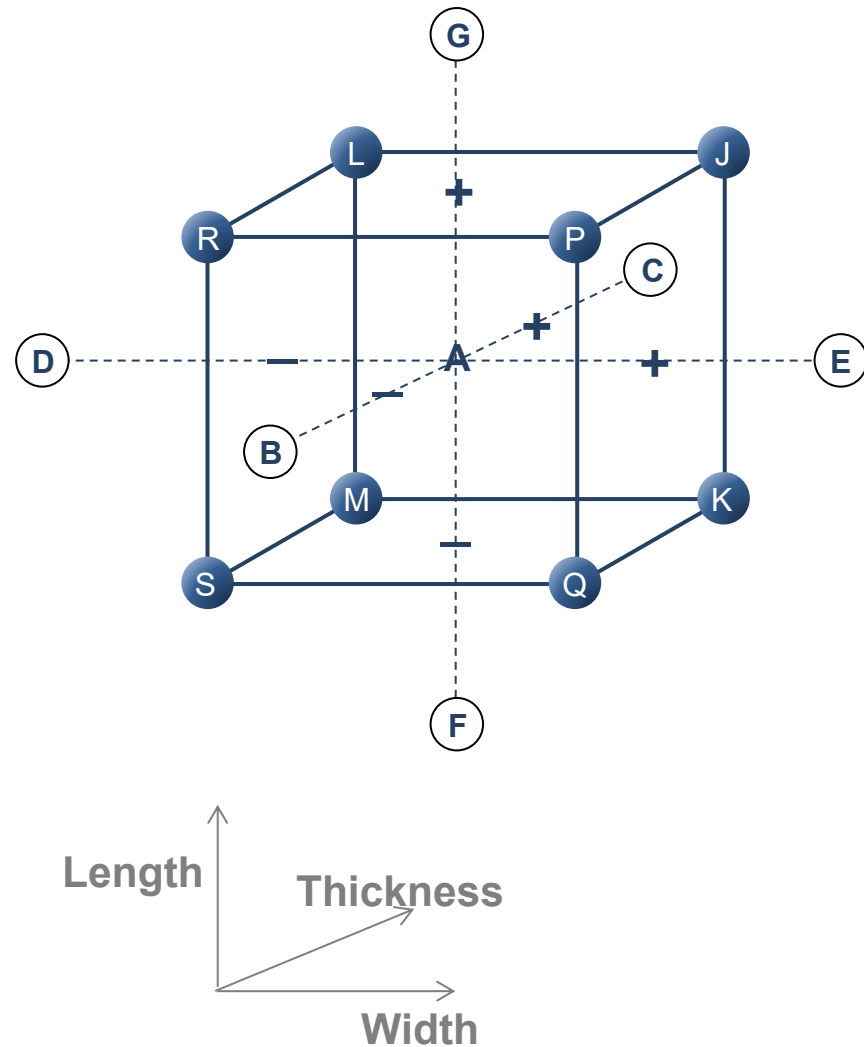
Participant Anthropometrics

- Participants were recruited to get an even split of users in the bottom 15%, middle 70% or top 15% of the US Adult height distribution
 - Short: 57-62 in (22 people)
 - Medium: 63-72 in (107 people)
 - Tall: 73-81 in (12 people)
- Thumb thickness and length were measured during the session to gain an understanding of distribution.



Study Approach

- Create a set of models that represent a range of length, width and thickness
 - The technique used to select points is called a “central composite design”
- The diagram to the right shows how the specific model dimensions were chosen
 - The model is of 3D space, with the dimensions being length, width and thickness
 - Each point represents one model created for the study
- Model A is the center point
 - Average on all dimensions
- The solid/blue points that form the square systematically vary the dimensions
 - Adjacent points share two of the same dimensions and vary only on the third. This helps isolate the effects of changing a particular dimension
 - The eight points represent the following model designs:
 - Short, thick, wide
 - Short, thick, narrow
 - Short, thin, wide
 - Short, thin, narrow
 - Long, thick, wide
 - Long, thick, narrow
 - Long, thin, wide
 - Long, thin, narrow
- The outline/white points that form the star represent points that are extreme on one dimension and average on the other two



Central Composite Design (CCD)

Theoretical Basis

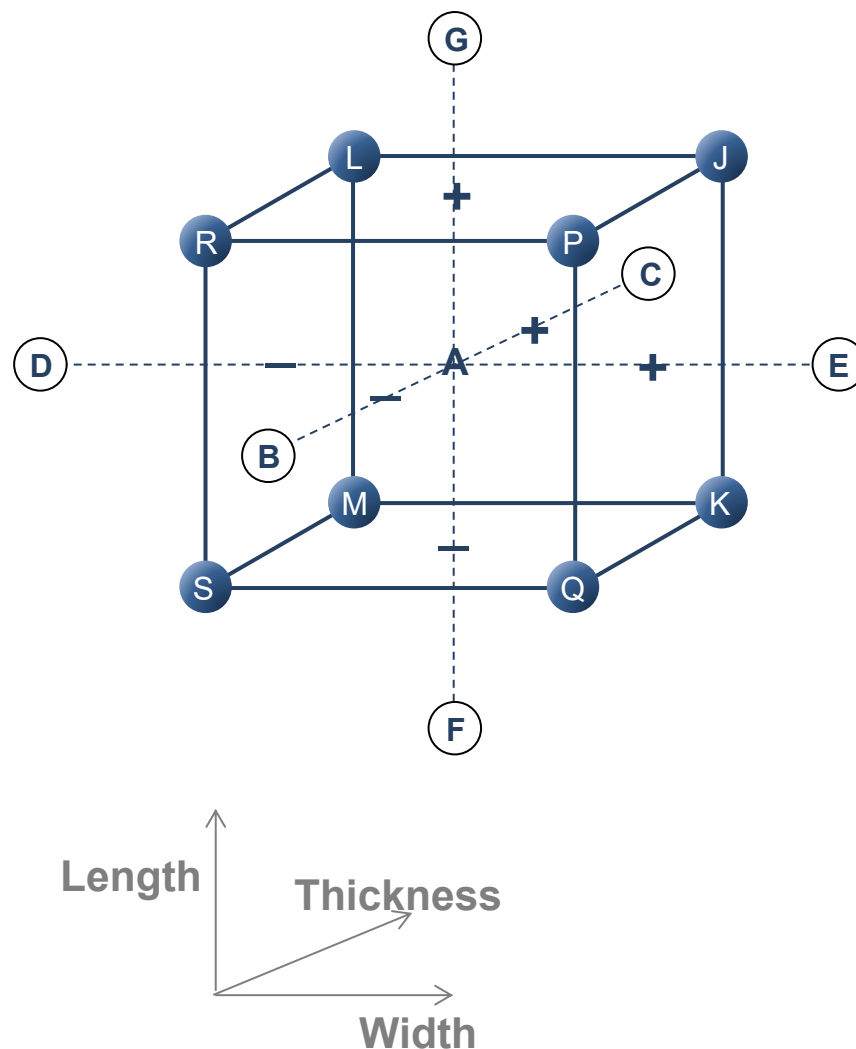
- CCD is a technique for selecting a subset of points to test in a factorial design
- The square points (solid blue dots) outline the primary area of interest
 - Adjacent points (connected by lines) share 2 of the 3 dimensions
 - By varying one dimension at a time, we can see the effects of each dimension independent of the others
- The star/axial points extend beyond the square
 - This allows us to check for quadratic relationships
- The five levels are usually coded as $-\alpha, -1, 0, 1, \alpha$
For 3 factors, $\alpha = 1.682$
- It is a very efficient technique, using only 15 points to test 3-factor space
 - A full factorial design of the same space would require $5 \times 5 \times 5 = 125$ points
 - A full factorial design with just 3 levels would require $3 \times 3 \times 3 = 27$ points and you would be unable to test for quadratic relationships

Analysis Approach

- Linear regression of all points tested
- Analysis can test for main effects, square effects, and two-way interactions (2nd order effects), but not three-way interactions

Additional Information

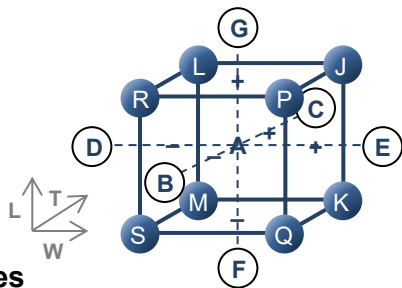
- <http://www.itl.nist.gov/div898/handbook/pri/section3/pri3361.htm>



Devices Mockup Models Tested

Device Selection

- 15 device sizes were selected based on points needed to execute the central composite design and 4 device sizes were selected based on research interests for a total of 19 devices for testing
- These devices covered a range of currently existing and experimental touchscreen smartphone dimensions



Prototypes

- 19 prototypes were created varying in length, width, thickness, and screen size
- Weights varied from 79g to 158g based on device size/volume (constant density)
- Device screens were represented throughout the majority of each session by gray panels to ensure participants focused on variations in physical size rather than display
- To evaluate device physical size with the additional context of screen size, screen overlays were printed for each device
 - Screen overlays simulated text and images in the form of both a full webpage and a mobile webpage in portrait and landscape orientations

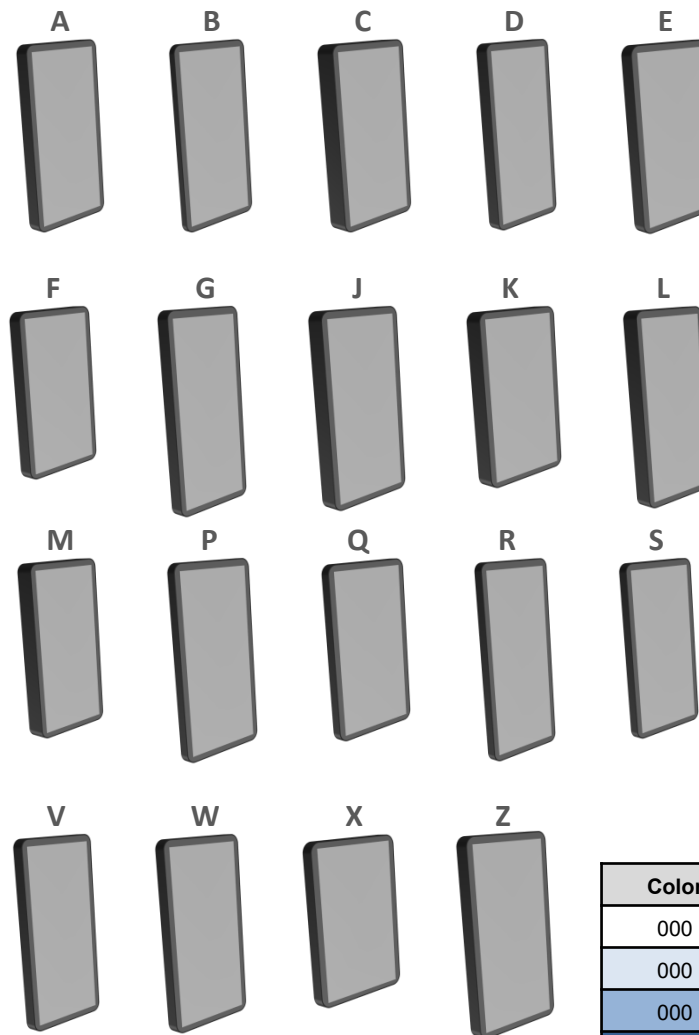
Color	Meaning
000	Minimum
000	
000	Average

Color	Meaning
000	
000	Maximum
000	Other

Letter	Point Assignment	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
A	Center	119	61	9	124mm (4.9")	118
J	Cube	127	66	12	132mm (5.2")	155
K	Cube	112	66	12	119mm (4.7")	142
L	Cube	127	57	12	129mm (5.1")	140
M	Cube	112	57	12	114mm (4.5")	129
P	Cube	127	66	6	132mm (5.2")	101
Q	Cube	112	66	6	119mm (4.7")	95
R	Cube	127	57	6	130mm (5.1")	94
S	Cube	112	57	6	114mm (4.5")	88
B	Axial (Star)	119	61	4	124mm (4.9")	79
C	Axial (Star)	119	61	14	124mm (4.9")	158
D	Axial (Star)	119	54	9	122mm (4.8")	110
E	Axial (Star)	119	69	9	127mm (5.0")	127
F	Axial (Star)	107	61	9	112mm (4.4")	111
G	Axial (Star)	132	61	9	135mm (5.3")	126
V	Other	123	60	8	127mm (5.0")	107
W	Other	123	66	8	127mm (5.0")	113
Z	Other	129	69	9	135mm (5.3")	133
X	Other	105	73	8	117mm (4.6")	109

Summary of Mockups

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal
A	119	61	9	124mm (4.9")
J	127	66	12	132mm (5.2")
K	112	66	12	119mm (4.7")
L	127	57	12	129mm (5.1")
M	112	57	12	114mm (4.5")
P	127	66	6	132mm (5.2")
Q	112	66	6	119mm (4.7")
R	127	57	6	130mm (5.1")
S	112	57	6	114mm (4.5")
B	119	61	4	124mm (4.9")
C	119	61	14	124mm (4.9")
D	119	54	9	122mm (4.8")
E	119	69	9	127mm (5.0")
F	107	61	9	112mm (4.4")
G	132	61	9	135mm (5.3")
V	123	60	8	127mm (5.0")
W	123	66	8	127mm (5.0")
Z	129	69	9	135mm (5.3")
X	105	73	8	117mm (4.6")



Color	Meaning
000	Minimum
000	
000	Average
000	
000	Maximum
000	Other

Prototype Construction

19 prototypes were constructed according to Samsung's specifications

Construction parameters

- 15 devices were chosen for the 15 points of the Central Composite Design, to systematically vary the device dimensions (this selection is described elsewhere)
- 4 devices were made to specifications provided by Samsung
- All screens had a 2mm border around the edge
- All devices were weighted to simulate a constant density which was slightly higher than the foam used

Construction process

- Each prototype was created from high density foam, the edges and corners were slightly rounded and all surfaces were sanded smooth
- Holes were drilled into the foam prototypes and filled with the appropriate weights (Photo A)
- The open area around weights was filled with epoxy and trimmed (Photo B)
- The prototypes were finished with multiple coats of black spray paint and a protective clear coat (Photo C)
- Gray screens were pasted on top of the devices, (Photo D)



Photo A



Photo B



Photo C



Photo D

* The construction photos are from a previous study but the process was similar

Overlays

Printed overlays were created to help users think about using the screen of the device.

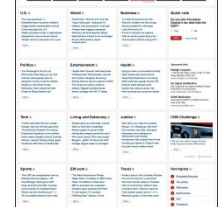
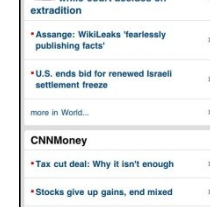
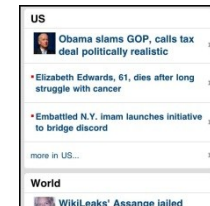
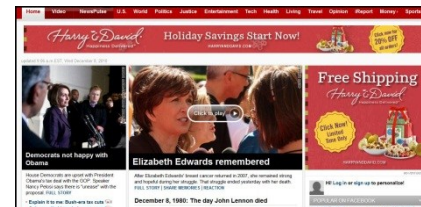
Design

- A screen capture of the CNN homepage was taken, pasting together several captures to get a very tall page
- A second capture of the mobile page was taken on a mobile device
- For each dimension, the screen capture was scaled to the width of the screen and then cropped to the height
- The overlays were printed on a high-resolution color printer

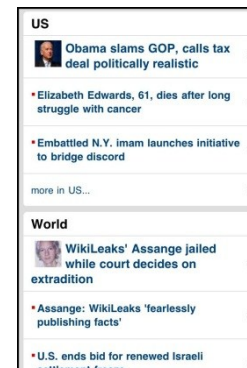
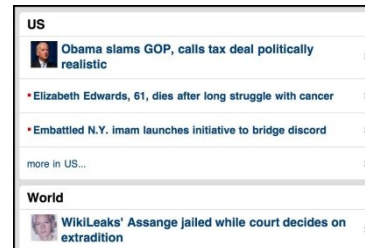
Alternate Approaches

- We considered simulating pixel density of a real screen
 - Pixel densities change over time; we did not want current technology to limit the study results
 - The screen sizes varied considerably, and we were unable to come up with a fair algorithm that could be applied consistently
- We considered a combination of zooming and cropping to the width of the device.
 - For example, a user would be unlikely to view a full web page unzoomed on a small screen
 - We were unable to come up with a fair algorithm that could be applied consistently

Device A Overlays



Device X Overlays



Study Setup

Room Layout

- Two tables were setup with 6 participants at each table (Photo A). Participants remained at the same seat for the entire session.
- The front of the room included a table where all 19 prototype devices were placed for the overall most and least preferred ranking portion and a white board for facilitating focus group discussions.
- Prototype devices were stored in 4 bins at the front of the room during the moderator's initial instructions and explanations.
- On the wall opposite to that was the observation 2 way mirror (Photo B)



Photo A



Photo B

Camera Recording Layout

- Two cameras were set up to record the videos of the 12 sessions (Photo C)
- Each camera captured 6 of the 12 participants from the front in order to record their interactions with each of the 19 devices



Photo C

Materials Used

- A printed packet containing a user questionnaire, individual device questionnaire, and an overall device questionnaire were provided to each participant.
- A checklist and a pen was provided for tracking their rounds completed and device letters evaluated.

Session Overview

2-hour sessions

Moderator provides introduction, session overview, and instructions

User Questionnaire

- Participants answered questions about themselves and their current phone
- Completed during free time during session

Thumb Documentation

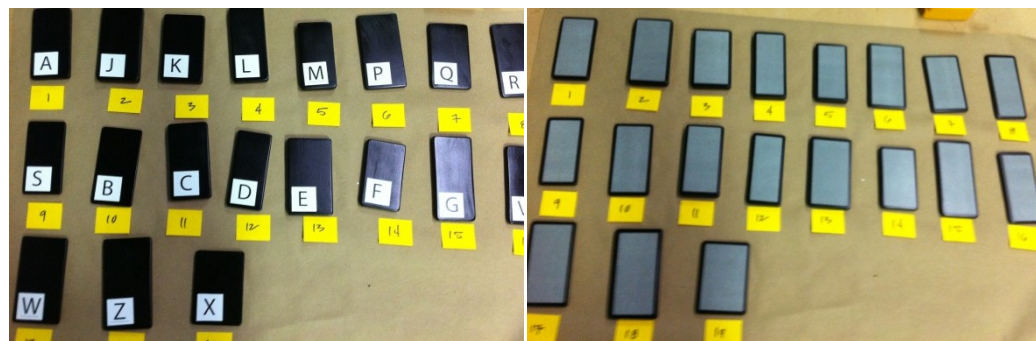
- Participants' thumb length and thickness were measured using electronic calipers

Device Questionnaire & Overall Questionnaire

- Timeframe: 1 hour 30 minutes - Self Paced
- Participants spent time independently with each device filling out questionnaires about their preferences and opinions on device size as well as their ability to easily and comfortably reach areas on the screen, hold, and carry the devices
- Participants were asked to list top 3 most preferred and bottom 3 least preferred devices
- Screen overlays were provided and they were asked to rerate their top 3 most preferred devices after seeing text and images

Group Discussion

- Timeframe: 30 minutes - Moderator Guided
- Participants engaged in a 20-30 minute focus group discussion at the end of every session
- Participants were asked what device characteristics and anticipated usage influenced their decisions and if the screen overlays caused them to change their top 3 most preferred devices



User Questionnaire

User Questionnaire Activity

- Each participant had a questionnaire packet that was given to them at check in
- The questionnaire was completed within the participants' free time throughout the session

User Questionnaire Content

- Electronic devices owned
- Current Phone
 - Brand and model
 - Features
 - Method of carrying
 - Likelihood of purchasing a phone with full physical QWERTY keyboard
 - Decision factors to purchase phone
- Hands & Fingers
 - Hand used for writing
 - Ear used for phone calls
 - Hand used for one-handed tasks with small electronic devices (button pressing)
 - Thumbnaill length
 - Using fingers or thumbs to press buttons
- Tradeoffs & Comparisons
 - Touchscreen phone owned/like to own/considered buying due to size, weight and screen resolution

USER QUESTIONNAIRE – SECTION 1

GENERAL

- Day (circle one): MON TUES WED THURS FRI
- Session Start Time (circle one): 9:00A 11:30A 12:00P 1:00P 2:30P 3:00P 5:00P
- Hanging Letter provided at check-in: _____
- What is your first name and last initial? _____

YOUR CURRENT ELECTRONICS & PHONE

- Which of the following types of electronic devices do you currently own?
(Mark all that apply.) Does it have a physical keyboard? (circle Y or N)

☐ Smartphone	Y / N
☐ Ultraportable/Ultrabook (6-7" screen)	Y / N
☐ Tablet computer	Y / N
☐ Notebook computer	
☐ Laptop computer	
☐ Desktop Computer	
- What is the brand name and model of the phone you currently own?
If you don't know the model, it's usually printed near or underneath the battery.

USER QUESTIONNAIRE – SECTION 1

- Think about how you typically use your phone. What percentage of time do you spend using your phone in the following orientations? *(circle percentages adding up to 100%)*

One-hand hold with same hand interaction	One-hand hold with other hand interaction	Two-hand hold with two-handed interaction	Sitting on the table
- How likely is it that your next phone will have a full physical QWERTY keyboard? *(mark one)*

Definitely a full physical QWERTY 1	Probably a full physical QWERTY 2	Unlikely 3	Definitely touchscreen only 4	Probably touchscreen only 5
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CARRYING YOUR PHONE

- How frequently do you carry your phone in the following locations? *(rate each)*

Most Days 75%	Frequently 50%	Sometimes 25%	Rarely 10%	Never
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USER QUESTIONNAIRE – SECTION 1

HANDEDNESS

- Mitch hand do you write with? *(mark one)*
 - ☐ Left
 - ☐ Right
- When you talk on the phone, which ear are you more likely to use? *(mark one)*
 - ☐ Left
 - ☐ Right
 - ☐ Both equally
- When doing one-handed tasks such as dialing numbers on a phone, using a calculator, ATM, microwave, or small digital music player such as an iPod, which hand do you tend to use to press the buttons? *(mark one)*

Always left 1	Usually left 2	More often both 3	Both equally 4	More often right 5	Usually right 6	Always right 7
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HANDS & FINGERS

- How long do your fingernails tend to be? *(mark one)*
 - ☐ 1. Shorter than my fingertip
 - ☐ 2. About even with my fingertip
 - ☐ 3. A little past the end of my fingertip
 - ☐ 4. Very long
- When holding a small touchscreen device in the same hand that is tapping buttons, how often do you tap the screen with the tip of your finger versus the flat/pad of your finger? *(mark one)*
 - ☐ 1. Never use the tip - my nails are too long
 - ☐ 2. Never use the tip - other reasons
 - ☐ 3. Seldom use the tip - 10 %
 - ☐ 4. Sometimes use the tip - 25 %
 - ☐ 5. Frequently use the tip - 50 %
 - ☐ 6. Usually use the tip - 75 %
 - ☐ 7. Always use the tip - 100 %
- When using a QWERTY touchscreen keyboard to type on a phone, do you tend to use your fingers or thumbs to press the buttons? *(mark one)*

Always use fingers 1	Usually use fingers 2	More often use fingers 3	Both equally 4	More often use thumb 5	Usually use thumb 6	Always use thumb 7
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USER QUESTIONNAIRE – SECTION 1

TRADEOFFS & COMPARISONS

- Is there a touchscreen phone out on the market that you would like to own if it were a different size or shape? *(If you have not or do not research phones on the market, you can leave this question blank)*

- Is there a touchscreen phone you seriously considered buying but didn't because it was too big? *(If you have not or do not research phones on the market, you can leave this question blank)*

- Have you owned any touchscreen or smartphones before your current phone? How do they compare? Why did you switch? We're particularly interested in any issues related to size, weight, and screen resolution.

PLEASE STOP HERE until asked to continue

Individual Device Evaluation and Questionnaire

Individual Device Evaluation

- The sessions involved 12 people working simultaneously.
- The devices were placed in 4 bins
 - 3 bins contained 5 models
 - 1 bin contained 4 models
- Devices were randomly placed into the bins with the intent to achieve a random yet even distribution of lengths, widths, and thicknesses.
- 12 participants were broken into groups of 3, each group was randomly assigned a bin.
- Once all groups had completed evaluating devices in their assigned bins, they were asked to pass the bin to another group.
- Using a device checklist, participants tracked bin numbers and device letters as they completed their evaluations.

Device Questionnaire Activity

- Participants spent time independently with each device filling out questionnaires about their preferences and opinions on device size as well as their ability to easily and comfortably reach areas on the screen, hold, and carry the devices.

DEVICE ORDER	DEVICE LETTER	BIN
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		

Cross out the device letters as you start them
Sample: *AC*

A
B
C
D
E
F
G
J
K
L
M
P
Q
R
S
V
W
X
Z



Individual Device Evaluation and Questionnaire

Device Questionnaire Content

- First Impressions
 - Consider carrying the device in several different locations (front pants pocket, back pants pocket, belt clip, shirt breast pocket) and rate each device's fit: Just right, big but acceptable, too big, or far too big
- Detailed Evaluation
 - Is the device comfortable to hold with one/two hands in portrait/landscape orientation? Consider device size, reach, and secureness when holding
 - What would you think of this device as your everyday phone if it were a smartphone? Consider your experiences holding/carrying and reaching across the device.
- Device Overall
 - Rate each device based on its dimensions: Length, Width, and Thickness as being just right, too big, or too small and your overall like/dislike: Really Like, Like, Dislike or Really Dislike

DEVICE QUESTIONNAIRE - SECTION 2

Example:
For each question, write the letter on the back of the device in a box.
You can put just one letter in a box or many letters into the same box.

	Poor 1	Neutral 2	Good 3	Excellent 4
EXAMPLE	A K	G	D B P	

FIRST IMPRESSIONS
We'd like you to try a variety of grips and tell us what you think of the size of this device. Take a moment to hold the device in several positions. Get a feel for its size.

1. What do you think about carrying this device in the following locations?
Write the device letter in one of the boxes for each of the four carrying locations below:

	Just Right 1	Big but Acceptable 2	Too Big 3	Far Too Big 4
Front pants pocket				
Back pants pocket				
Belt clip				
Shirt breast pocket				

DEVICE QUESTIONNAIRE - SECTION 2

DETAILED EVALUATION PART 1
Hold the device one-handed with your preferred hand and two-handed in both landscape/wide and portrait/tall orientation. Consider your impressions of the following:

- Device **size** (width, length, thickness)
- Reach** distance and ability to tap a button if it were located at the farthest edge of the screen (only consider edge/corner, don't worry about the position of the screen)
- Secureness** of your hold on the device given its size

Based on your overall impressions of each device, rate your overall comfort from Very Uncomfortable to Very Comfortable. Write each device letter in one of the boxes for each of the hold types (one-handed or two-handed) and device orientations (portrait/tall or landscape/wide)

	Very Uncomfortable 1	Uncomfortable 2	Somewhat Uncomfortable 3	Neither 4	Somewhat Comfortable 5	Comfortable 6	Very Comfortable 7
One-Hand Hold							
Portrait/Tall							
One-Hand Hold							

DEVICE QUESTIONNAIRE - SECTION 2

DETAILED EVALUATION PART 2
Consider your experiences holding/carrying and reaching across the device. What do you think of this device as your everyday phone if it were a smartphone, a phone that make calls, view/edit documents, browse the web, and compose/read email?

Hold the device as you would typically hold and use your phone (any and all positions you commonly use) and rate your impressions for each of the following:

	Far Too Narrow 1	Too Narrow 2	Narrow but Acceptable 3	Just Right 4	Wide but Acceptable 5	Too Wide 6	Far Too Wide 7
Device Width							

	Far Too Short 1	Too Short 2	Short but Acceptable 3	Just Right 4	Tall but Acceptable 5	Too Tall 6	Far Too Tall 7
Device Length							

	Far Too Thin 1	Too Thin 2	Thin but Acceptable 3	Just Right 4	Thick but Acceptable 5	Too Thick 6	Far Too Thick 7
Device Thickness							

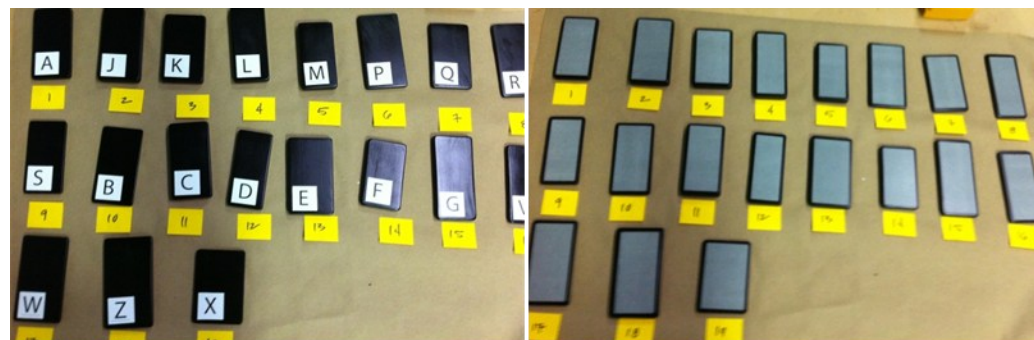
	Really Dislike 1	Dislike 2	Like 3	Really Like 4
Device Overall				

PLEASE STOP HERE until called to continue

Overall Questionnaire

Overall Questionnaire Activity

- Devices were gathered at the front of the room
- Participants revisited and interacted with each of their highest and lowest rated devices
- Each participant rated their top 3 most preferred (1st, 2nd, and 3rd) and bottom 3 least preferred devices (17th, 18th, and 19th)
- They were then given screen overlays for the devices to simulate viewing with text and images, and asked to rerate their top 3 most preferred choices
- Participants came to the front to interact with the devices if needed



Overall Questionnaire Content

- List the top 3 most preferred and bottom 3 least preferred devices that you would buy as your primary touchscreen phone
- Place screen overlays on the devices. Now that you can see what the device might be like with text and images, what are your top 3?
- Choose characteristics that are important while selecting a cell phone

OVERALL QUESTIONNAIRE - SECTION 3		OVERALL QUESTIONNAIRE - SECTION 3	COMMENTS, QUESTIONS, & NOTES
1. List the top 3 <u>most preferred</u> and bottom 3 <u>least preferred</u> devices that you would buy as your primary touchscreen phone. Please, no ties!			
First (Best):	_____ Why? _____	2. Place screen overlays on the devices. Now that you can see what the device might be like with text and images, what are your top 3? First (Best): _____ Why? _____ Second: _____ Why? _____ Third: _____ If your ranks changed from the previous page, please explain: 3. For each of the following comparisons, choose which characteristic is more important to you when selecting a cell phone? (Mark one for each paired comparison.) ☐ Long battery life OR ☐ Easy to fit in my pocket ☐ Long battery life OR ☐ Light weight ☐ Thin device OR ☐ Full physical keyboard ☐ Bigger screen OR ☐ Full physical keyboard ☐ Bigger screen OR ☐ Easily reach across the screen	
Second:	_____ Why? _____		
Third:	_____ Why? _____		
17th:	_____ Why? _____		
18th:	_____ Why? _____		
19th (Worst):	_____ Why? _____		
Any additional comments:			
PLEASE SIGN			

Group Discussion

Group Discussion

- Participants were asked to list all devices that ended up being in their top 3 most preferred
 - Moderator made a list of letters on the screen
 - For each letter, moderator asked for a show of hands for users that had that device in their top 3
 - Moderator asked them to explain why
- Participants were asked to list all the devices that were in their bottom 3 least preferred
 - Moderator made a list of letters on the screen
 - For each letter, moderator asked for a show of hands for users that had that device in their bottom 3
 - Moderator asked them to explain why
- Moderator asked users to explain any changes they made due to the overlays





Results – User Ratings of Models

Overall Like/Dislike

Perception of Width, Length, Thickness

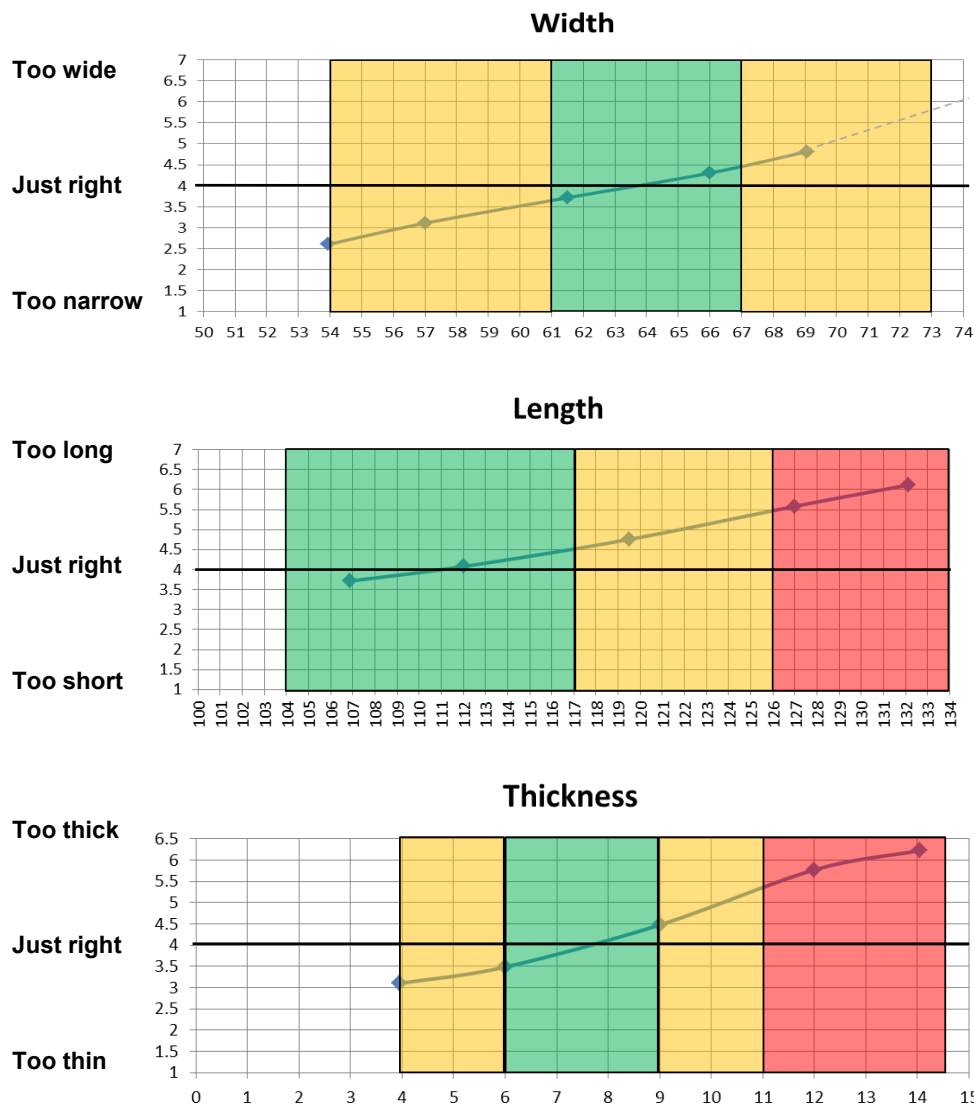
Summary of Mockup Device Dimension Ratings by Participants

Description of Graphs

- Each point represents a size tested in the study
- The graphs at the right represent average user ratings of each dimension
 - 7 = far too much (wide/long/thick)
 - 6 = too much
 - 5 = too much but acceptable
 - 4 = just right
 - 3 = too little but acceptable
 - 2 = too little
 - 1 = far too little (narrow/short/thin)
- The graphs are based only on the 15 devices where we consistently varied the size (excluding VWXZ)

Overview of results

- These ratings are for individual dimensions
 - In reality, they affect/depend on each other.
- These ratings are likely to be based more on the physicality of the device than screen size.
- Width
 - 61-67 optimal, 54-73 acceptable
 - The range tested were all acceptable
- Length
 - 104-117 optimal
 - ???-126 acceptable
 - No devices tested were too short
 - No devices tested were short but acceptable
- Thickness
 - 6-9 optimal
 - 4-11 acceptable
 - No devices tested were too thin



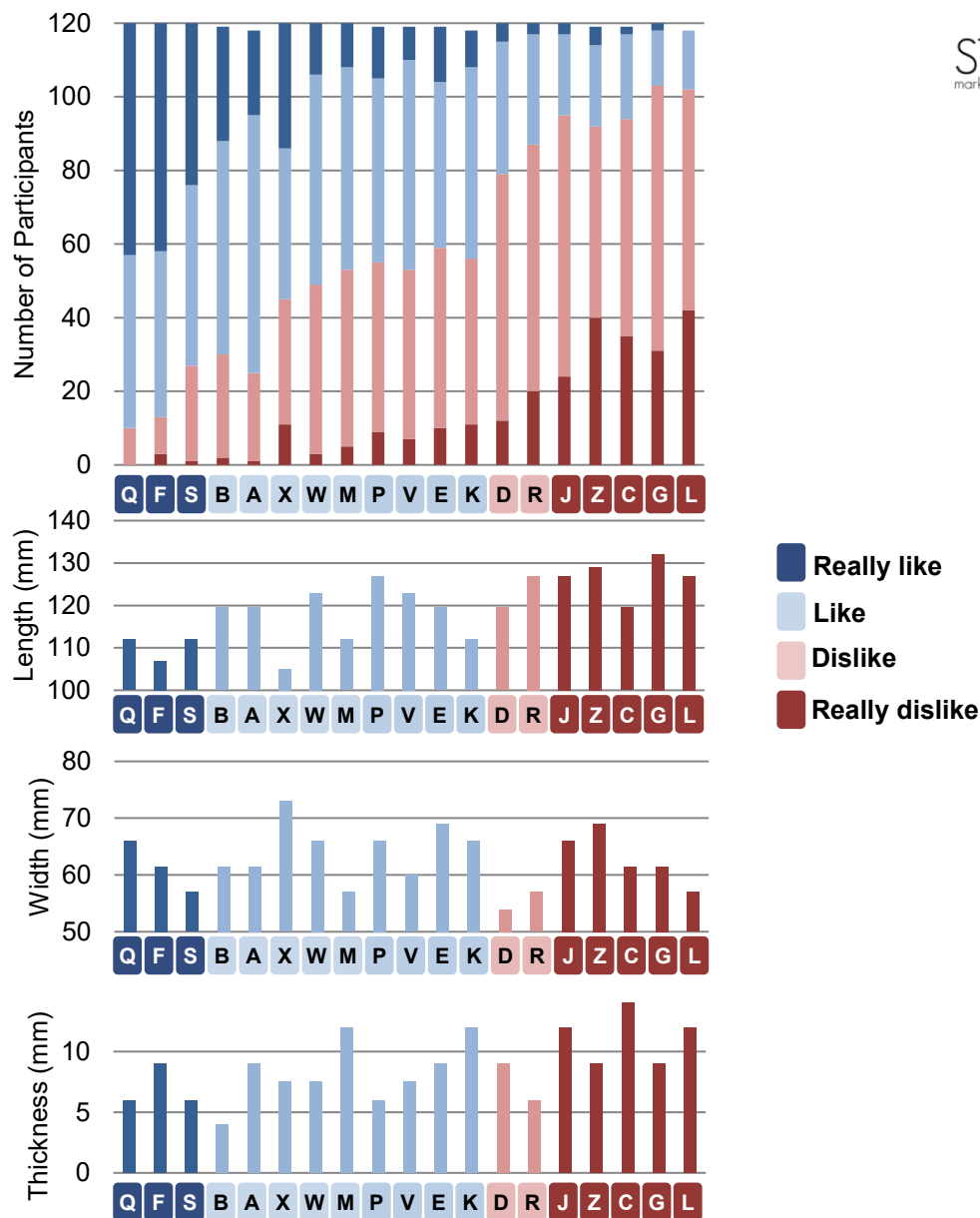
Mockup Overall: Like/Dislike

Chart Summary

- Q, F, and S were the most liked devices overall, followed by X, B, A, and W.
- Z, J, C, G, and L were the least liked devices overall.

Discussion

- The combination of length and thickness is a good predictor of how much users will like a device
 - Long/thick devices are strongly disliked
 - Short/thin devices are liked
 - The only thick device that was mildly disliked was M, which is both short and narrow, which would have decreased its weight and overall size
- Length is a good predictor of how much users will like a device
 - Short devices are liked
 - Mid-length devices can be liked or disliked
 - Long devices are disliked or neutral
- Width is not a good predictor of how much users will like a device



*Device selections in summary are based on ANOVA

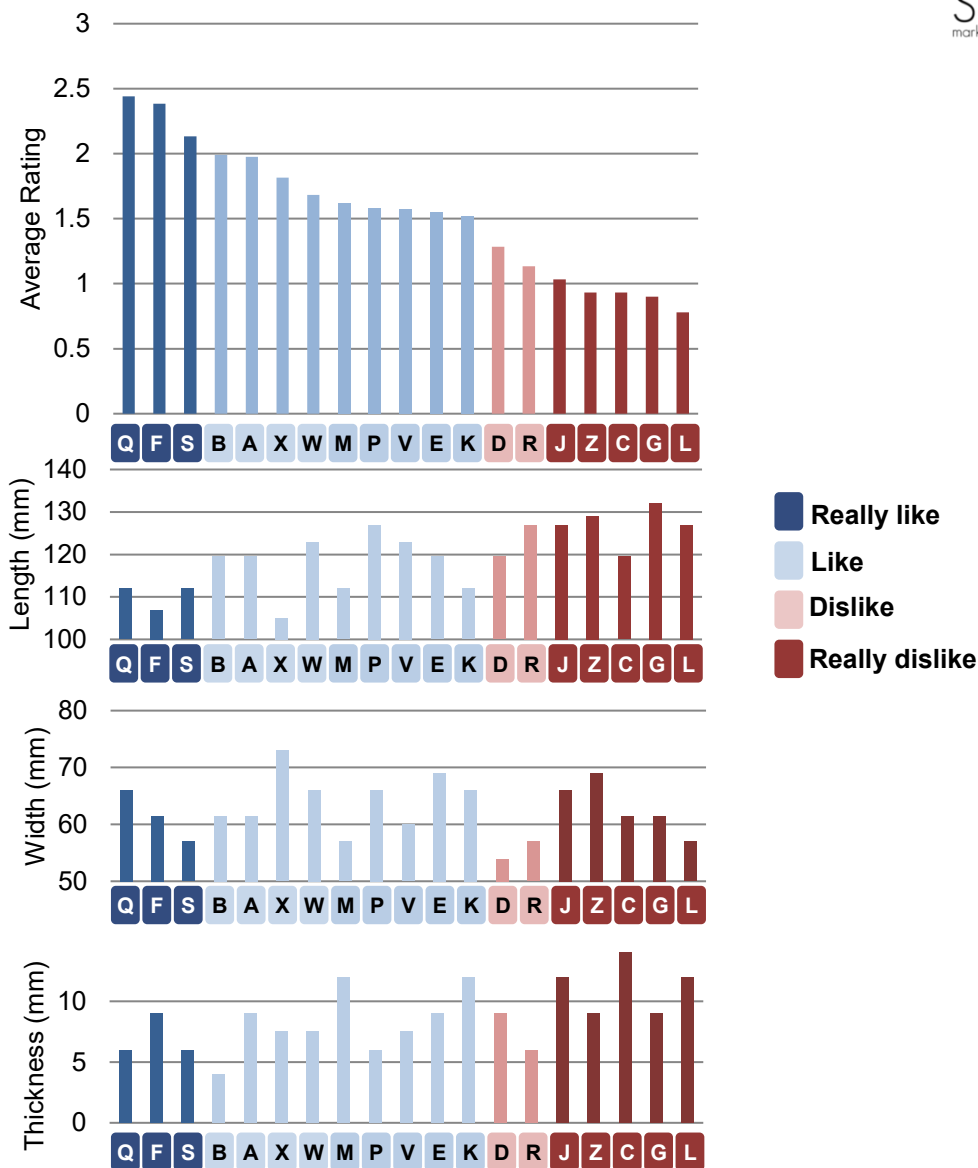
Mockup Overall: Like/Dislike

Chart Summary

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- Length is a good predictor of how much users will like a device
 - Short devices are liked
 - Mid-length devices can be liked or disliked
 - Long devices are disliked or neutral
- Width alone is not a good predictor of how much users will like a device



*Device selections in summary are based on ANOVA

Overall: Dimension Relationships

Description of the Graphs

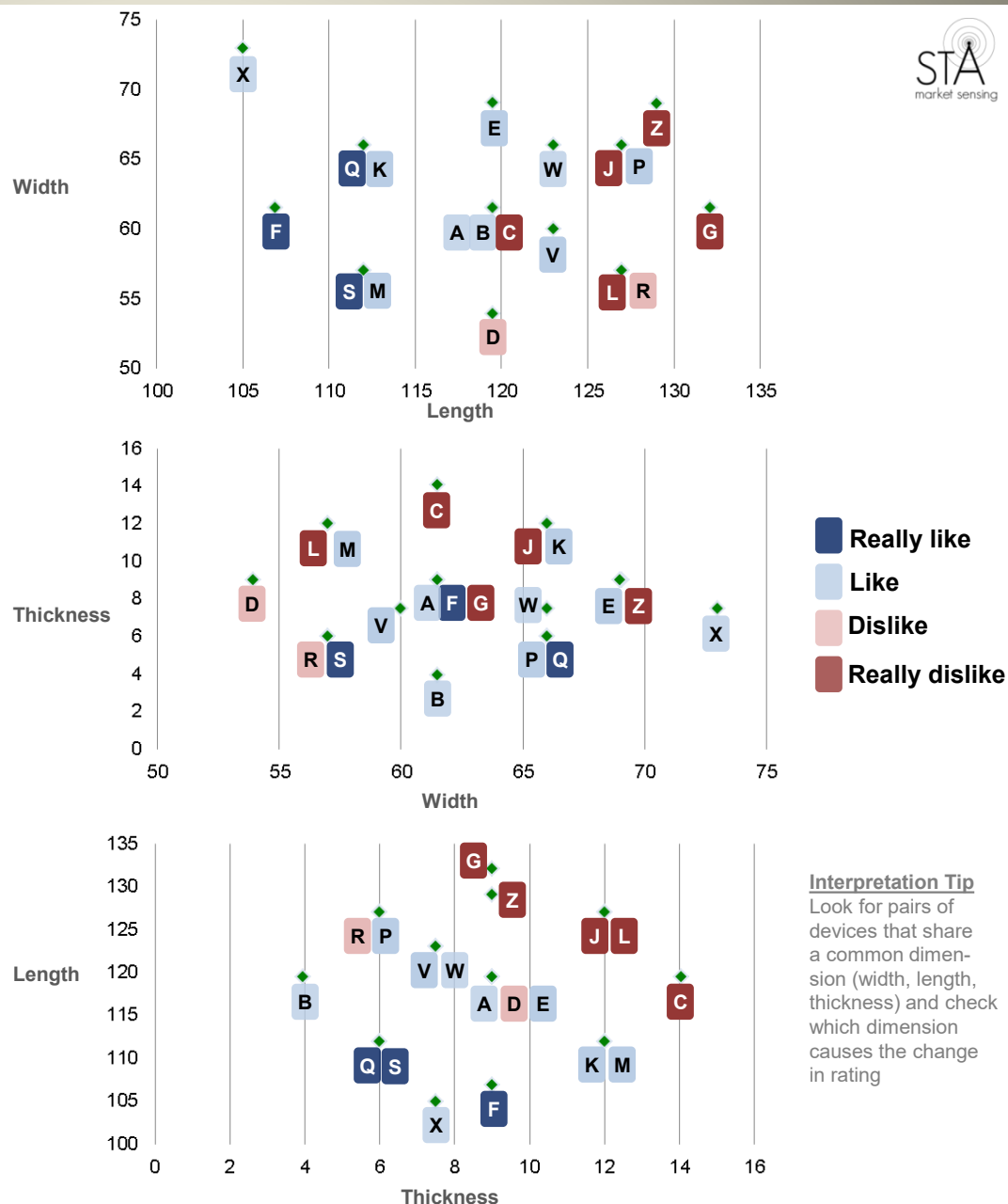
- Each graph at the right shows the devices by two of their three physical dimensions
- The most and least favorite devices are highlighted with color

Chart Summary

- Q, F, and S were the most liked devices overall, followed by X, B, A, and W.
- Z, J, C, G, and L were the least liked devices overall.

Discussion

- Length is a good predictor of how much users will like a device
 - Short devices are liked
 - Mid-length devices can be liked or disliked
 - Long devices are disliked or neutral
- Thickness is a good predictor of liking a device
 - Thicker devices are more disliked
- Width alone is not a good predictor
- There is an interaction among all three dimensions that is complex, but when looking for a single predictor, the combination of length and thickness is the best predictor of liking a device
 - Long, narrow, thick is least preferred
 - Long, wide, thick is somewhat disliked
 - Short/thin devices are liked



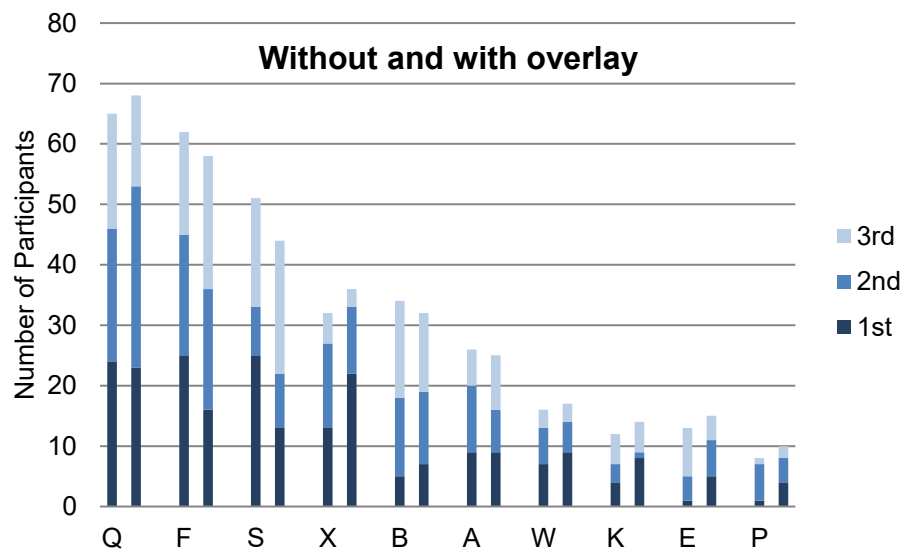
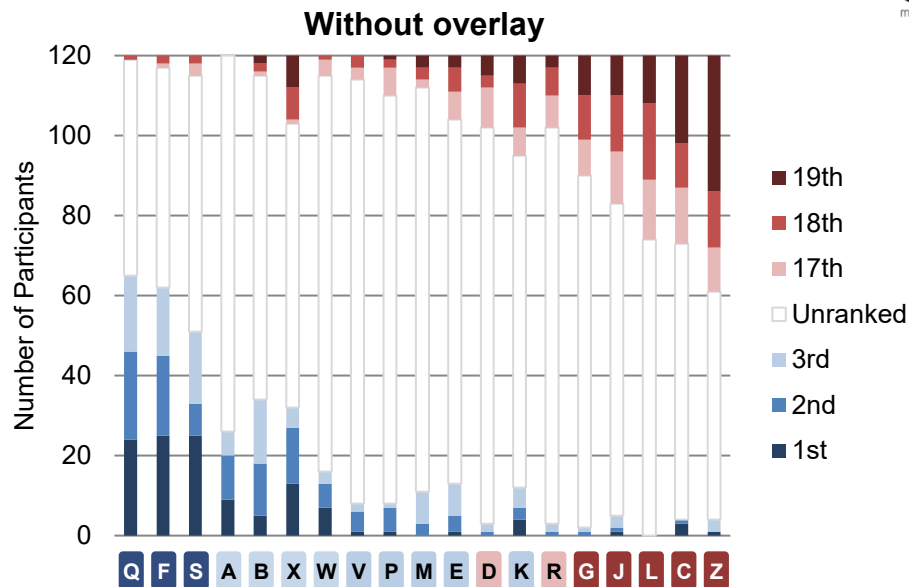
User Preferences: Most Preferred & Least Preferred

Chart Summary

- Users picked their Top 3 and bottom 3 devices (top chart)
- Users then placed an overlay with an example of a web page on top of the device, and marked their top 3 with the overlay (bottom chart)

Discussion

- X had many top and bottom ratings
 - People that really liked it
 - Better for watching videos or viewing photos
 - More width in portrait mode
 - More screen size for the overall device size
 - People that really disliked it
 - Too small overall
 - Might be good as a “device” but bad as a phone – too short
 - In portrait mode with an onscreen keyboard, there would be little space left above the keyboard
- K had many top and bottom ratings
 - People that really liked it
 - Want a large, durable phone with lots of features, and K was best of the larger/thicker devices
 - People that really disliked it
 - Thick and heavy
 - Too long for its width



Dimensions: Length Summary

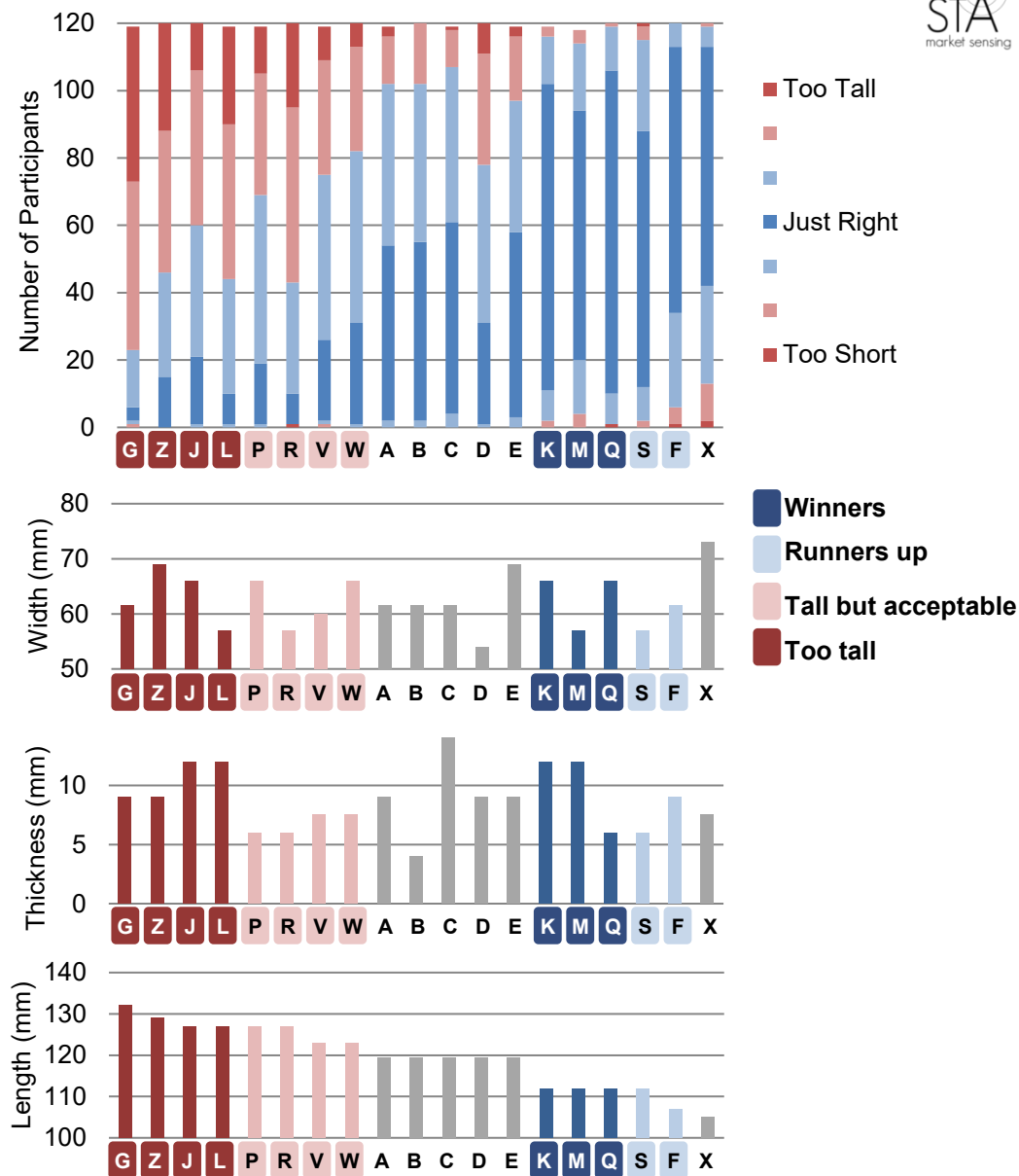
Chart Summary

- K, M, and Q were closest to being “Just Right” in terms of length.
- F followed closely behind K, M, and Q but was rated as being slightly shorter.
- S also followed closely behind K, M, and Q but was rated as being slightly taller.
- G, Z, L and R were rated more frequently than the other devices as being “Too Tall”
- X was rated “Short but Acceptable” more frequently than the other devices and “Too Short” by a few.

Discussion

- Ratings of “Just Right” varied somewhat based on user characteristics and usage behavior.
- Common reasons for “Just Right”:
 - *Aesthetically pleasing* when combined with other dimensions such as width
 - *Familiarity* with current or previously owned touchscreen phones
 - *Ability to reach* areas toward the top (portrait) or center (landscape) of the screen
- Common reasons for “Too Tall”:
 - Poor pocket fit
 - Unbalanced (one-handed portrait/landscape)
 - Thumb reach (two-handed landscape)
 - Disproportional when compared with other dimensions (width, thickness)
- Common reason for “Too Short”:
 - Screen size

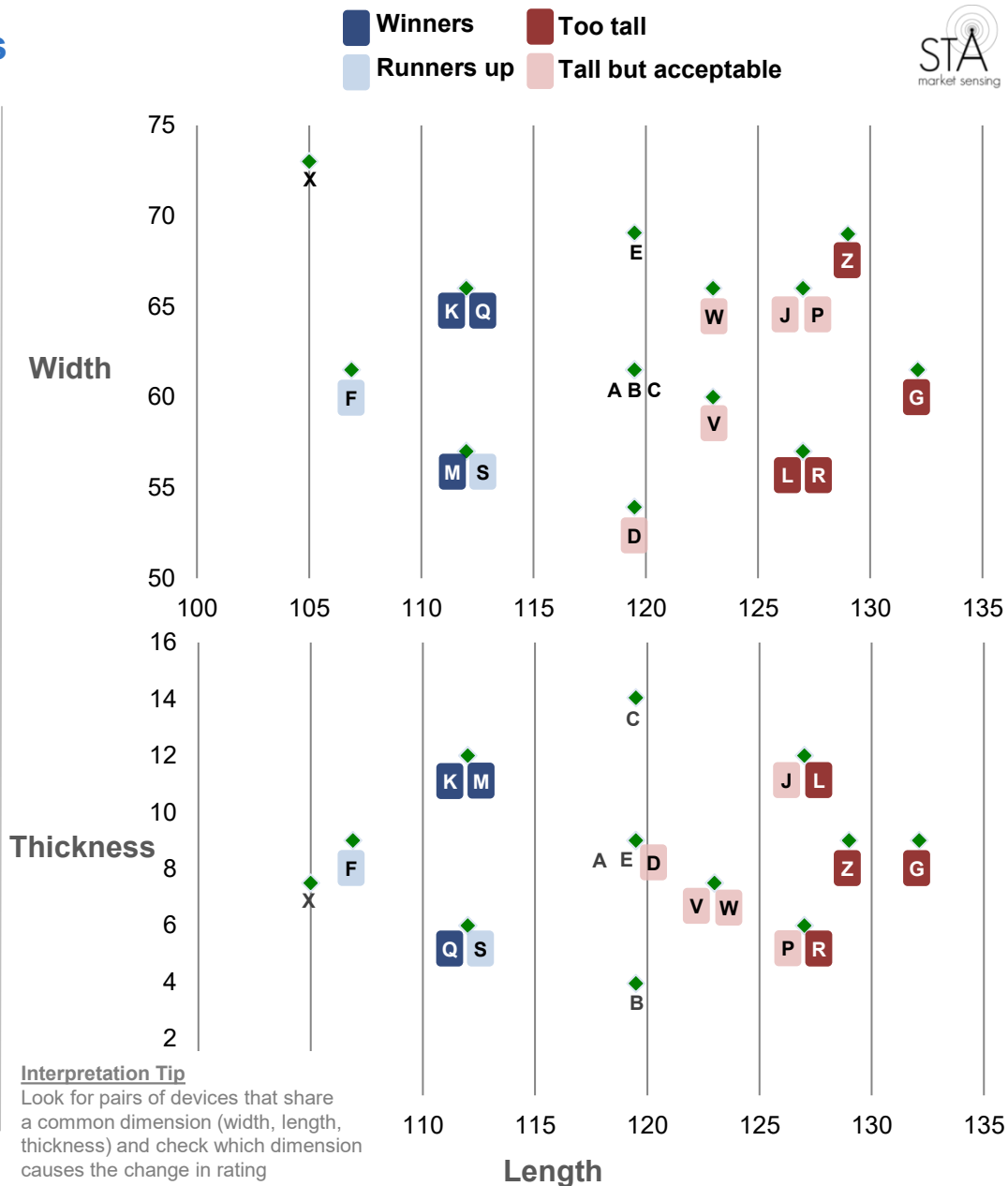
*Device selections in summary are based on ANOVA



Dimensions: Length Relationships

Relationship Discussion

- The 112mm devices tested the best, followed closely by the 107mm device (F)
- The only 120mm device that was rated tall but acceptable was D, which is the narrowest of that group
- Similarly, the two 127mm devices that were rated as tall but acceptable were L,R, which were also the two narrower devices
- Thickness did not seem to have clear effect on perception of length
- User perception of the length of a device is influenced by width; taller devices are more acceptable when they are also wide



Dimensions: Width Summary

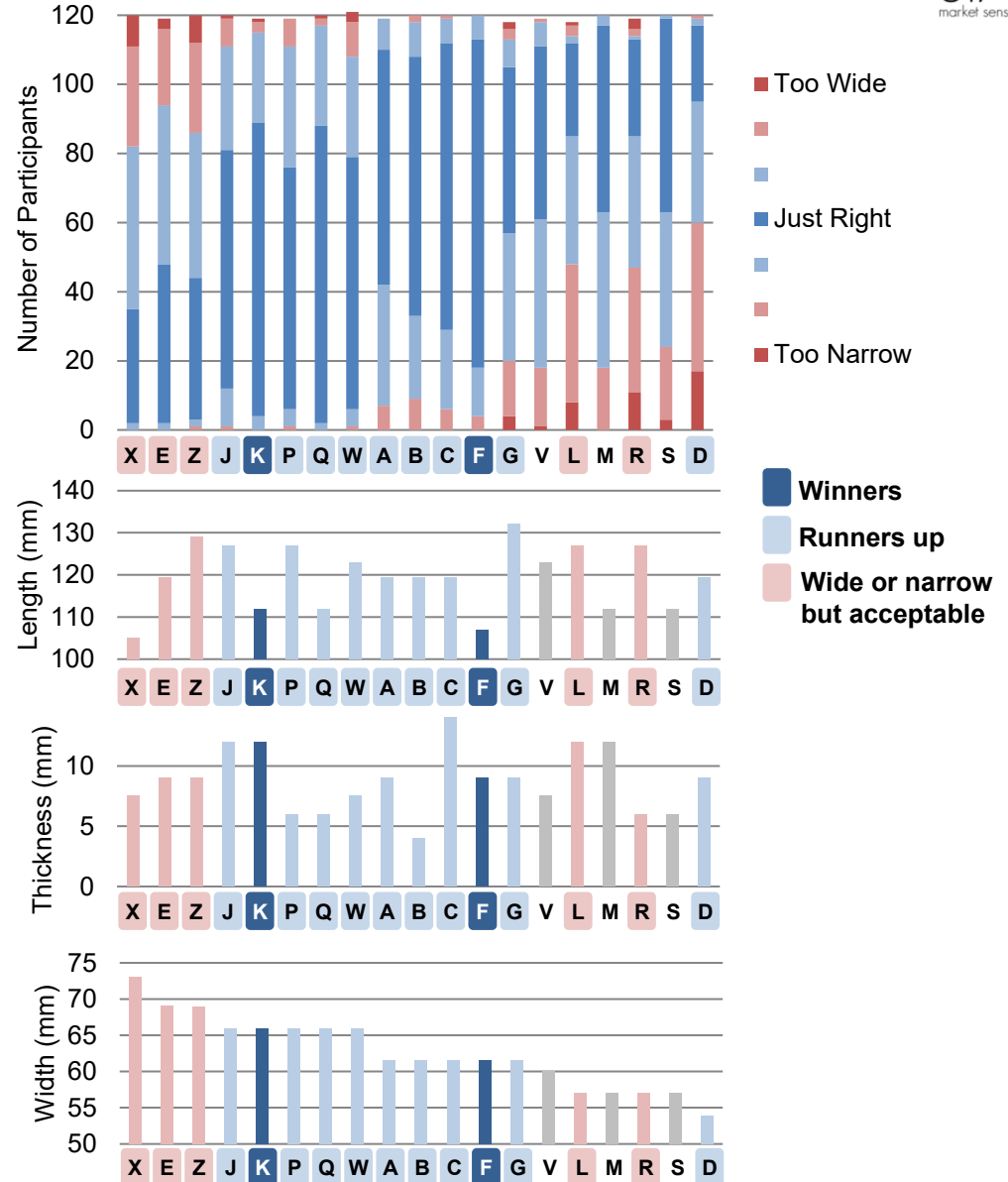
Chart Summary

- F and K were the closest to being “Just Right” in terms of width.
- A, B, C, and G followed closely behind F and K but were rated as being slightly narrower.
- Q, J, P, and W also followed closely behind F and K but were rated as being slightly wider.
- D, L and R were rated “Narrow but Acceptable” more frequently than the other devices and “Too Narrow” by some.
- E, Z, and X were rated “Wide but Acceptable” more frequently than the other devices and “Too Wide” by some.

Discussion

- Ratings of “Just Right” varied somewhat based on user characteristics and usage behavior.
- Common reasons for “Just Right”:
 - Good screen width* for viewing websites, videos, images, etc.
 - Less need to zoom* in order to read/view content easily
 - Ability to fit* and still reach more screen content
- Common reasons for “Too Wide”:
 - Uncomfortable fit in hand
 - Unsecure grip/hold
 - Disproportional when compared with other dimensions (length, thickness)
 - Blocks peripheral vision/seems obtrusive when using as a phone
- Common reason for “Too Narrow”:
 - Screen size
 - Unsecure grip/hold

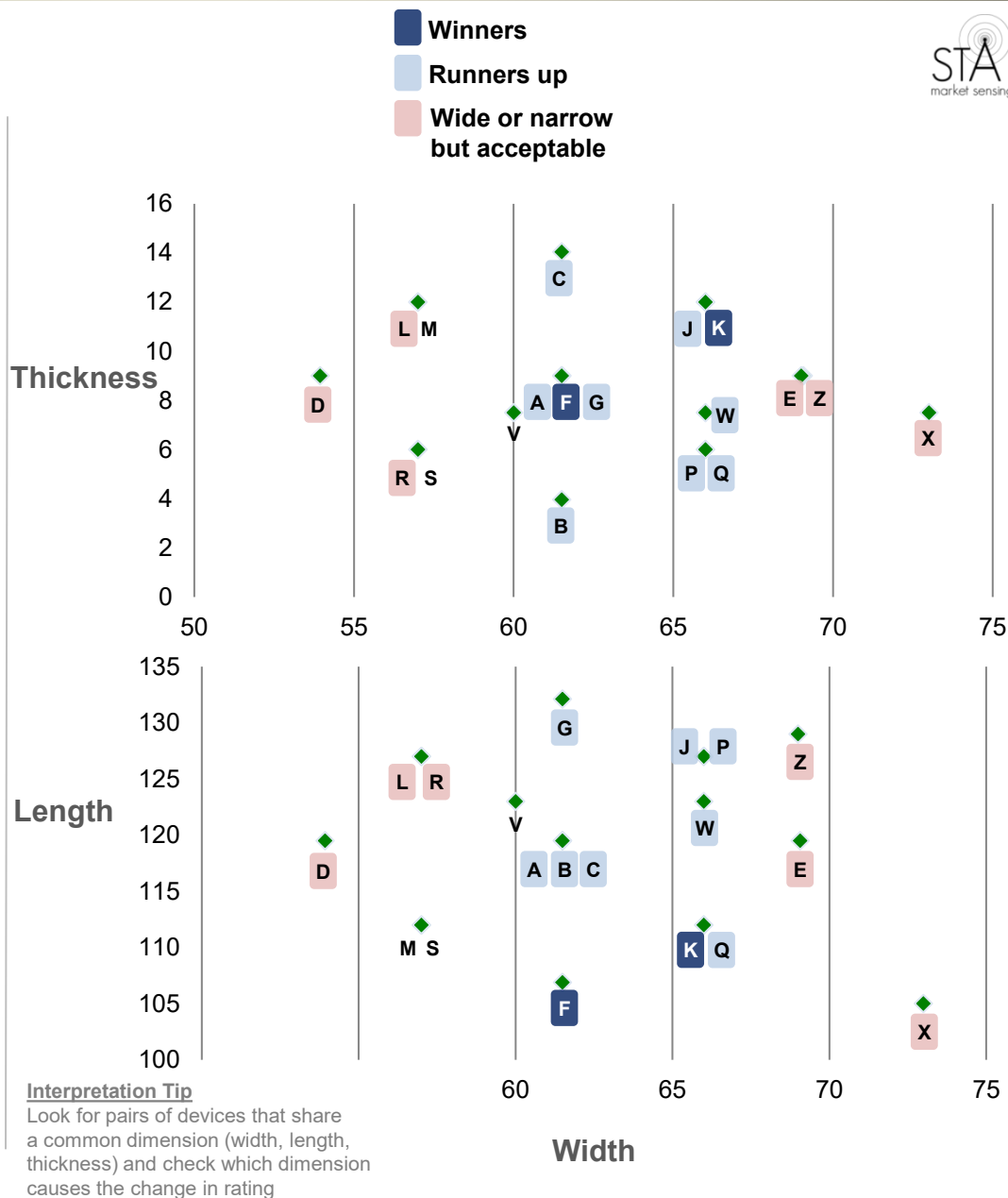
*Device selections in summary are based on ANOVA



Dimensions: Width Relationships

Relationship Discussion

- Thickness did not appear to have a clear influence on perception of width; length was more important
- L,R were considered narrow but acceptable, and both of these devices were longer devices
- M,R were rated closer to Just Right; both of these devices were shorter devices
- F,K were rated closest to Just Right; the most distinguishing characteristic of these two is that they are shorter than the rest
 - K is further distinguished from Q in that it is thicker than Q, suggesting that longer devices may be more acceptable when they are thicker



Dimensions: Thickness Summary

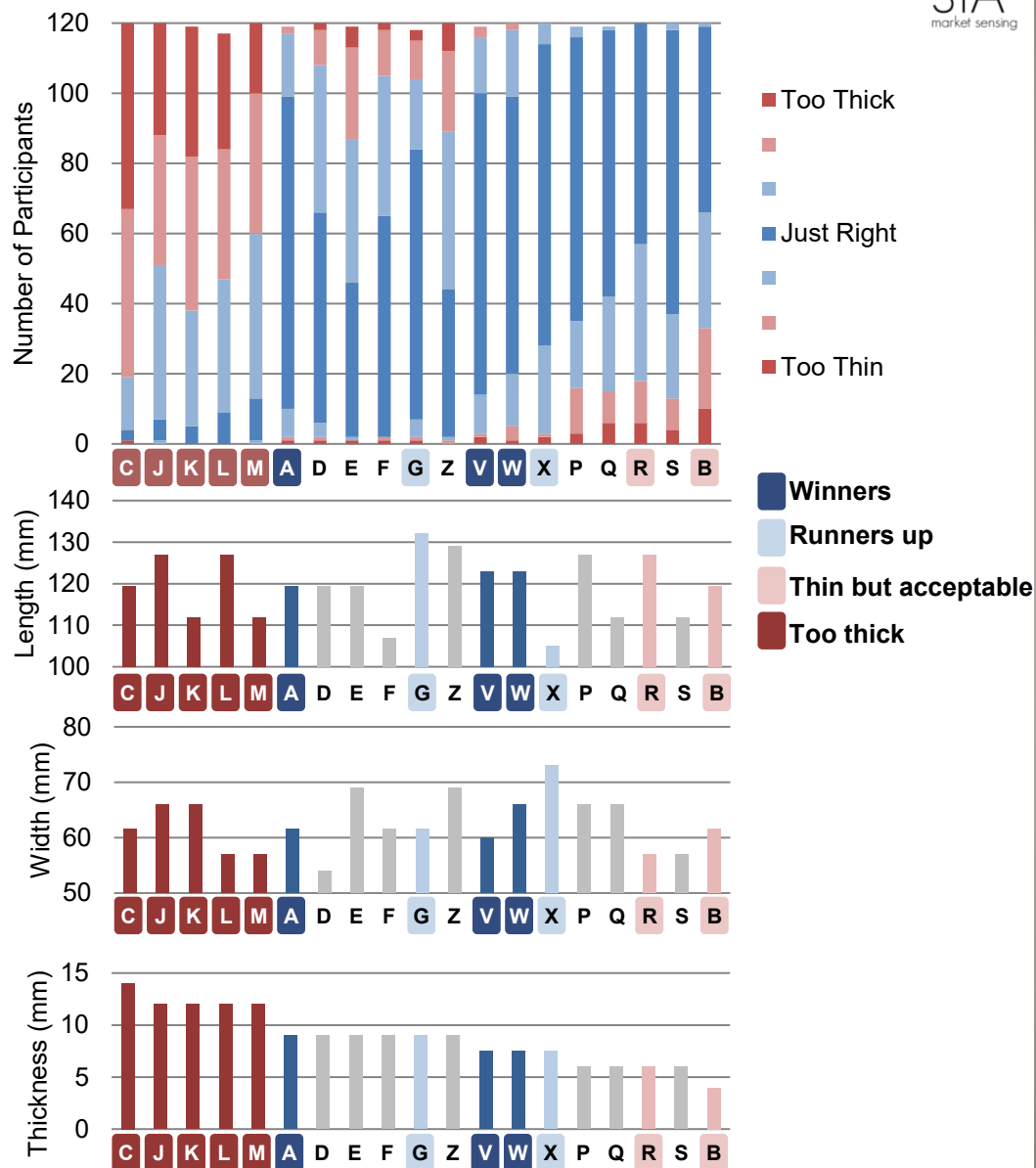
Chart Summary

- W, V, and A were the closest to being “Just Right” in terms of thickness.
- X followed closely behind W, V, and A but was rated as being slightly thinner.
- G also followed closely behind W, V, and A but was rated as being slightly thicker.
- C, J, K, L, and M were rated more frequently than the other devices as being “Too Thick”
- Comparatively, very few devices were rated as frequently being “Too Thin”, however devices B and R were rated as “Thin but Acceptable” more frequently than the other devices.

Discussion

- Ratings of “Just Right” varied somewhat based on user characteristics and usage behavior.
- Common reasons for “Just Right”:
 - Aesthetically pleasing* when combined with other dimensions
 - Comfortable fit in the hand*
 - Good pocket fit* without being bulky/obtrusive (pocketability)
 - Protective covers* can be used without making the phone too thick
 - Perceptions of good durability*
- Common reasons for “Too Thick”:
 - Poor aesthetics – disproportional, dated
 - Poor pocket fit
 - Excessive weight when holding/carrying
- Common reason for “Too Thin”:
 - Not noticeable when carried in the pocket
 - Perceptions of poor durability
 - Perceptions of poor battery life

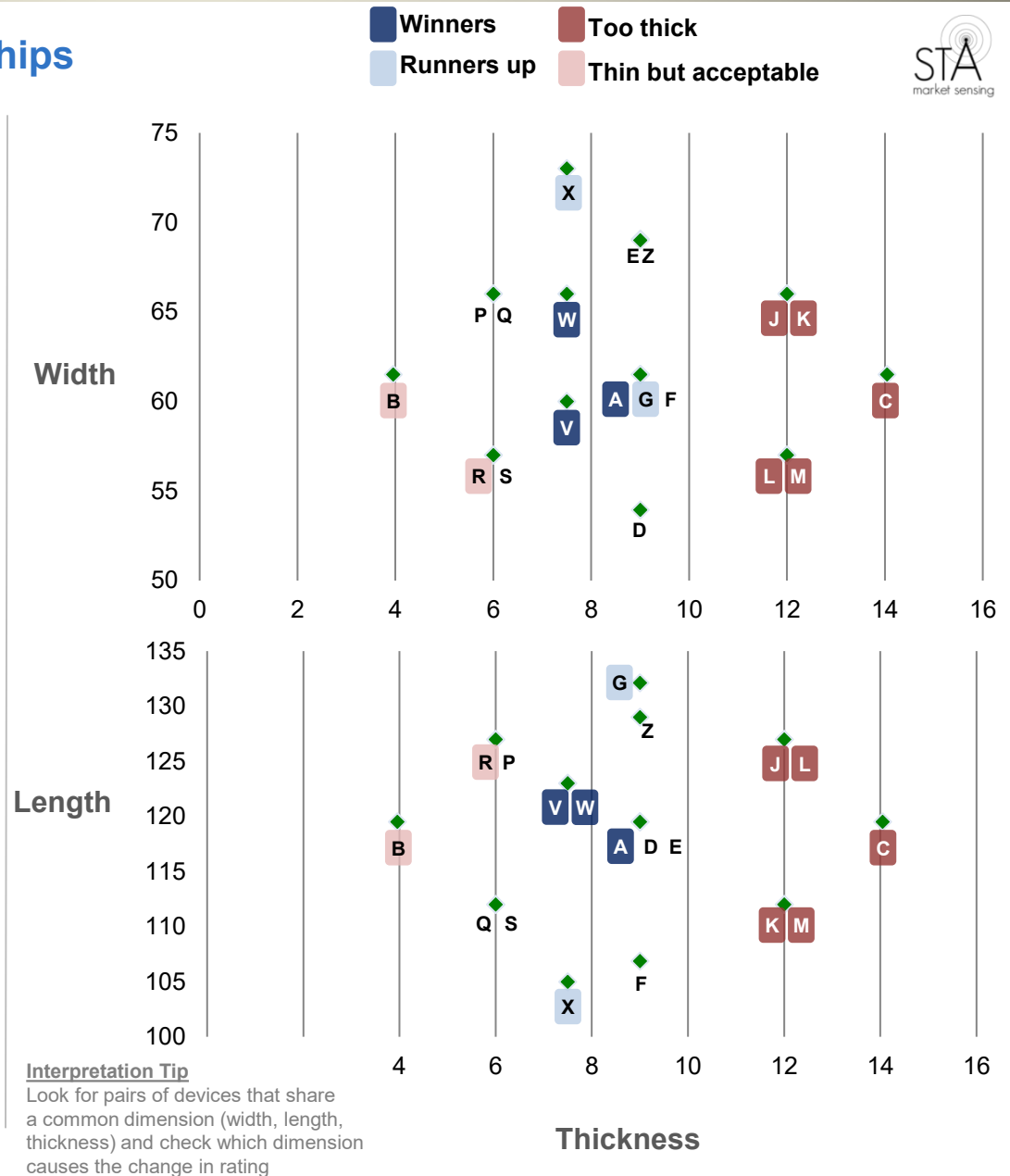
*Device selections in summary are based on ANOVA



Dimensions: Thickness Relationships

Relationship Discussion

- Users have strong agreement about which devices are too thick, but there is less agreement about which devices are too thin
- Perception of thickness appears to be somewhat unaffected by width or length of the device
 - R,S have the same thickness and width; R is rated as closer to too thin because it is longer than S
 - R,P have the same thickness and length; R is rated as closer to too thin because it is narrower than P
 - A,G,F have the same width and thickness
 - A is of middle length and is just right for thickness
 - G is very long and is close to just right
 - F is short and is on the thick side of just right



Dimensions: Overall % Responses

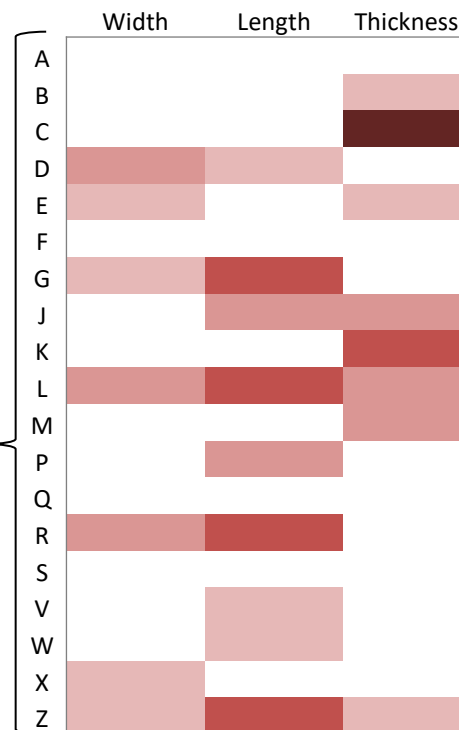
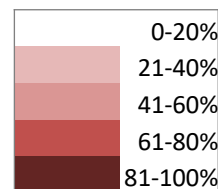
Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal
A	119	61	9	124mm (4.9")
J	127	66	12	132mm (5.2")
K	112	66	12	119mm (4.7")
L	127	57	12	129mm (5.1")
M	112	57	12	114mm (4.5")
P	127	66	6	132mm (5.2")
Q	112	66	6	119mm (4.7")
R	127	57	6	130mm (5.1")
S	112	57	6	114mm (4.5")
B	119	61	4	124mm (4.9")
C	119	61	14	124mm (4.9")
D	119	54	9	122mm (4.8")
E	119	69	9	127mm (5.0")
F	107	61	9	112mm (4.4")
G	132	61	9	135mm (5.3")
V	123	60	8	127mm (5.0")
W	123	66	8	127mm (5.0")
Z	129	69	9	135mm (5.3")
X	105	73	8	117mm (4.6")

*44

ratings (coded 1-2 + 6-7) for each dimension.



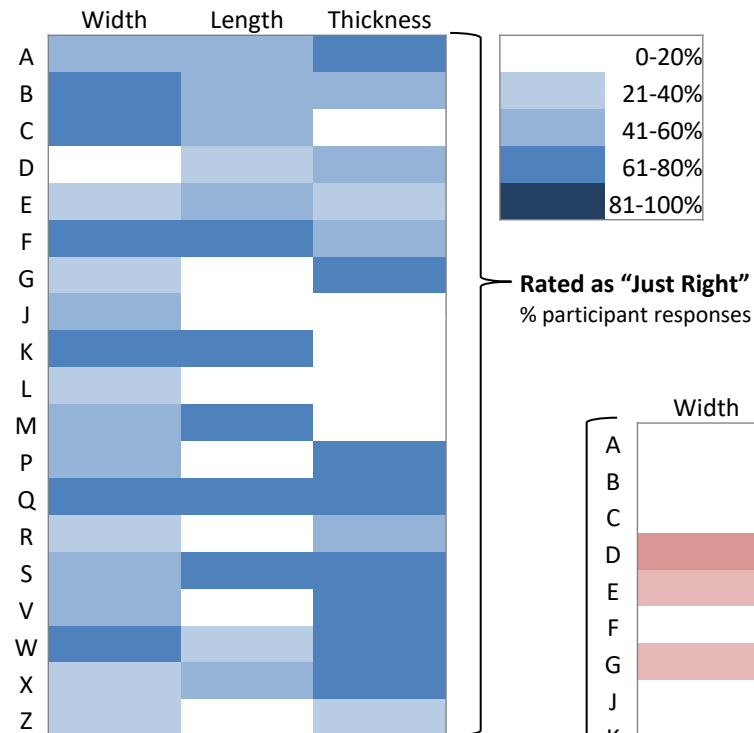
Rated as "Too Big" & "Too Small"
% participant responses



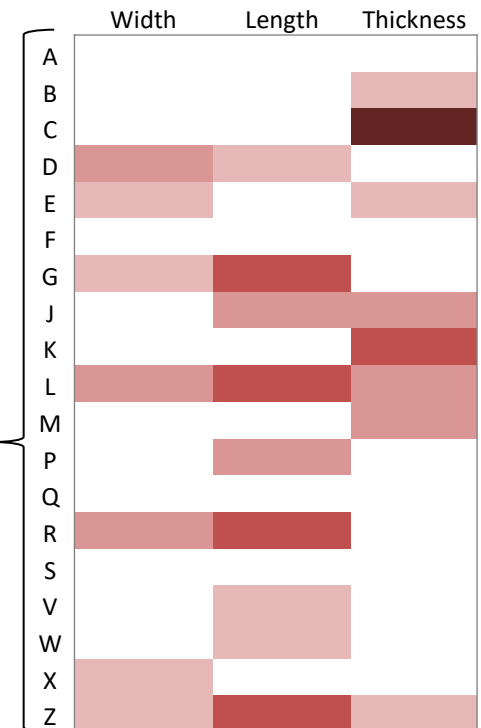
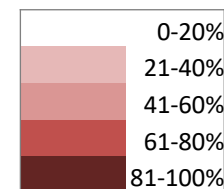
Dimensions: Overall % Responses

Chart Summary

- Devices B, C, F, K, Q, and W were rated by 61-80% of people as having a “Just Right” width.
- Devices F, K, M, Q, and S were rated by 61-80% of people as having a “Just Right” length.
- Devices A, G, P, Q, S, V, W, and X were rated by 61-80% of people as having a “Just Right” thickness.
- 40-60% of people felt devices D, L, and R were too skinny.
- More than 61-80% of people felt that devices G, L, R, and Z were too tall.
- More than 81% of people felt that device C was too thick and 61-80% felt that device K was too thick.



Rated as “Too Big” & “Too Small”
% participant responses



*“Too Big” and “Too Small” include combined extreme frequency ratings (coded 1-2 + 6-7) for each dimension.



Appendix A: Statistical Output

Output from Statistical Packages

ANOVAs for Width, Length, Thickness and Overall

Analysis overview

- Four ANOVAs were conducted for each of the four major dependent measures
 - Type III Sum of Squares to account for uneven sample sizes
 - Width (1 = too narrow, 7 = too wide)
 - Length (1 = too short, 7 = too tall)
 - Thickness (1 = too thin, 7 = too thick)
 - Overall
 - 1 = really dislike
 - 2 = dislike
 - 3 = like
 - 4 = strongly like
- All four ANOVAs were significant ($p < .01$)
- Post-hoc tests were conducted using Tukey's HSD ($\alpha = .05$)

Tukey HSD Tables

- The numbers in the body of the table represent the mean rating for a device
- Each "subset" represents a group of devices that are NOT statistically different from each other
 - i.e. two devices that appear in the same subset are similar to each other

Interpretation

- "Winners" are the group that are always statistically similar to the group that has a "Just Right" rating = 4
- "Runners Up" are devices that are sometimes in a similarity group with the Just Right group

Length



Tukey HSD^{a,b,c}

DEVICE	N	Subset								
		1	2	3	4	5	6	7	8	9
X	120	3.59								
F	120	3.72	3.72							
Q	119		4.02	4.02						
M	118		4.03	4.03						
K	119		4.06	4.06						
S	120			4.20	4.20					
C	119				4.56	4.56				
B	120					4.68				
A	119					4.70				
E	119					4.70				
W	120						5.11			
D	120						5.16			
V	119						5.21			
P	118						5.37	5.37		
J	120						5.43	5.43	5.43	
R	119							5.74	5.74	5.74
Z	120								5.76	5.76
L	117								5.80	5.80
G	117									6.11
Sig.		1.000	.143	.975	.083	.999	.214	.075	.068	.065

Means for groups in homogeneous subsets are displayed.

Based on Type III Sum of Squares

The error term is Mean Square(Error) = .696.

a. Uses Harmonic Mean Sample Size = 119.097.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = .05.

ANOVAs for Width, Length, Thickness and Overall

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- "Runners Up" are devices that are sometimes in a similarity group with the Just Right group

Width



Tukey HSD^{a, b, c}

DEVICE	N	Subset									
		1	2	3	4	5	6	7	8	9	10
D	120	2.60									
L	117	2.91	2.91								
R	119	2.92	2.92								
S	120		3.26	3.26							
M	118			3.34	3.34						
V	119			3.40	3.40	3.40					
G	117			3.50	3.50	3.50	3.50				
A	119				3.66	3.66	3.66	3.66			
B	120					3.77	3.77	3.77			
C	119					3.78	3.78	3.78			
F	120						3.88	3.88			
K	119							4.26	4.26		
Q	119								4.29		
J	120								4.30		
P	118								4.37		
W	120								4.42		
E	119									4.42	
Z	120										4.82
X	120										4.95
Sig.		.310	.143	.820	.262	.074	.072	.060	.994	.053	.629

Means for groups in homogeneous subsets are displayed.

Based on Type III Sum of Squares

The error term is Mean Square(Error) = .738.

a. Uses Harmonic Mean Sample Size = 119.097.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = .05.

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Thickness



Tukey HSD^{a, b, c}

DEVICE	N	Subset								
		1	2	3	4	5	6	7	8	9
B	120	3.10								
R	119	3.32	3.32							
Q	119	3.48	3.48	3.48						
S	120		3.57	3.57						
P	118		3.57	3.57						
X	120			3.77	3.77					
W	120				3.97	3.97				
V	119				4.03	4.03				
A	119				4.08	4.08				
G	117					4.35	4.35			
D	120					4.49	4.49			
F	120						4.56	4.56		
E	119							4.90	4.90	
Z	120								5.57	5.57
M	118								5.78	5.78
J	120								5.80	5.80
L	117								5.95	5.95
K	119									5.95
C	119									6.22
Sig.		.070	.753	.430	.399	.077	.935	.078	.065	.618

Means for groups in homogeneous subsets are displayed.

Based on Type III Sum of Squares

The error term is Mean Square(Error) = .734.

a. Uses Harmonic Mean Sample Size = 119.097.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = .05.

Q was excluded from "thin but acceptable" because it is in subset 3 with X and the average rating was above 3.5

Thickness Range

Description of the Table

- We predicted how much users will like every combination of width, length and thickness
- Values in the table represent the range of thickness that the model predicts will be “liked” (≥ 2.5 on a scale of 1, 2, 3, or 4)

Discussion

- The maximum liked width is 127mm (we tested up to 132mm)
- The range of liked thickness is largest for width ~ 64 , and decreases as the width increases or decreases
- The range is largest for the shorter lengths and decreases as the device gets longer, especially for very narrow or wide devices
- $>13\text{mm}$ is not predicted to be acceptable for any combination of width/length
- The range of liked thickness is small at the edges of the colored areas below

	Thickness
Range of Interest	4-14mm
Range that users “really like”	min-6mm
Range that users “like”	min-13mm

[illegible]

ANOVAs for Width, Length, Thickness and Overall

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Tukey HSD Tables

- The numbers in the body of the table represent the mean rating for a device
- Each "subset" represents a group of devices that are NOT statistically different from each other
 - i.e. two devices that appear in the same subset are similar to each other

Interpretation

- Really like ≥ 3
- Like ≥ 2.5
- Dislike < 2.5
- Really Dislike ≤ 2

Overall



Tukey HSD^{a, b, c}

DEVICE	N	Subset							
		1	2	3	4	5	6	7	8
L	117	1.78							
G	120	1.90	1.90						
C	119	1.93	1.93						
Z	119	1.93	1.93						
J	120	2.03	2.03	2.03					
R	119		2.13	2.13					
D	120			2.28	2.28				
K	118				2.52	2.52			
E	119				2.55	2.55			
V	119				2.57	2.57			
P	119				2.58	2.58			
M	120				2.62	2.62			
W	120					2.68	2.68		
X	120					2.82	2.82	2.82	
A	118						2.97	2.97	
B	119						2.99	2.99	
S	120							3.13	3.13
F	120								3.38
Q	120								3.44
Sig.		.444	.679	.488	.062	.167	.133	.104	.132

Means for groups in homogeneous subsets are displayed.

Based on Type III Sum of Squares

The error term is Mean Square(Error) = .556.

a. Uses Harmonic Mean Sample Size = 119.257.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = .05.

Regression Results

Process Used

- Centered and scaled from -1 to 1 values for width, length, thickness
- Created pseudo variables for squared terms by multiplying the dimension with itself
- Created pseudo variables for interaction terms by multiplying the two dimensions
- Conducted stepwise linear regression using all 9 terms (p-in = .05, p-out = .10); none of the terms dropped out (all are significant)

Description of the Table

- Terms are sorted from highest to lowest impact
- Standardized coefficients can be used to compare effect size from one to another
- Unstandardized coefficients can be used to create a predictive equation
 - Note: the equation will work on the centered scaled variables, as described above. They will not work on the raw values without translation

Discussion

- Length and thickness have the highest impact
- Width also has a high impact
- All 9 terms, including interactions, are in the model
 - This means that understanding the relationship between the variables will be difficult because the relationship is complex

Optimum

- The optimum point for this equation within the range tested is
 - width = 62.6mm, length = 107mm, thickness = 5.1mm
 - predicted value = 3.6 out of 4

	Standardized Coefficients	Unstandardized Coefficients
Model	Beta	B
(Constant)		2.96
Length	-0.45	-0.70
Thickness	-0.32	-0.50
Width ²	-0.18	-0.51
Thickness ²	-0.16	-0.46
Width	0.11	0.17
Length ²	-0.10	-0.29
Width * Thickness	-0.06	-0.21
Length * Thickness	0.05	0.19
Width * Length	0.05	0.18

Regression Analysis for CCD

Analysis overview

- Only analyzed the 15 points in the CCD
 - Dropped VWXZ
- Centered, scaled width, length, height from -1 to 1
- Created pseudo variables for the squared terms
- Created pseudo variables for the interactions terms by multiplying the terms with each other
- Conducted stepwise linear regression using all 9 terms (p-in = .05, p-out = .10)

Interpretation tips

- Standardized coefficients can be compared to estimate effect sizes
- Unstandardized coefficients can be used to build a model from centered scaled dimensions
 - The coefficients and constant need to be adjusted for an equation that works with raw dimensions

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	284.030	1	284.030	442.011	.000 ^a
	Residual	1148.945	1788	.643		
	Total	1432.975	1789			
2	Regression	431.215	2	215.607	384.741	.000 ^b
	Residual	1001.660	1787	.561		
	Total	1432.975	1789			
3	Regression	447.891	3	149.297	270.682	.000 ^c
	Residual	985.084	1786	.552		
	Total	1432.975	1789			
4	Regression	456.895	4	114.224	208.886	.000 ^d
	Residual	976.080	1785	.547		
	Total	1432.975	1789			
5	Regression	468.906	5	93.781	173.541	.000 ^e
	Residual	964.069	1784	.540		
	Total	1432.975	1789			
6	Regression	476.464	6	79.411	148.027	.000 ^f
	Residual	956.511	1783	.536		
	Total	1432.975	1789			
7	Regression	481.678	7	68.811	128.899	.000 ^g
	Residual	951.297	1782	.534		
	Total	1432.975	1789			
8	Regression	485.951	8	60.744	114.237	.000 ^h
	Residual	947.024	1781	.532		
	Total	1432.975	1789			
9	Regression	489.761	9	54.418	102.896	.000 ⁱ
	Residual	943.213	1780	.530		
	Total	1432.975	1789			

a. Predictors: (Constant), DL

b. Predictors: (Constant), DL, DT

c. Predictors: (Constant), DL, DT, DW

d. Predictors: (Constant), DL, DT, DW, DW2

e. Predictors: (Constant), DL, DT, DW, DW2, DT2

f. Predictors: (Constant), DL, DT, DW, DW2, DT2, DL2

g. Predictors: (Constant), DL, DT, DW, DW2, DT2, DL2, DWT

h. Predictors: (Constant), DL, DT, DW, DW2, DT2, DL2, DWT, DLT

i. Predictors: (Constant), DL, DT, DW, DW2, DT2, DL2, DWT, DLT, DWL

j. Dependent Variable: RATING

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.445 ^a	.198	.198	.802
2	.549 ^b	.301	.300	.749
3	.559 ^c	.313	.311	.743
4	.565 ^d	.319	.317	.739
5	.572 ^e	.327	.325	.735
6	.577 ^f	.332	.330	.732
7	.580 ^g	.336	.334	.731
8	.582 ^h	.339	.336	.729
9	.585 ⁱ	.342	.338	.728

a. Predictors: (Constant), DL

b. Predictors: (Constant), DL, DT

c. Predictors: (Constant), DL, DT, DW

d. Predictors: (Constant), DL, DT, DW, DW2

e. Predictors: (Constant), DL, DT, DW, DW2, DT2

f. Predictors: (Constant), DL, DT, DW, DW2, DT2, DL2

g. Predictors: (Constant), DL, DT, DW, DW2, DT2, DL2, DWT

h. Predictors: (Constant), DL, DT, DW, DW2, DT2, DL2, DWT, DLT

i. Predictors: (Constant), DL, DT, DW, DW2, DT2, DL2, DWT, DLT, DWL

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
9	(Constant)	2.960	.066		44.622	.000
	DL	-.698	.030	-.445	-23.161	.000
	DT	-.503	.030	-.321	-16.673	.000
	DW	.168	.030	.107	5.584	.000
	DW2	-.514	.076	-.179	-6.757	.000
	DT2	-.462	.076	-.161	-6.078	.000
	DL2	-.287	.076	-.100	-3.778	.000
	DWT	-.206	.065	-.061	-3.149	.002
	DLT	.186	.065	.054	2.834	.005
	DWL	.176	.065	.052	2.682	.007



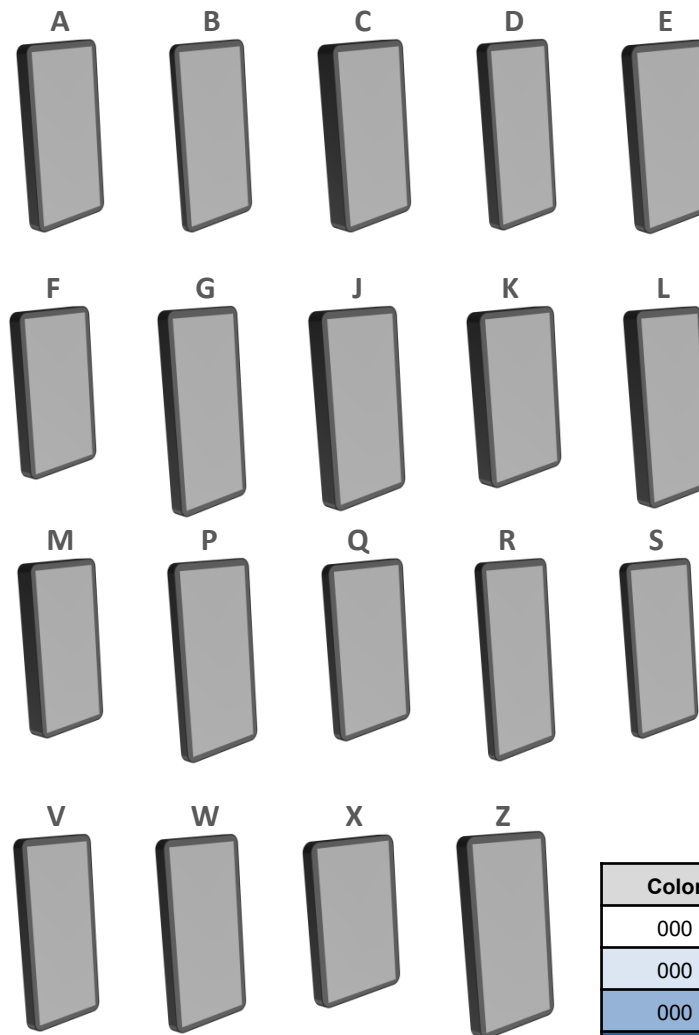
Appendix B: Model Ratings, Comments

Summary of Individual Model Ratings & Comments

Testing the specific model sizes was not a primary focus for the study, and so the analysis of each model is not to the depth of previous studies.

Summary of Mockup Devices

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal
A	119	61	9	124mm (4.9")
J	127	66	12	132mm (5.2")
K	112	66	12	119mm (4.7")
L	127	57	12	129mm (5.1")
M	112	57	12	114mm (4.5")
P	127	66	6	132mm (5.2")
Q	112	66	6	119mm (4.7")
R	127	57	6	130mm (5.1")
S	112	57	6	114mm (4.5")
B	119	61	4	124mm (4.9")
C	119	61	14	124mm (4.9")
D	119	54	9	122mm (4.8")
E	119	69	9	127mm (5.0")
F	107	61	9	112mm (4.4")
G	132	61	9	135mm (5.3")
V	123	60	8	127mm (5.0")
W	123	66	8	127mm (5.0")
Z	129	69	9	135mm (5.3")
X	105	73	8	117mm (4.6")



Color	Meaning
000	Minimum
000	
000	Average
000	
000	Maximum
000	Other

Legend

OVERALL

-  Really Like
-  Like
-  Dislike
-  Really Dislike

DIMENSION

-  Just Right
-  Small But Acceptable, Big But Acceptable
-  Too Small, Too Big
-  Far Too Small, Far Too Big

COMFORT

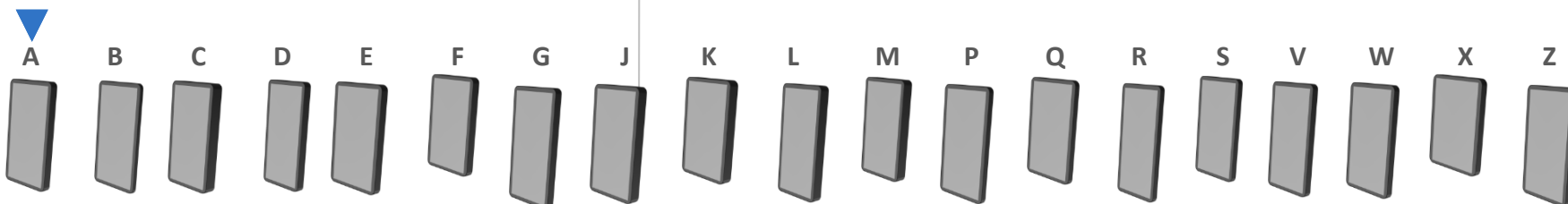
-  Very Comfortable
-  Comfortable, Somewhat Comfortable
-  Uncomfortable, Somewhat Uncomfortable
-  Very Uncomfortable
-  Neither

POCKETABILITY

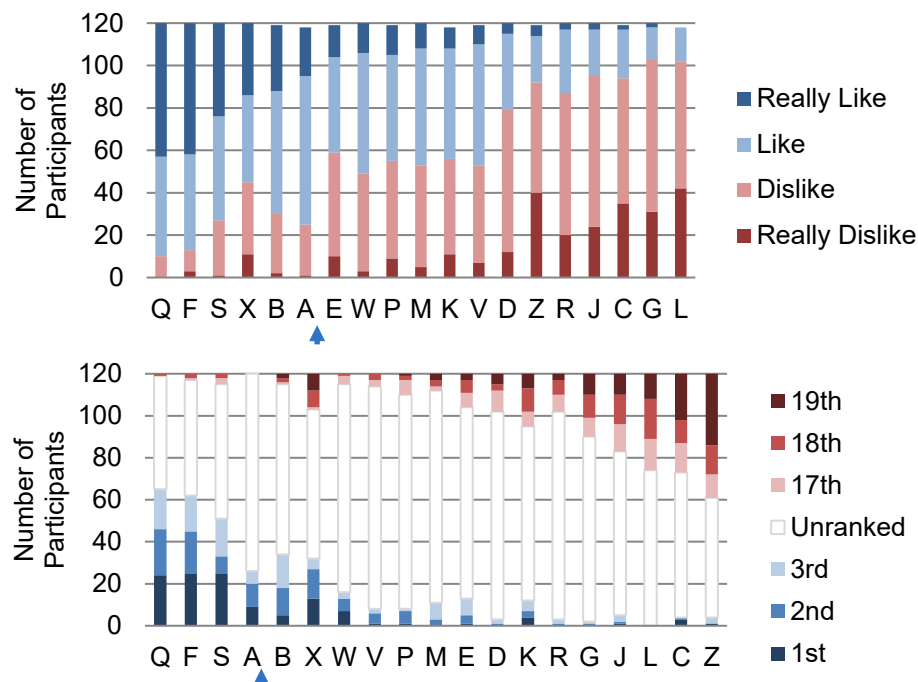
-  Just Right
-  Big But Acceptable
-  Too Big
-  Far Too Big

Device A: 4.9" Screen

+ Overall	Like
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Somewhat Comf.
+/- Comfort: 1Hand Land.	Neither
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable



Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
A	119	61	9	124mm (4.9")	118



Device A: 4.9" Screen – Focus Group

+ Overall	Like
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Somewhat Comf.
+/- Comfort: 1Hand Land.	Neither
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable

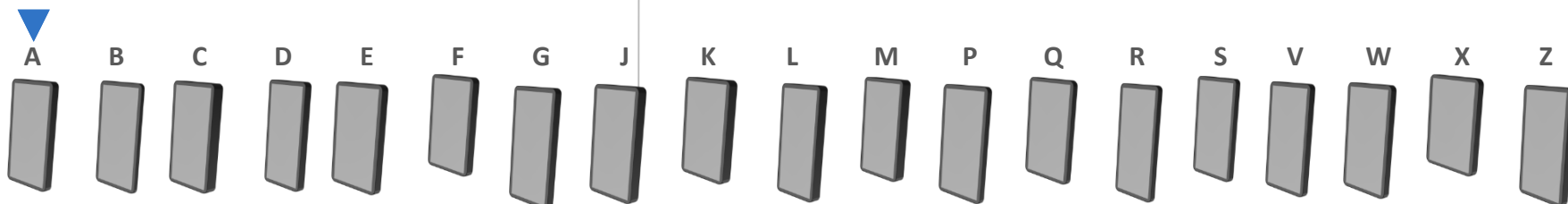
Most Preferred

People like the length and width proportion of this phone. They felt it was thin with a good weight.

- "Easier to see the writing, reach & fit in hand"
- "The thickness is dead on & feels sturdy, fits well in pocket & head"
- "I like the size, wasn't too thin or thick but still had a good size screen"
- "It was comfortable holding with 1 or 2 hands, when I type I go fast and this isn't too narrow, I wouldn't hit the wrong key as often. It wasn't too thick or thin, just the right weight"

Least Preferred

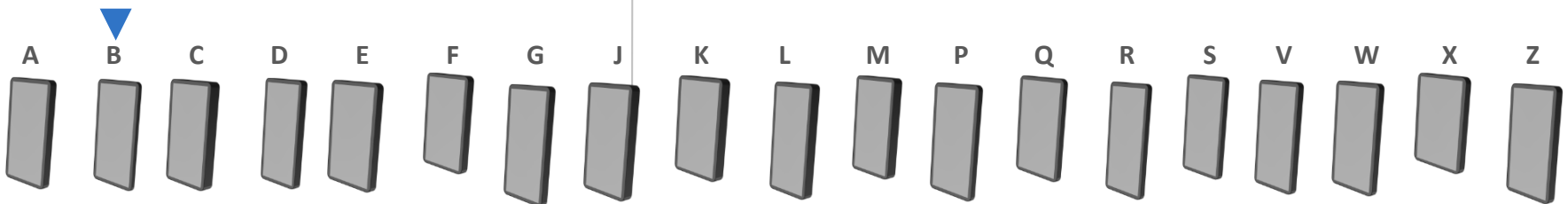
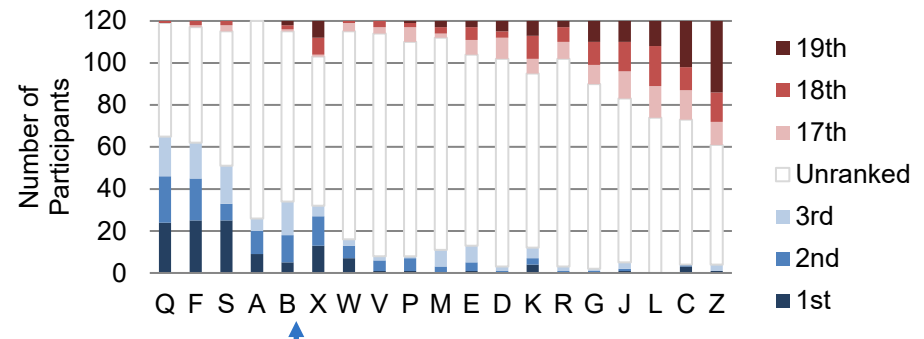
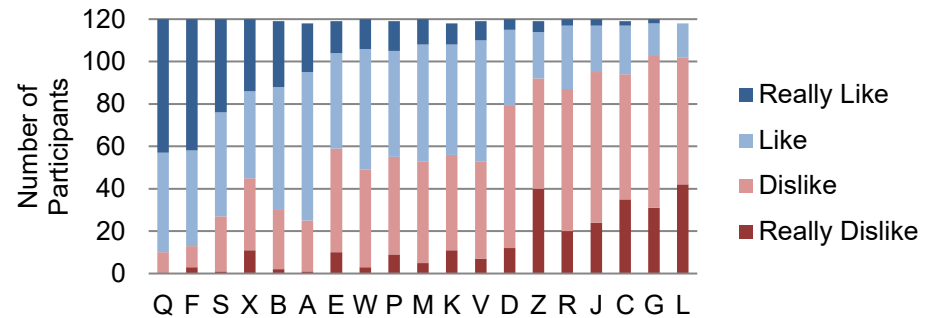
no comments



Device B: 4.9" Screen

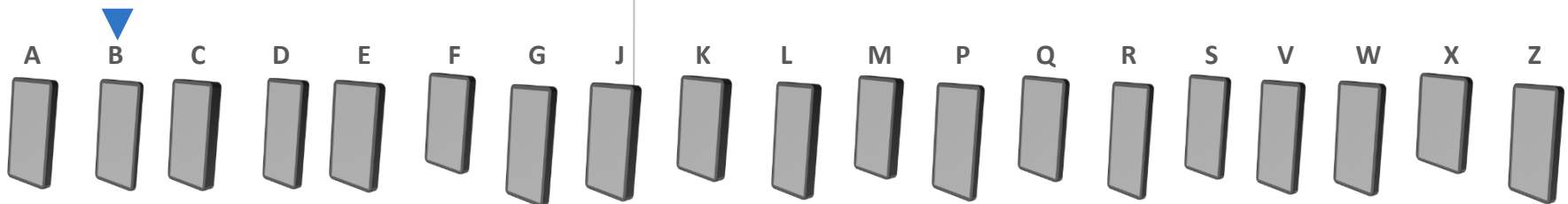
+ Overall	Like
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Sm But Acceptable
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Comfortable
+/- Comfort: 1Hand Land.	Neither
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
B	119	61	4	124mm (4.9")	79



Device B: 4.9" Screen – Focus Group

+ Overall	Like
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Sm But Acceptable
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Comfortable
+/- Comfort: 1Hand Land.	Neither
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable



Most Preferred

The screen size was good on this phone. Some people liked how thin this phone was which made it lightweight and slim in pockets.

- "It's thin, small & light. I like thin because it's easier to hold/maneuver, fits in my pocket easier, plus smartphones have a case which adds width so I want it thin"
- "In landscape you could get both thumbs across easily and they weren't overlapping too much. I was a little worried about how fragile it might be"
- "It fit well in my hand, it lay in my palm well, it was easier to hold, went well in my pocket. The thinness was a large factor in that. If it were any thicker at that ratio it would be clunky"
- "3rd, balanced, not too thick, fit good in your pocket. It was a little long but acceptable"

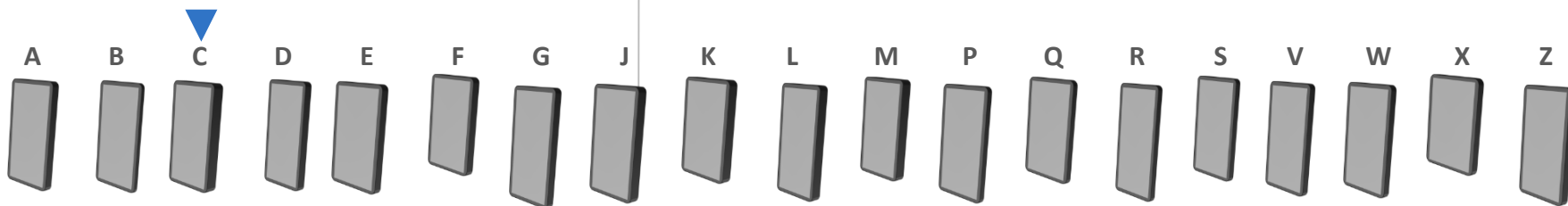
Least Preferred

This phone feels breakable because it is too thin. It is too long for pockets and holding in landscape orientation.

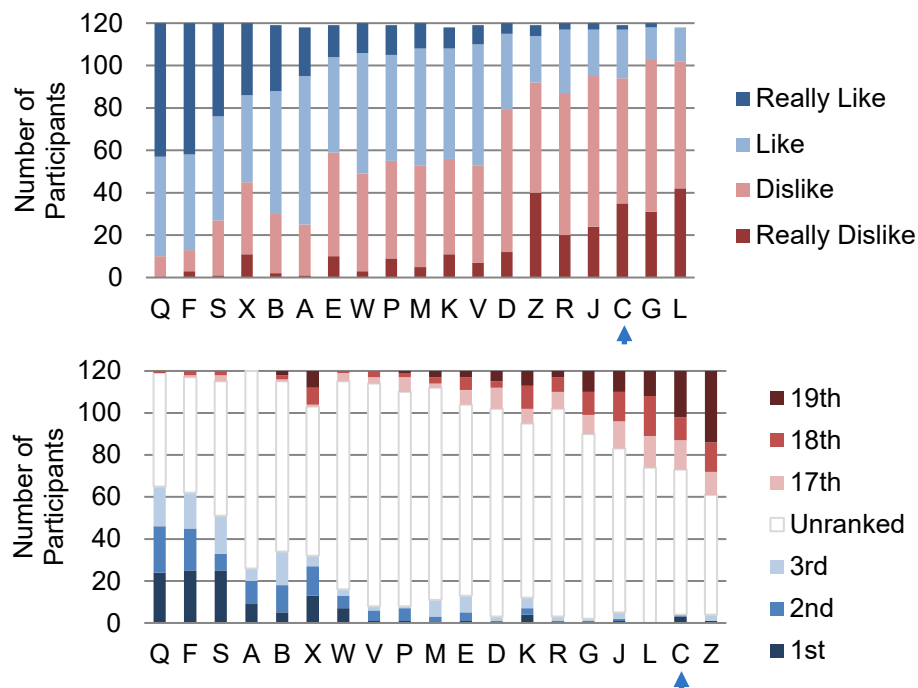
- "Too long, can't put in back pant pocket, could break if I sit on it, too wide in landscape for 2 hands"
- "It's too thin, it looks like it would crack"
- "It was thin, it was hard to pick up"

Device C: 4.9" Screen

- Overall	Really Dislike
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
- Dimensions: Thickness	Too Big
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big



Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
C	119	61	14	124mm (4.9")	158



Device C: 4.9" Screen – Focus Group

- Overall	Really Dislike
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
- Dimensions: Thickness	Too Big
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Most Preferred

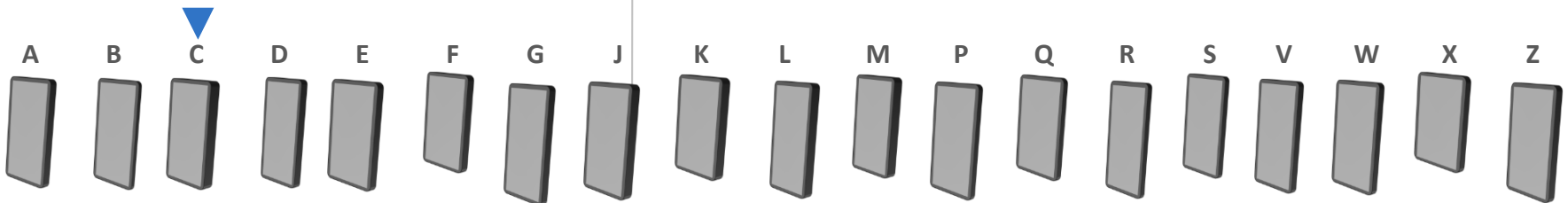
Only 3 people liked this device and were looking for a larger or heavier phone.

- "Felt better 1-handed; It's a little longer which is not best but it's narrow & fits my hand"
- "I like C better because it's thicker & weighs more"
- "I have big banana hands, so I like the thickness & the weight; I'm worried the smaller ones will flip out of my hands"

Least Preferred

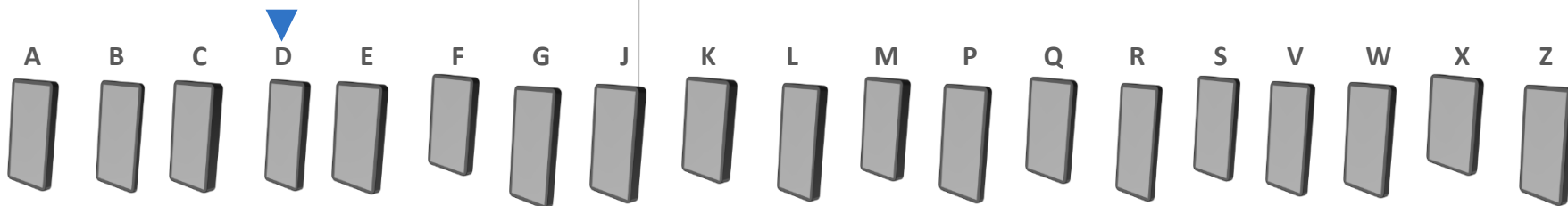
The phone feels breakable because it's too thin. It's too long for pockets and landscape use.

- "Too long, can't put in back pant pocket, could break if I sit on it, too wide in landscape for 2 hands"
- "It's too thin, it looks like it would crack"
- "It was thin, it was hard to pick up"

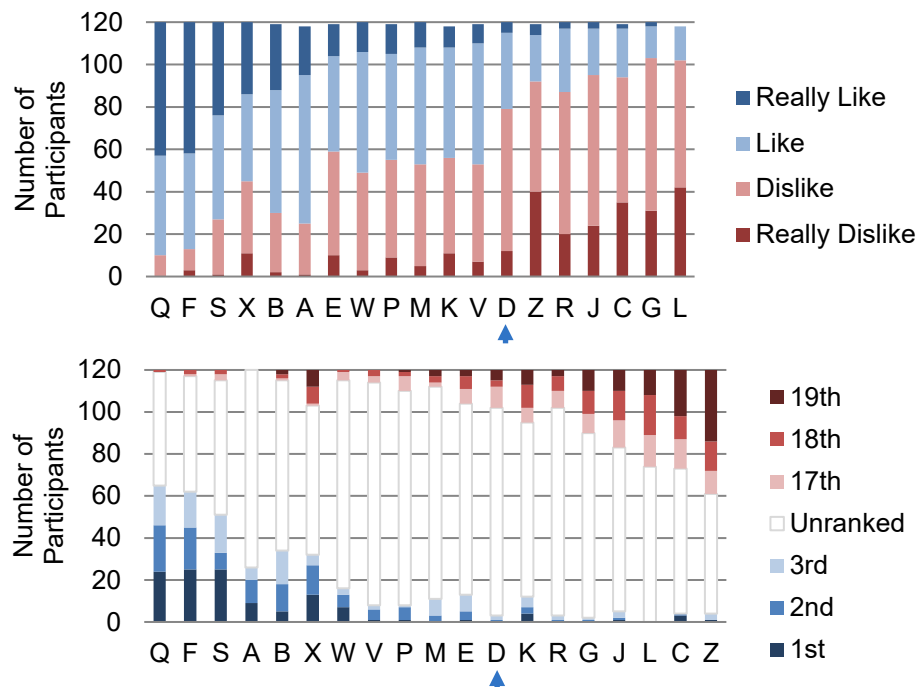


Device D: 4.8" Screen

- Overall	Dislike
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Sm.But Acceptable
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+/- Comfort: 2Hand Port.	Neither
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Somewhat Comf.
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
- Shirt Pocket	Too Big



Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
D	119	54	9	122mm (4.8")	110



Device D: 4.8" Screen – Focus Group

- Overall	Dislike
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Big But Acceptable
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+/- Comfort: 2Hand Port.	Neither
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Somewhat Comf.
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
- Shirt Pocket	Too Big

Most Preferred

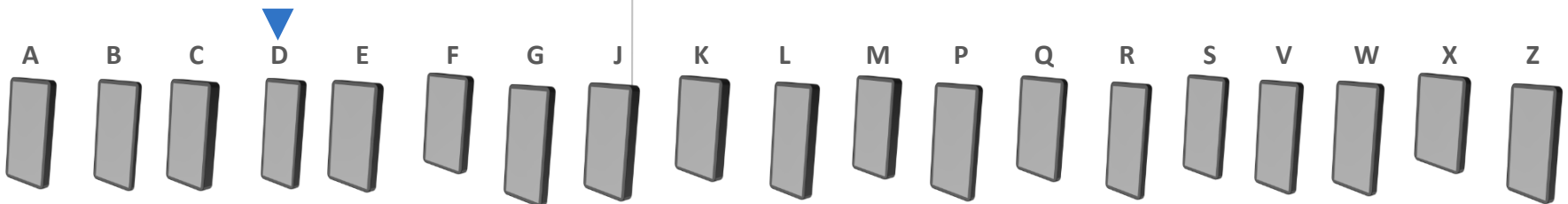
One person felt this phone was narrow which was good for calling and texting (not multimedia).

- "It's a little narrower, since I don't browser the internet all I do is phone calls & text messaging with a full keyboard, this would have a good keyboard"

Least Preferred

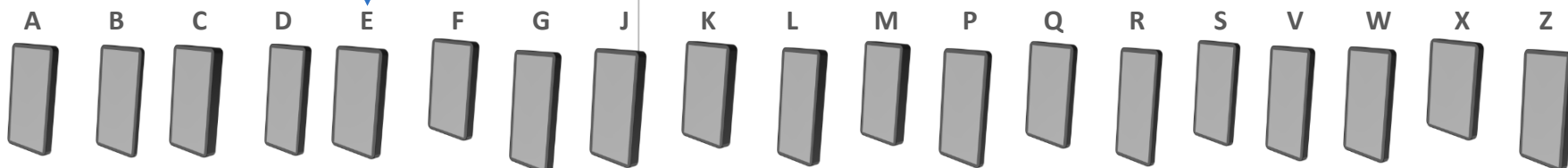
This phone is too tall and narrow. It creates an awkward shape especially using it in portrait orientation

- "Totally too narrow, too much scrolling to read"
- "Too narrow & long; Wouldn't fit in my pocket it would stick out (of back pocket)"
- "Too long. It just didn't look right; It looked like a remote control, not a phone. It's not big enough"
- "Doesn't seem like you could fit a keypad across it in portrait"

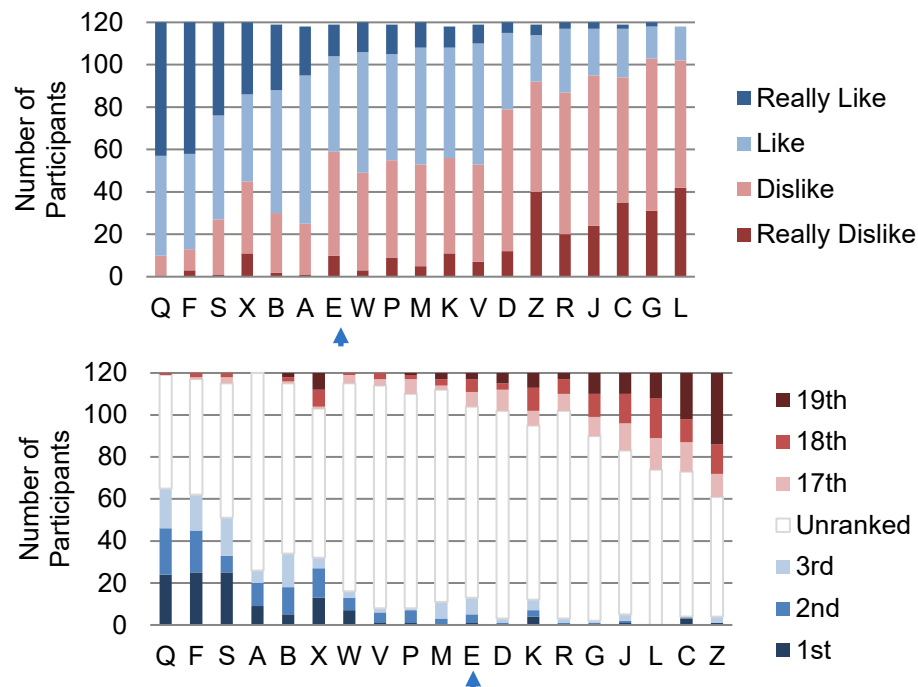


Device E: 5.0" Screen

+ Overall	Like
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Big But Acceptable
+ Dimensions: Thickness	Just Right
+/- Comfort: 1Hand Port	Neither
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Too Big
+ Back Pocket	Too Big
+ Belt Clip	Too Big
+ Shirt Pocket	Too Big



Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
E	119	69	9	127mm (5.0")	127



Device E: 5.0" Screen – Focus Group

+	Overall	Like
+	Dimensions: Length	Big But Acceptable
+	Dimensions: Width	Big But Acceptable
+	Dimensions: Thickness	Just Right
+/-	Comfort: 1Hand Port	Neither
+	Comfort: 2Hand Port.	Somewhat Comf.
-	Comfort: 1Hand Land.	Somewhat Uncomf.
+	Comfort: 2Hand Land.	Comfortable
+	Front Pocket	Too Big
+	Back Pocket	Too Big
+	Belt Clip	Too Big
+	Shirt Pocket	Too Big

Most Preferred

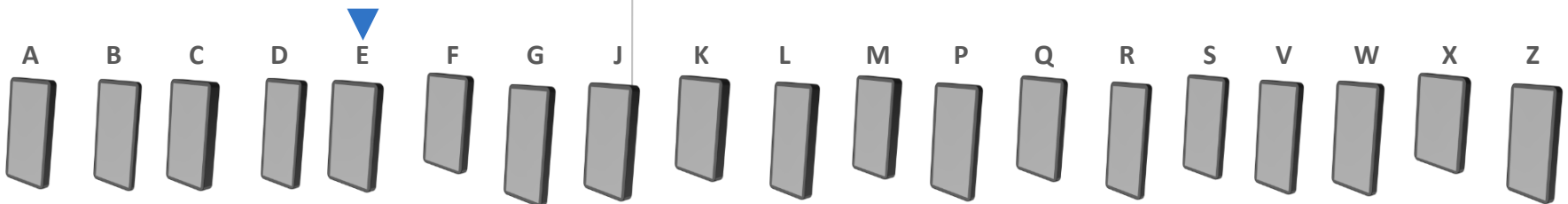
People felt it was slightly thick but that added to it feeling of durability and not easy to lose. Also this screen size was slightly larger which people liked for multimedia use.

- "Good height & width, a little too thick but thin phones are easy to lose, forget, fall out pocket unnoticed"
- "I preferred E because it's similar to what I have now. It's convenient; I don't want to lose it"
- "I like big phones so I can see what I am reading on the internet. E is a little bigger which makes it easier to read things"
- "It was my 3rd, prescreen size. After screen size, E became my top. Games, browsing, etc. are very important to me. So E prevailed. It felt durable, not too thick"

Least Preferred

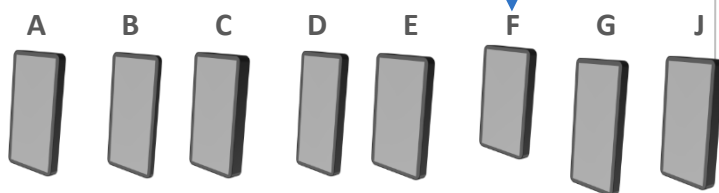
This phone is too thick, bulky and heavy in your hands.

- "Too big for grip, uncomfortable in pocket, very thick, too bulky"
- "Too big & clunky"
- "Proportions not feel right in my hand"
- "I like a thicker phone, but that's just unattractive, it's a big thick phone"

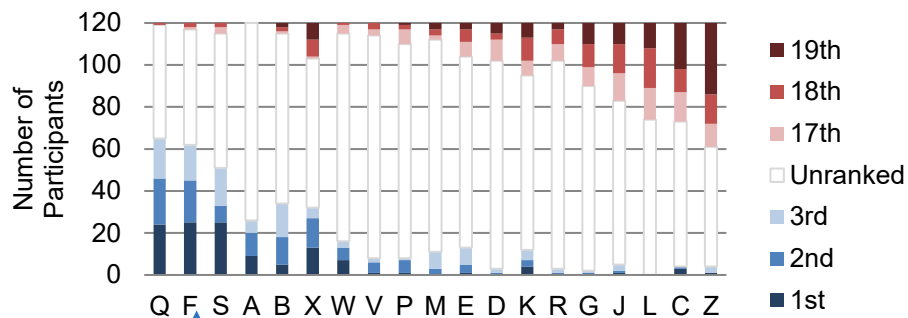
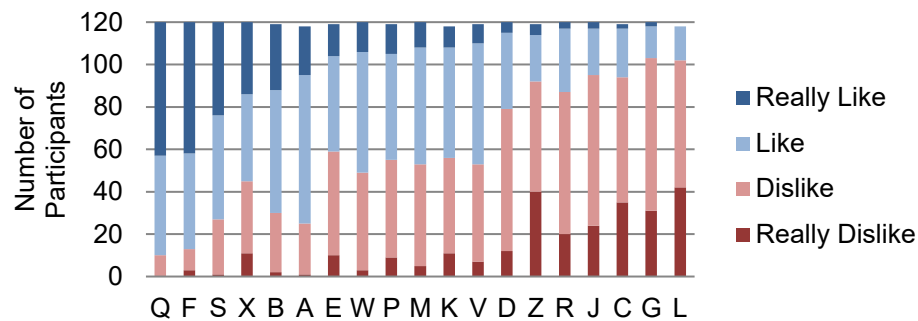


Device F: 4.4" Screen

+ Overall	Really Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Comfortable
+ Comfort: 1Hand Land.	Somewhat Comf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Just Right
+ Back Pocket	Just Right
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable



Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
F	107	61	9	112mm (4.4")	111



Device F: 4.4" Screen – Focus Group

+ Overall	Really Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Comfortable
+ Comfort: 1Hand Land.	Somewhat Comf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Just Right
+ Back Pocket	Just Right
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable

Most Preferred

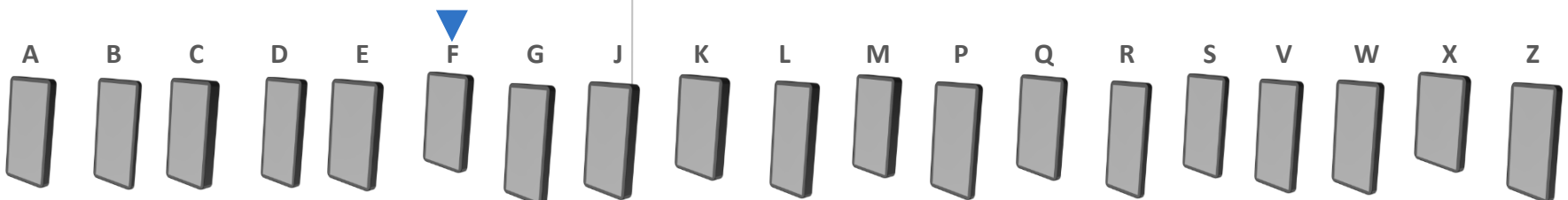
This phone has a great screen size while still feeling compact and fitting in the hand. It's thin but still feels durable. Many people mentioned that it felt similar to current phone. This phone felt useable for people with small hands as well.

- "It seemed to fit my hand perfectly, others were too big or thin; Fit in your pocket easy but not too skinny. Others were too narrow and the screen seemed like you wouldn't fit enough text on it"
- "It was comfortable in the hand, not too thin, fit just right, the weight was good"
- "Most proportionate screen size. Closest to what I own, portable"
- "It was small, I liked the reach because of text messaging, good length (like current), width, screen; I have small hands it was comfortable"
- "Like the length & width proportion, others were too long; Thinner than I'm used to (good)"
- "It wasn't too super thin, so it'll be a little more durable, especially if you used it without a case"

Least Preferred

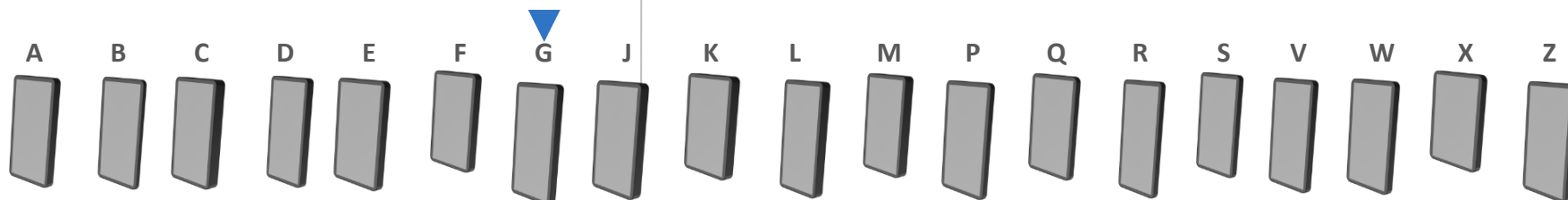
This phone was slightly thicker and shorter which bothered a few people.

- "Thicker & more substantial, not quite to the point where it ruled it out, but it was a little on that top side"
- "Too short & not comfortable, It was too wide in my hand, I have small hands"

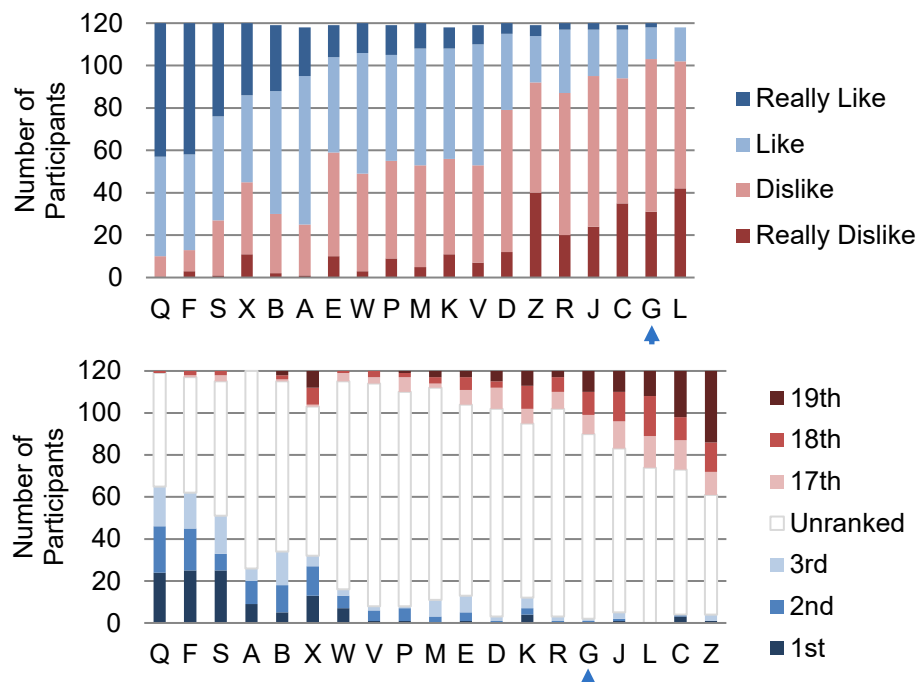


Device G: 5.3" Screen

- Overall	Really Dislike
- Dimensions: Length	Too Big
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big



Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
G	132	61	9	135mm (5.3")	126



Device G: 5.3" Screen – Focus Group

- Overall	Really Dislike
- Dimensions: Length	Too Big
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

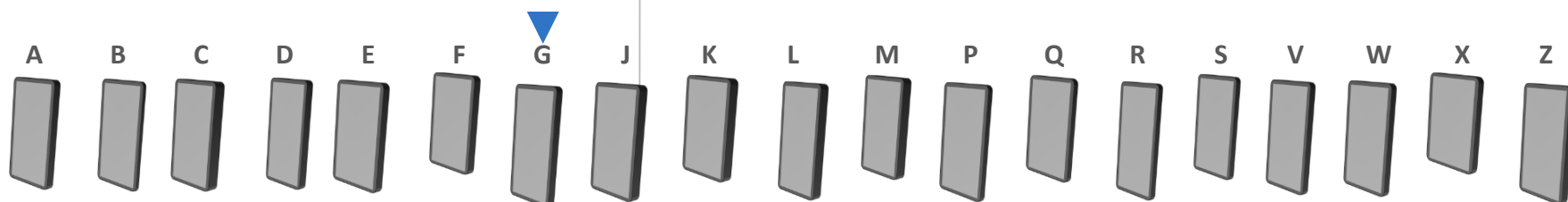
Most Preferred

no comments

Least Preferred

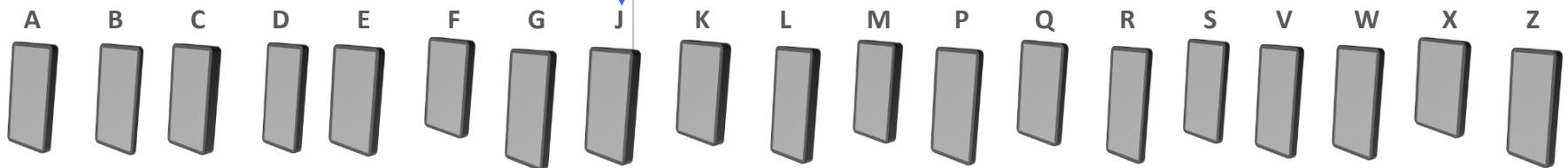
The worst part is the height in portrait for storing and reaching with thumbs. Some people also felt it was too thick.

- "Long & narrow. Thickness was so-so. It wouldn't fit in your pocket"
- "Way too long, least favorite, too long, bad overall feel & proportions, poor for browsing & reading"
- "It was too tall. When I hold it landscape, my hands would be too far apart"
- "Too tall. I hold it portrait, but it would be hard to reach up & down"
- "Can't use 1 hand in landscape"

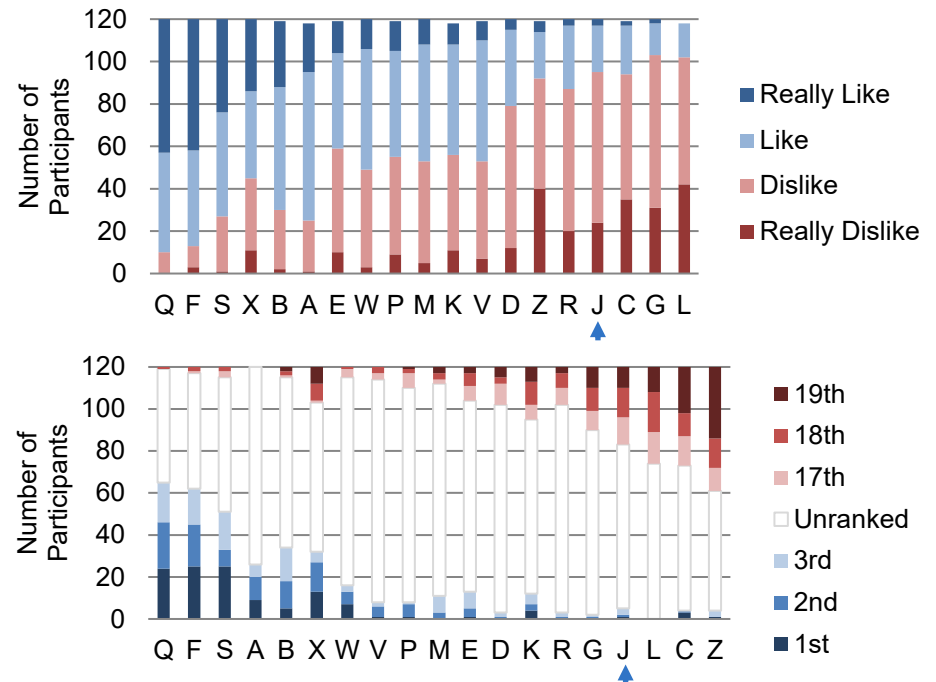


Device J: 5.2" Screen

- Overall	Really Dislike
- Dimensions: Length	Too Big
+ Dimensions: Width	Just Right
- Dimensions: Thickness	Too Big
+/- Comfort: 1Hand Port	Neither
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Far Too Big



Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
J	127	66	12	132mm (5.2")	155



Device J: 5.2" Screen – Focus Group

- Overall	Really Dislike
- Dimensions: Length	Too Big
+ Dimensions: Width	Just Right
- Dimensions: Thickness	Too Big
+/- Comfort: 1Hand Port	Neither
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Far Too Big

Most Preferred

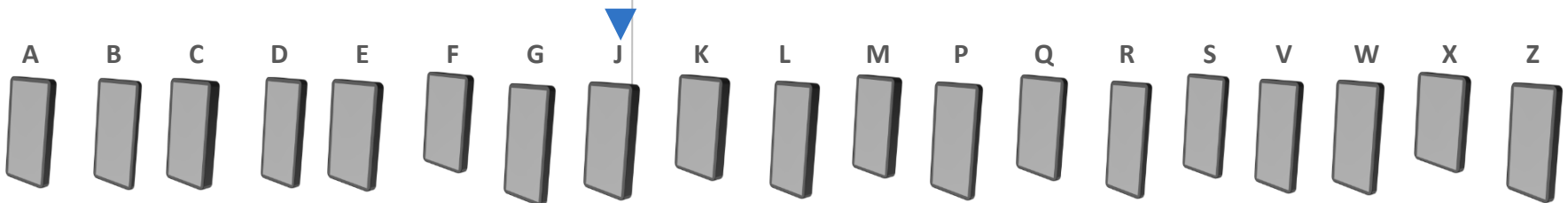
This phone was taller and people like the bigger landscape view it gave them.

- "I like size & weight; I'm more of a traditional phone talker as opposed to a browser. I like a heavy phone I can use anywhere"
- "Similar to C, I have big hands, this was slightly thinner but taller"
- "It was taller so it felt better in landscape for me and the images on it were pretty good. It was definitely thicker than the other ones I liked, but it wasn't too thick to bother me that much"

Least Preferred

This phone was too long and too thick which made it bulky and difficult to use with one hand.

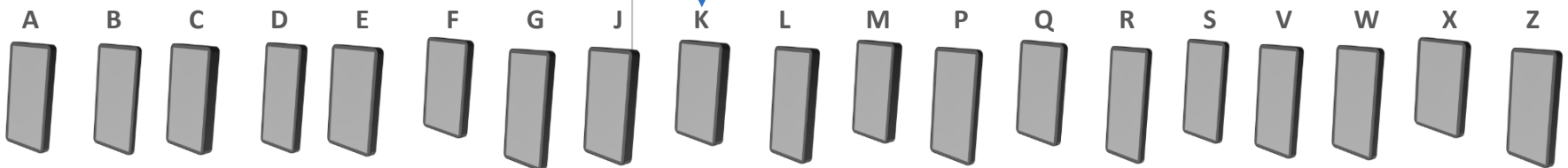
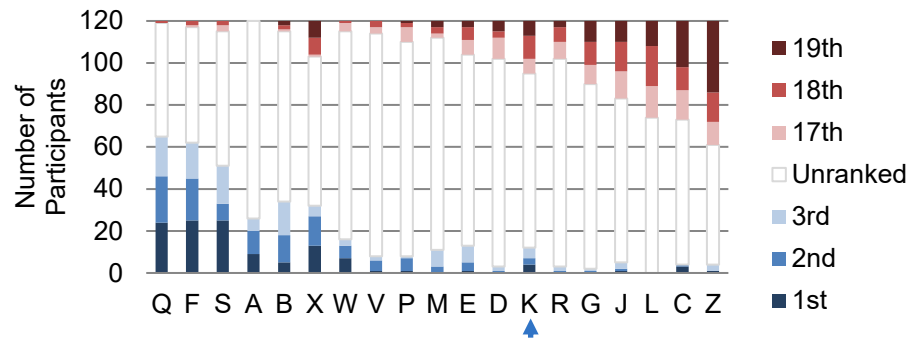
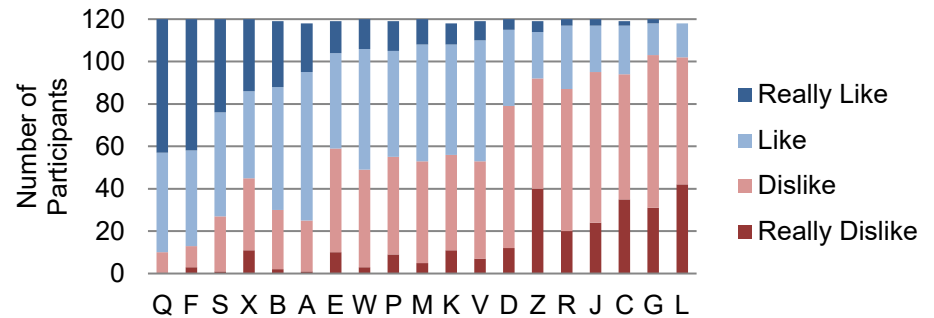
- "Way too thick. And it was too wide for its thickness, that made it seem cumbersome"
- "I feel like I could put a bunch of those together and build a wall"
- "Too thick for hand (portrait or landscape), too tall for thumbs, height & weight create poor balance"
- "It would be better if it were wider, the balance would be better, less lopsided"



Device K: 4.7" Screen

+ Overall	Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Just Right
- Dimensions: Thickness	Too Big
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Comfortable
+/- Comfort: 1Hand Land.	Neither
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
K	112	66	12	119mm (4.7")	142



Device K: 4.7" Screen – Focus Group

+ Overall	Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Just Right
- Dimensions: Thickness	Too Big
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Comfortable
+/- Comfort: 1Hand Land.	Neither
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Most Preferred

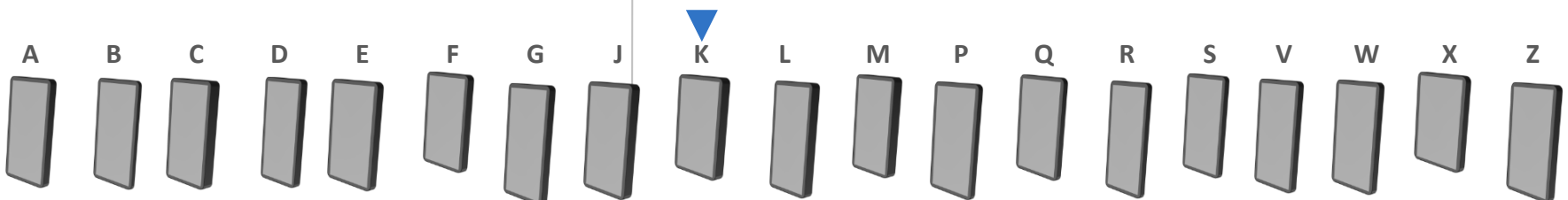
This phone had a good screen size and was a little thicker which made it seem more durable.

- "I like the thicker phones they feel sturdier & good in hands, thin feels like it might break if I drop it"
- "I like thick phones that have some weight to them so I don't forget which pocket they're in so I don't break them. It seems like you could see a decent amount whichever way you turned it"
- "I text landscape, on screen keyboard would be big enough to see when I'm typing at the same time"
- "I like that the phone is thicker because you can pack more into it, more features"

Least Preferred

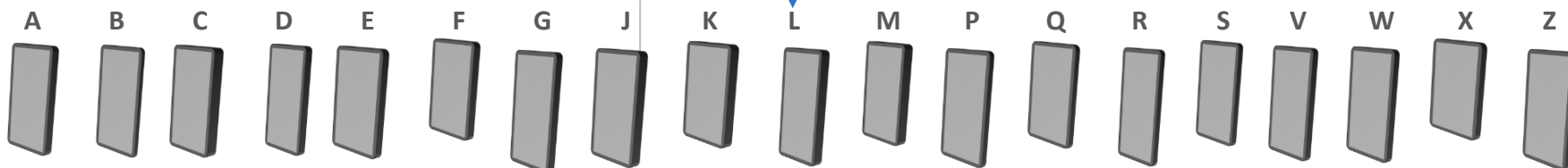
Overall this phone was too thick which made it very heavy for holding and pocketing.

- "Looks ancient. Feels like an older, candy bar phone"
- "Thick & wide. It looks like a deck of playing cards"
- "Too fat & chunky"
- "You could use it for a hockey puck"

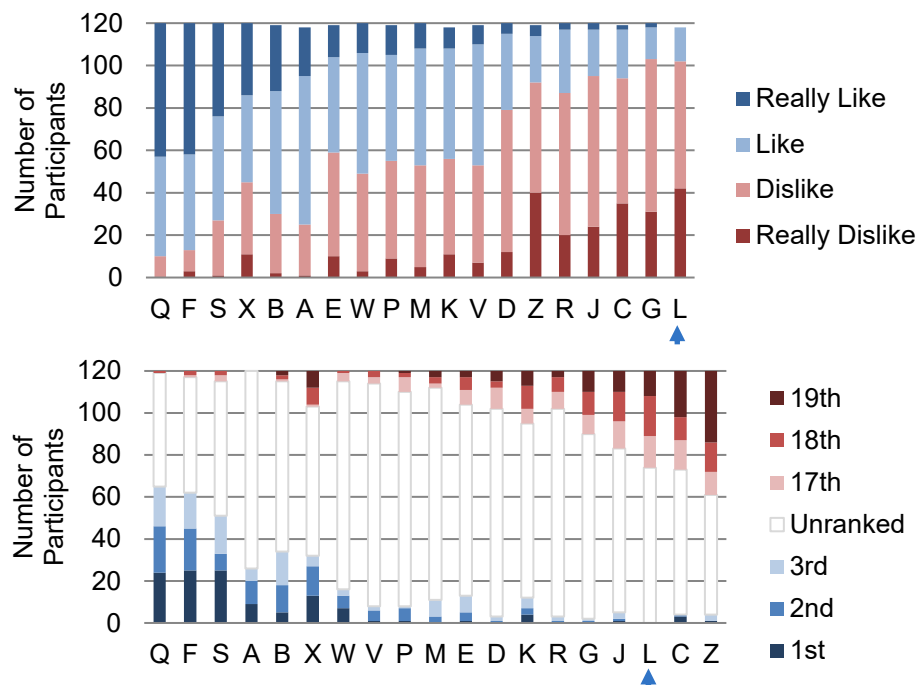


Device L: 5.1" Screen

+ Overall	Really Dislike
+ Dimensions: Length	Too big
+ Dimensions: Width	Sm. But Acceptable
- Dimensions: Thickness	Too Big
+ Comfort: 1Hand Port	Somewhat Comf.
+/- Comfort: 2Hand Port.	Neither
+ Comfort: 1Hand Land.	Uncomfortable
+ Comfort: 2Hand Land.	Somewhat Comf.
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big



Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
L	127	57	12	129mm (5.1")	140



Device L: 5.1" Screen – Focus Group

+ Overall	Really Dislike
+ Dimensions: Length	Too big
+ Dimensions: Width	Sm. But Acceptable
- Dimensions: Thickness	Too Big
+ Comfort: 1Hand Port	Somewhat Comf.
+/- Comfort: 2Hand Port.	Neither
+ Comfort: 1Hand Land.	Uncomfortable
+ Comfort: 2Hand Land.	Somewhat Comf.
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

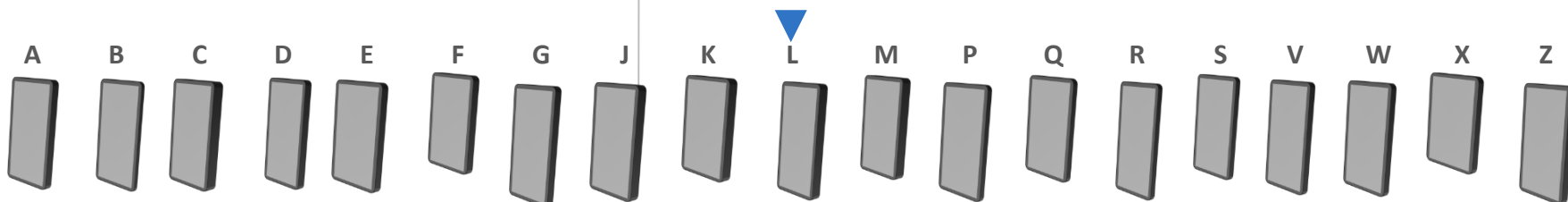
Most Preferred

No comments – L was never picked as one of the three most preferred devices.

Least Preferred

Overall the phone is too narrow and thick which makes a bad proportion.

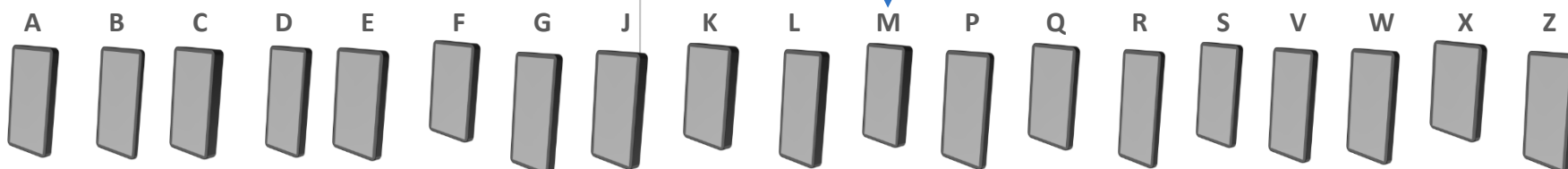
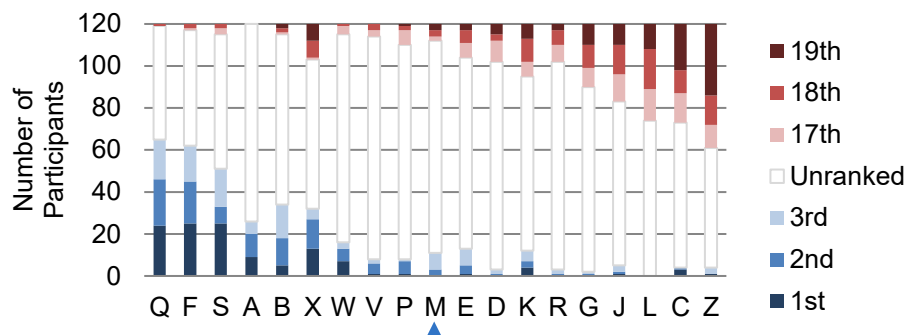
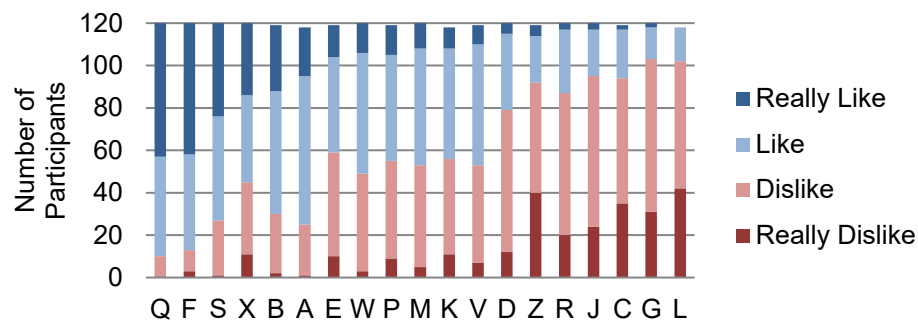
- “Too long, thick & super heavy. In portrait you could never use 2 thumbs they would hit each other”
- “It seemed like in portrait you’d only get 2-3 words and it would wrap”
- “Too narrow, too thick; Hard to slip in and out of pockets; Too narrow for watching a video”
- “Unusual aspect ratio- can’t reach 1 handed; Thicker than current sliders; A touch keyboard in landscape would lose $\frac{3}{4}$ of your vertical space”



Device M: 4.5" Screen

+ Overall	Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Just Right
- Dimensions: Thickness	Too Big
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Somewhat Comf.
+ Comfort: 1Hand Land.	Somewhat Comf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
M	112	57	12	114mm (4.5")	129



Device M: 4.5" Screen – Focus Group

+	Overall	Like
+	Dimensions: Length	Just Right
+	Dimensions: Width	Just Right
-	Dimensions: Thickness	Too Big
+	Comfort: 1Hand Port	Comfortable
+	Comfort: 2Hand Port.	Somewhat Comf.
+	Comfort: 1Hand Land.	Somewhat Comf.
+	Comfort: 2Hand Land.	Comfortable
+	Front Pocket	Big But Acceptable
+	Back Pocket	Big But Acceptable
+	Belt Clip	Big But Acceptable
+	Shirt Pocket	Big But Acceptable

Most Preferred

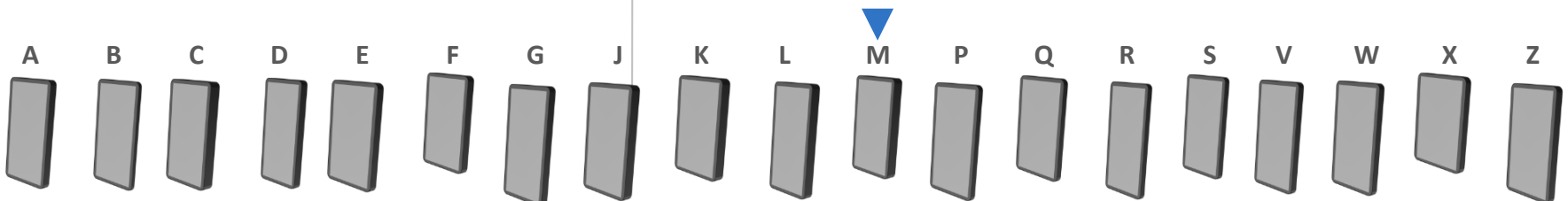
Most people who picked this mentioned that they liked the weight in their hand and that the phone felt solid.

- "I felt the most solid without being a brick. It's not thick & huge; it's thick and just the right size"
- "Similar to F, maybe narrower, but M felt more solid in my hand. It was the weight in my hand, I felt like I had a good grasp on it, as opposed to feeling like it could slip out of my hand because it was too thin"
- "It's the closest to my phone, it's a little thicker than I like so F was 1st, but this felt sturdy"
- "A little thinner than current phone, my thumb can touch everything on the screen, 3rd choice"

Least Preferred

Overall this phone was too small and extremely thick which made it bulky.

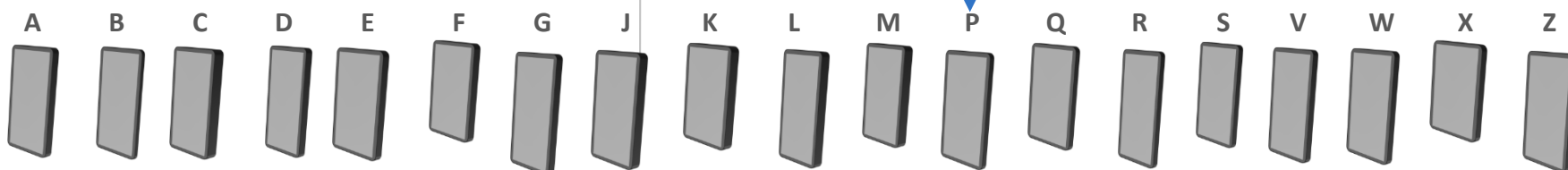
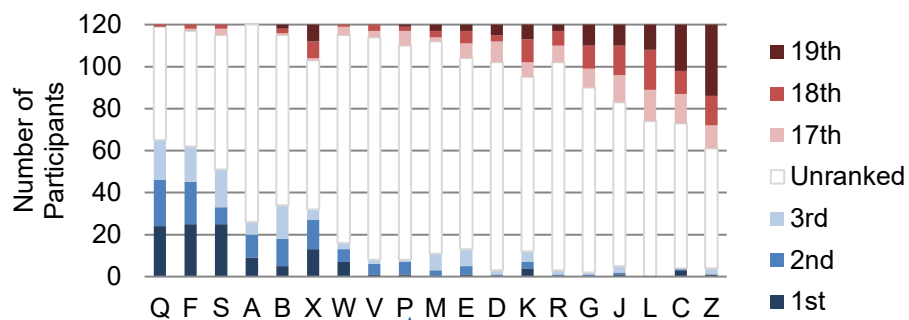
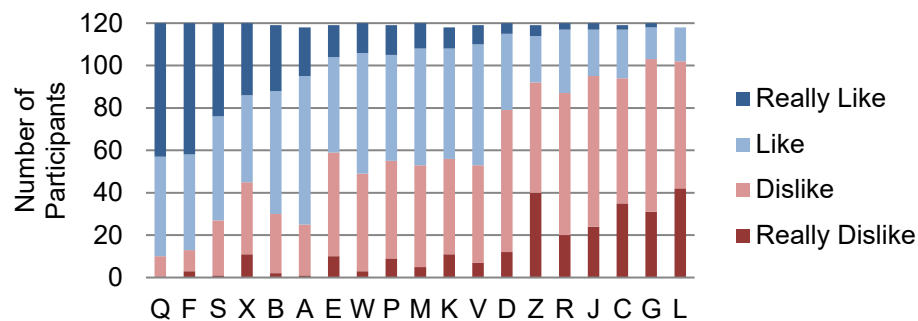
- "M was so thick I could hit someone and they would get hurt"
- "Thick & narrow, and not even that tall. I marked it well on holding in my hand, but it didn't fit well in the pocket because of the thickness, and I didn't like the screen size"
- "That one confused me, the screen size would have been my favorite, but it was so thick I threw it out"



Device P: 5.2" Screen

+ Overall	Like
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
P	127	66	6	132mm (5.2")	101



Device P: 5.2" Screen – Focus Group

+ Overall	Like
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Most Preferred

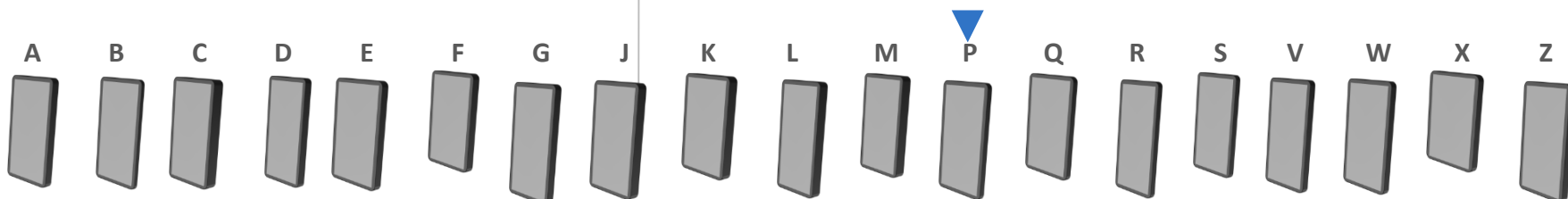
People liked the thinness and large screen, but they felt other phones might be slightly easier to reach.

- "It had enough width & length, in portrait you'd get enough on the web page"
- "I liked the thinness"
- "Little farther to reach"
- "It was just a little bit bigger than Q"
- "Thin & tall, Very similar to B but I liked that B was just a little bit smaller"

Least Preferred

This phone is too tall which makes it awkward. Some thought it was too thin and/or bulky in general.

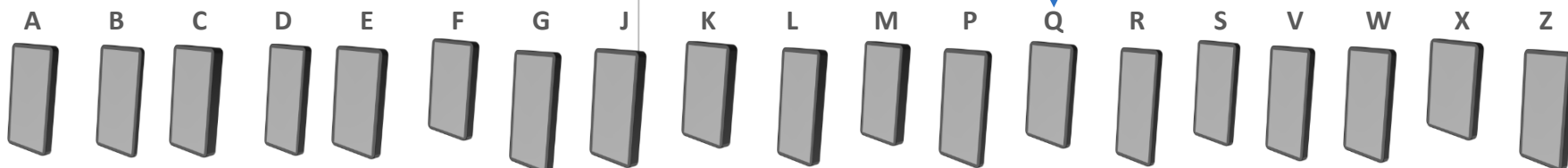
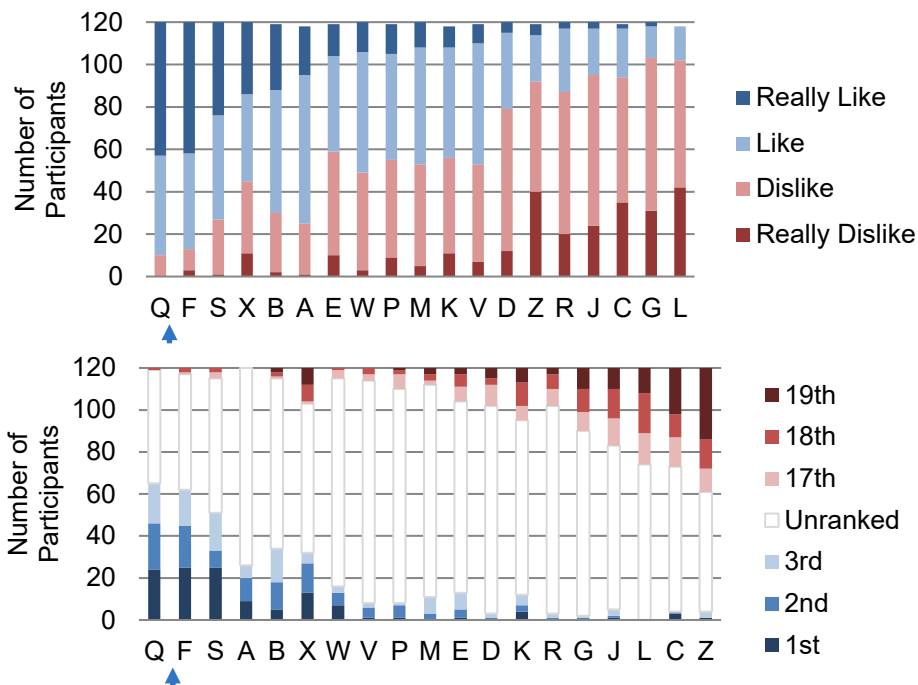
- "It was too narrow & tall, can't use landscape...out of control to do it with one hand. Too bulky"
- "Don't like the long phone"
- "Too thin for the length & width. Worried if I were holding it that it would flip out and fall to the ground"
- "Way too tall & big overall, especially on the height"



Device Q: 4.7" Screen

+ Overall	Really Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Comfortable
+ Comfort: 1Hand Land.	Somewhat Comf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
Q	112	66	6	119mm (4.7")	95



Device Q: 4.7" Screen – Focus Group

+ Overall	Really Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Comfortable
+ Comfort: 1Hand Land.	Somewhat Comf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable

Most Preferred

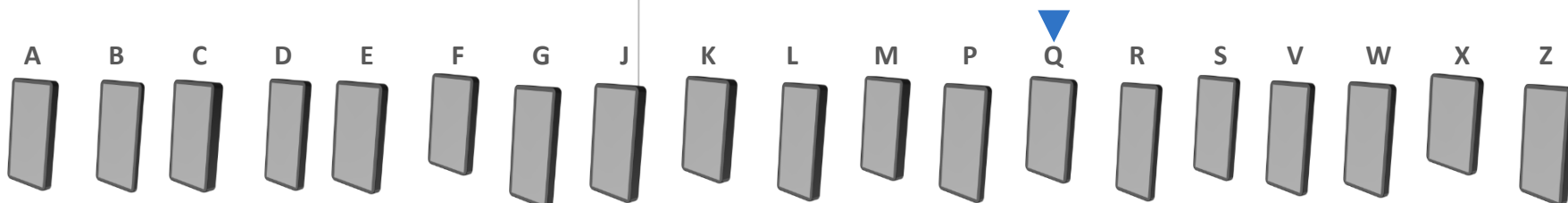
The screen was slightly wider than the other favorites which people liked for more viewing. It was thin and not too tall which made it fit well in people's hands.

- "The screen was great, good size, not too big, easy to hold"
- "Little shorter but still wider than most, good because it's wide enough to read across without jumping lines; Holding up to head may feel like holding a square & awkward. Good for things other than phone"
- "Like the screen dimensions, feels more like a portable multimedia device, good as my only device"
- "When it's too thin you can add a case to make it bulkier, but you start with a smaller foundation"
- "It's a good width for web pages, I can see lots without scrolling as much as I do on my phone"
- "Surface area for visibility & touch functionality; thumb could easily reach any icon; felt balanced"
- "You can reach anywhere, thin. Thin is good because easy to get in pockets and not too heavy"

Least Preferred

The only negative for this phone was that it was too thin and felt fragile.

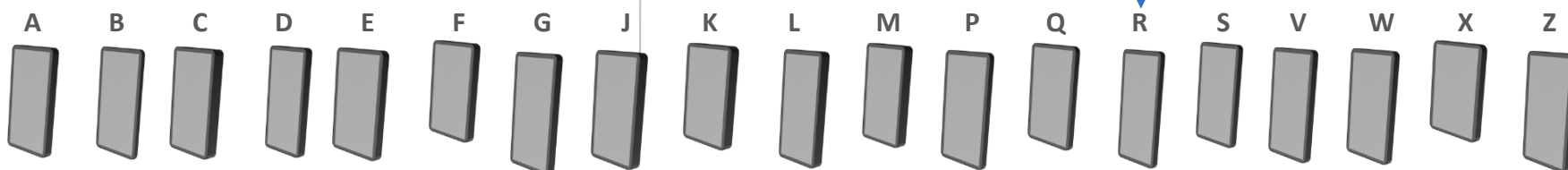
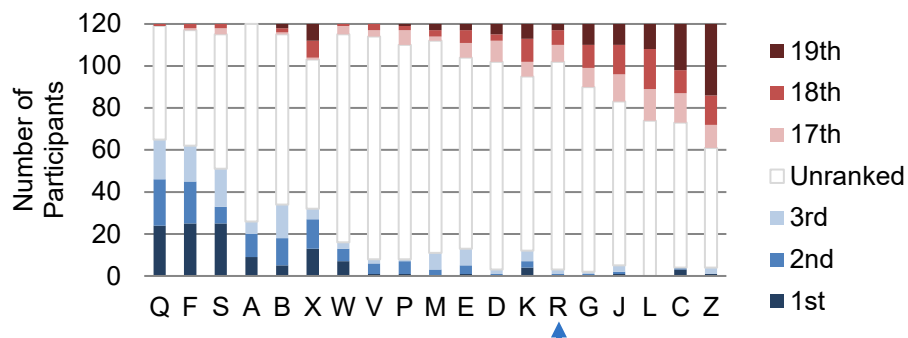
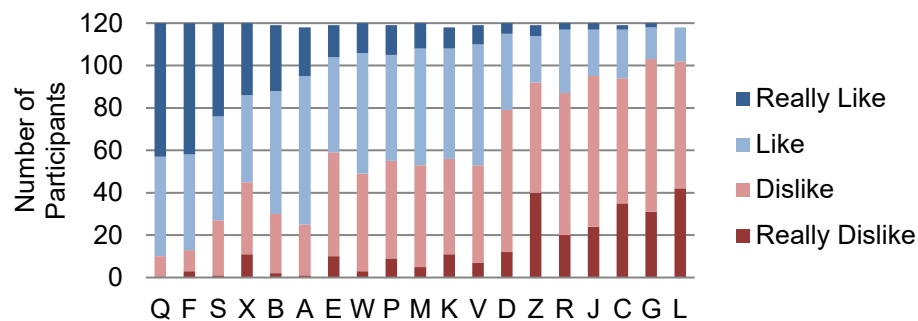
- "I thought it was a little too thin for the same reasons. If it's thicker it's safer"
- "I felt it was too thin, I don't like thin phones because I've broken too many of them"



Device R: 5.1" Screen

- Overall	Dislike
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Sm But Acceptable
+ Dimensions: Thickness	Sm But Acceptable
+ Comfort: 1Hand Port	Somewhat Comf.
+/- Comfort: 2Hand Port.	Neither
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
R	127	57	6	130mm (5.1")	94



Device R: 5.1" Screen – Focus Group

- Overall	Dislike
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Sm But Acceptable
+ Dimensions: Thickness	Sm But Acceptable
+ Comfort: 1Hand Port	Somewhat Comf.
+/- Comfort: 2Hand Port.	Neither
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Most Preferred

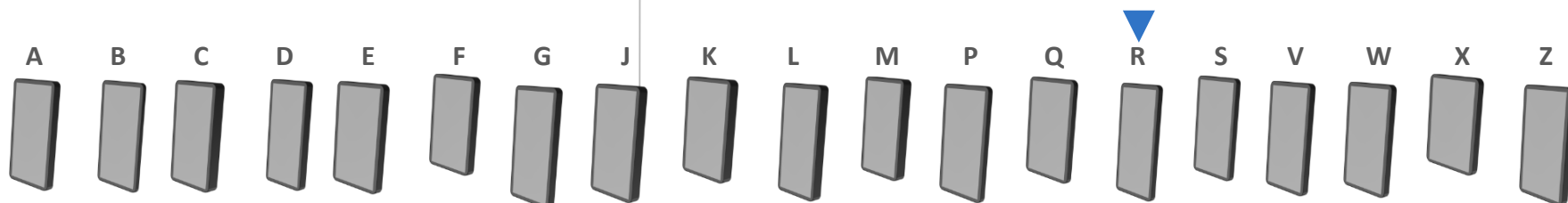
Two people commented that they liked the dimensions and did not feel it was too thin.

- "It was thin; It fit in my pocket; Comfortable to hold"
- "It had a lot of screen on it, and it felt good as a phone"

Least Preferred

This phone is so thin it feels like you can easily break it. It is too narrow especially in portrait orientation.

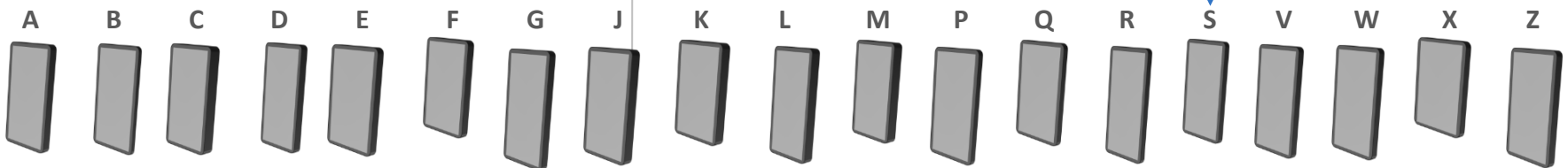
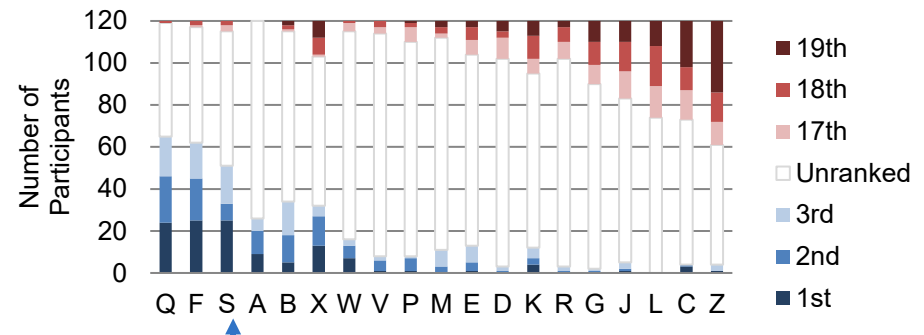
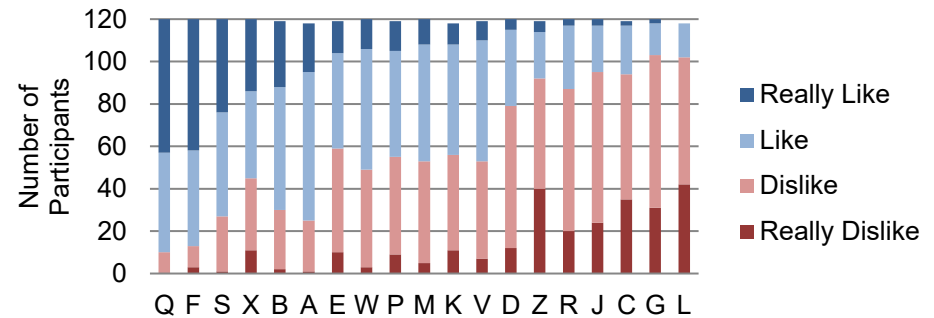
- "Feels like you're going to break it"
- "Yeah, I thought I might break it just by holding it, also too narrow"
- "I carry it mostly in front pocket, and R is too big & tall; sitting down would be uncomfortable"
- "It felt too narrow for me, if I were holding it in landscape I feel like I'd run out of vertical screen space"



Device S: 4.5" Screen

+ Overall	Really Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Just right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Somewhat Comf.
+ Comfort: 1Hand Land.	Somewhat Comf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Just Right
+ Back Pocket	Just Right
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
S	112	57	6	114mm (4.5")	88



Device S: 4.5" Screen – Focus Group

+ Overall	Really Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Just right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Comfortable
+ Comfort: 2Hand Port.	Somewhat Comf.
+ Comfort: 1Hand Land.	Somewhat Comf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Just Right
+ Back Pocket	Just Right
+ Belt Clip	Big But Acceptable
+ Shirt Pocket	Big But Acceptable

Most Preferred

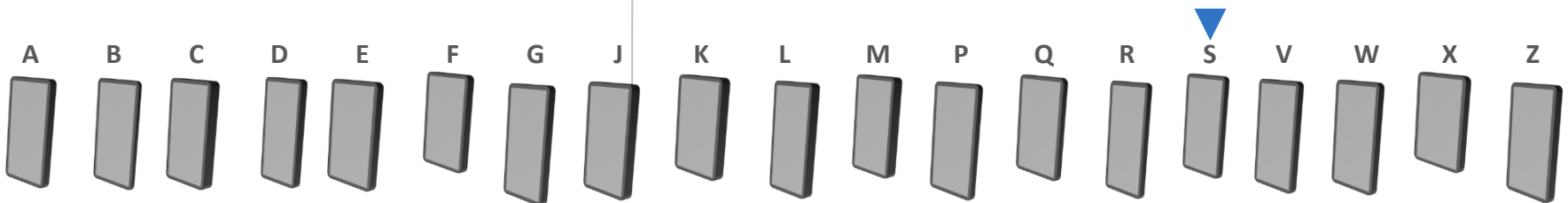
This phone is thin and can fit in more places easily and unnoticeably. It has good screen dimensions without being too big.

- "I'm a runner and bulk is bad. You can always put a cover on to make it thicker"
- "It's so thin - most important factors because it fits in more places, not holding something heavy"
- "Good size, thin, ease of holding in my hand & putting in pocket, won't fall out of your hand because it's too big. I carry in my front pocket and I like this because it's small"
- "Easy for little hands to hold, fit anywhere- pocket & purse, not so thin that I'd worry about it breaking"
- "Good dimensions for holding & screen for content, I don't like it covering half my face"

Least Preferred

This phone is so small and narrow that it feels cheap and low quality.

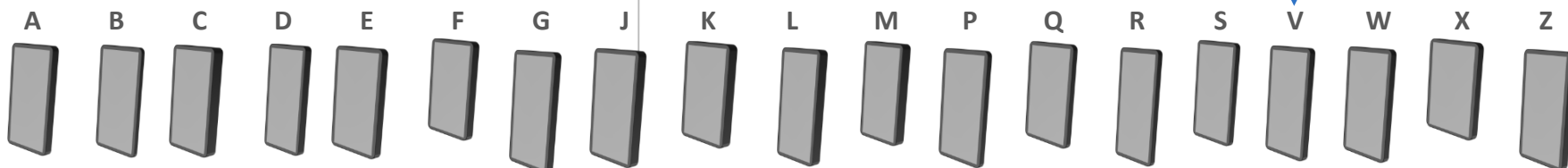
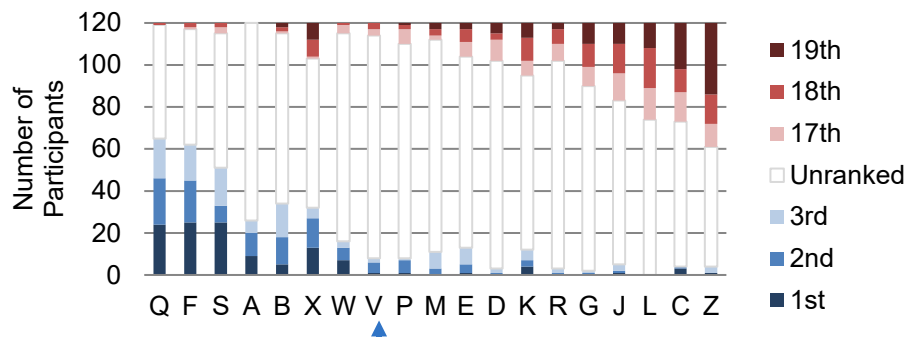
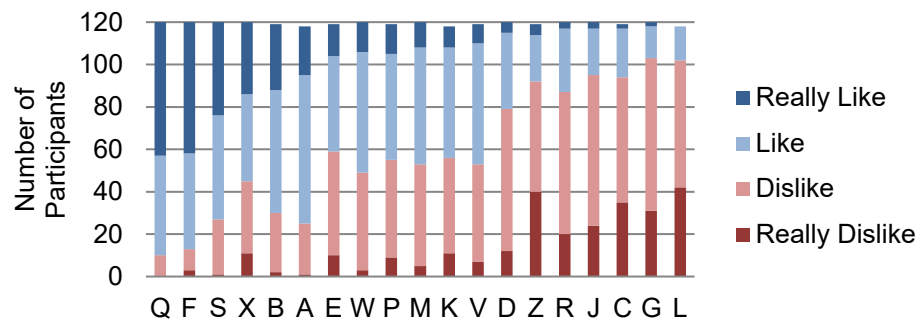
- "Size seems like a disposable phone for short-term use, would limit what you would do"
- "You're not going to enjoy getting a phone that size"
- "It felt so small it felt like you could snap it in half, so it didn't feel that durable to me"



Device V: 5.0" Screen

+ Overall	Like
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
+ Belt Clip	Big But Acceptable
- Shirt Pocket	Too Big

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
V	123	60	8	127mm (5.0")	107



Device V: 5.0" Screen – Focus Group

+	Overall	Like
+	Dimensions: Length	Big But Acceptable
+	Dimensions: Width	Just Right
+	Dimensions: Thickness	Just Right
+	Comfort: 1Hand Port	Somewhat Comf.
+	Comfort: 2Hand Port.	Somewhat Comf.
-	Comfort: 1Hand Land.	Somewhat Uncomf.
+	Comfort: 2Hand Land.	Comfortable
+	Front Pocket	Big But Acceptable
+	Back Pocket	Big But Acceptable
+	Belt Clip	Big But Acceptable
-	Shirt Pocket	Too Big

Most Preferred

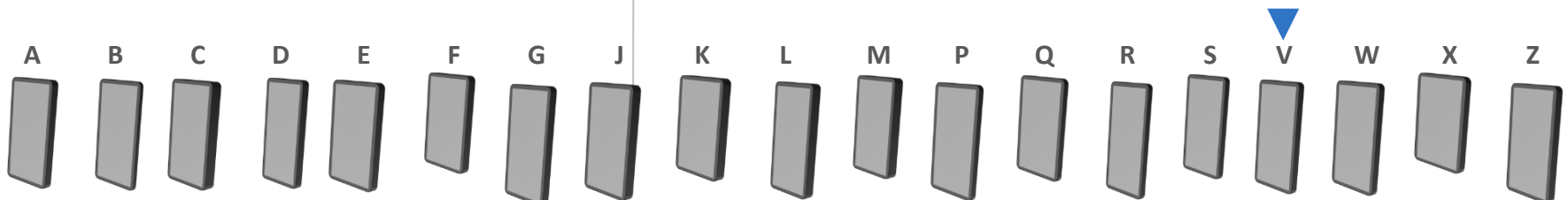
This phone was thin without feeling fragile. It was easy to type and reach all of the screen in portrait or landscape.

- "A little thicker which is good. I liked the height & width of V"
- "It would be easy to reach in portrait and good for typing in landscape"
- "Smaller, easier to reach, landscape felt good"
- "It was neck & neck for me with A, but A I chose as my top because it was a little bigger/thicker"

Least Preferred

The phone is too narrow to use with two hands and felt unsecure.

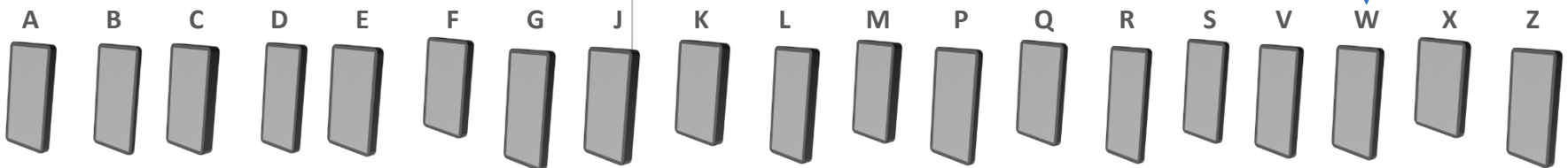
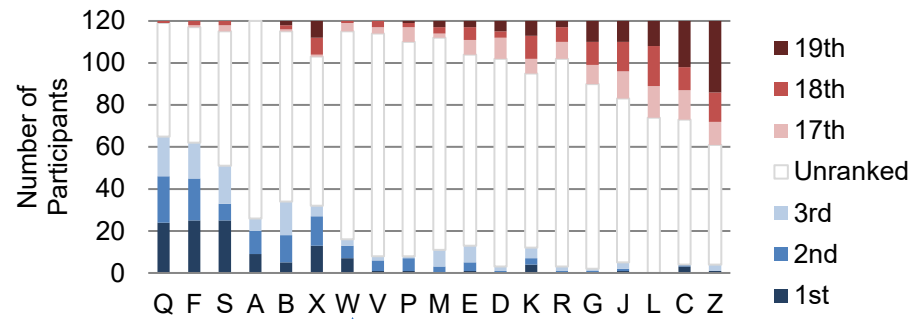
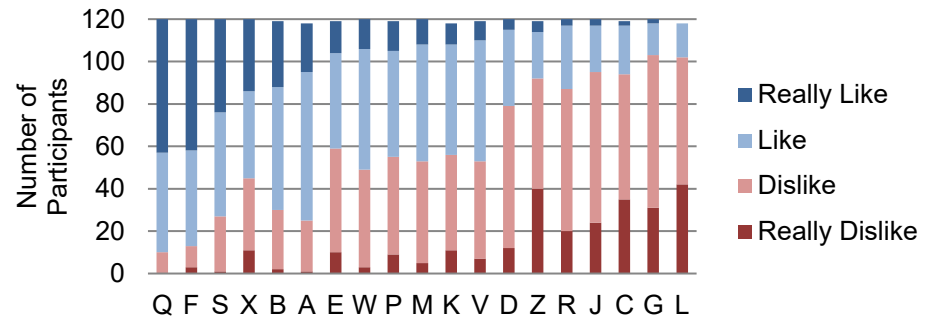
- "I did find it too narrow, awkward to use with 2 hands"
- "It was my 17th, it was too tall to get your thumb up & down"



Device W: 5.0" Screen

+ Overall	Like
+ Dimensions: Length	Big But Acceptable
+ Dimensions: Width	Just Right
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Somewhat Uncomf.
+ Comfort: 2Hand Land.	Comfortable
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
W	123	66	8	127mm (5.0")	113



Device W: 5.0" Screen – Focus Group

+	Overall	Like
+	Dimensions: Length	Big But Acceptable
+	Dimensions: Width	Just Right
+	Dimensions: Thickness	Just Right
+	Comfort: 1Hand Port	Somewhat Comf.
+	Comfort: 2Hand Port.	Somewhat Comf.
-	Comfort: 1Hand Land.	Somewhat Uncomf.
+	Comfort: 2Hand Land.	Comfortable
-	Front Pocket	Too Big
-	Back Pocket	Too Big
-	Belt Clip	Too Big
-	Shirt Pocket	Too Big

Most Preferred

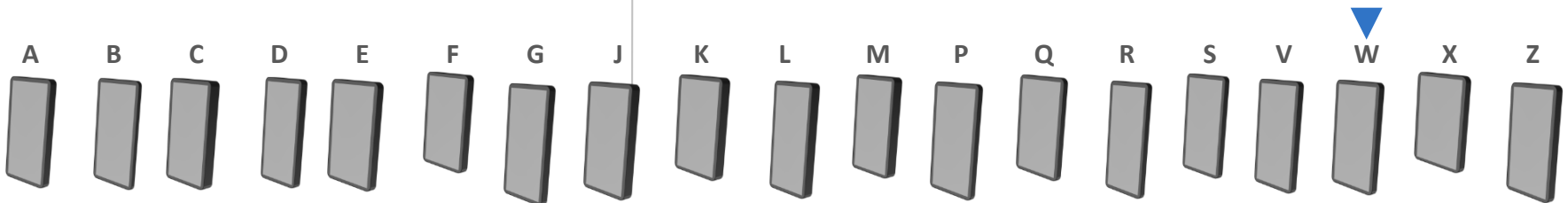
People liked the screen ratio of length to width and assumed the slight thickness could accommodate other hardware features (sliders, memory cards, etc.).

- "It's a nice big phone without being too big"
- "The ratio was what I liked; it had the right feel in all the dimensions, it felt good to me in my hands"
- "It felt like it could be a slider phone and it would be very comfortable using it in landscape"
- "Good size, weight, screen, the thickness should fit an external memory card like a Micro SD card"
- "Maybe okay for web & social media because of the larger screen"

Least Preferred

The screen is so big in width and height that the phone feels too big overall.

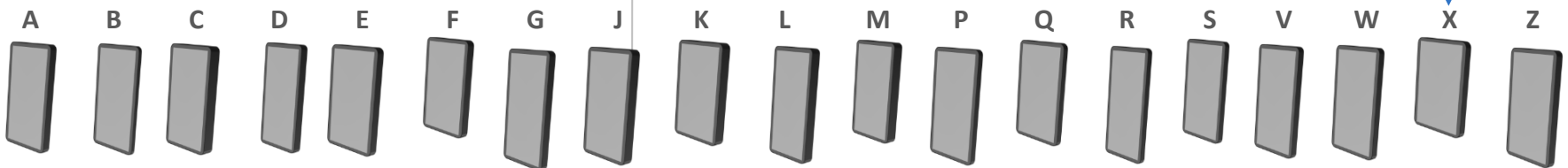
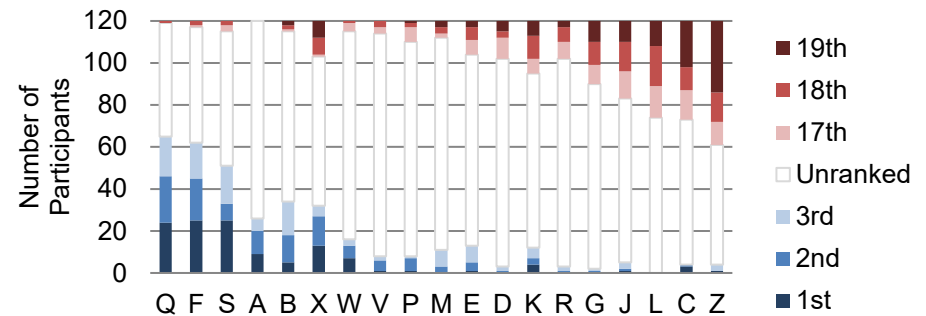
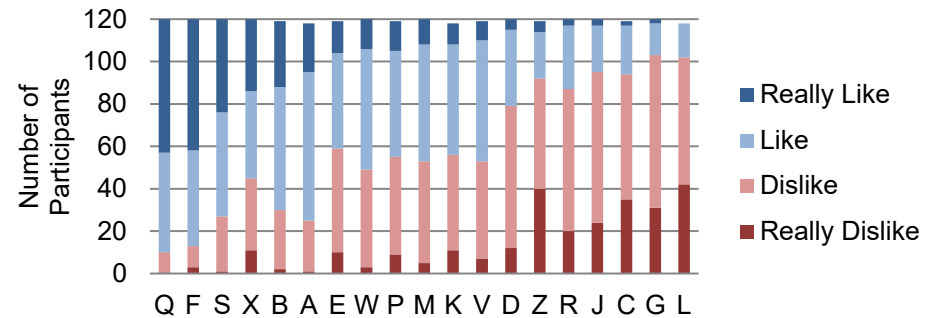
- "I liked the thickness, it was OK, but I felt it was too wide & too tall; big screen is too hard to hold"
- "Too wide, thin & tall; I think with a phone that big it needs to be thicker"



Device X: 4.6" Screen

+ Overall	Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Big But Acceptable
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Comfortable
+/- Comfort: 1Hand Land.	Neither
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
X	105	73	8	117mm (4.6")	109



Device X: 4.6" Screen – Focus Group

+ Overall	Like
+ Dimensions: Length	Just Right
+ Dimensions: Width	Big But Acceptable
+ Dimensions: Thickness	Just Right
+ Comfort: 1Hand Port	Somewhat Comf.
+ Comfort: 2Hand Port.	Comfortable
+/- Comfort: 1Hand Land.	Neither
+ Comfort: 2Hand Land.	Comfortable
+ Front Pocket	Big But Acceptable
+ Back Pocket	Big But Acceptable
- Belt Clip	Too Big
- Shirt Pocket	Too Big

Most Preferred

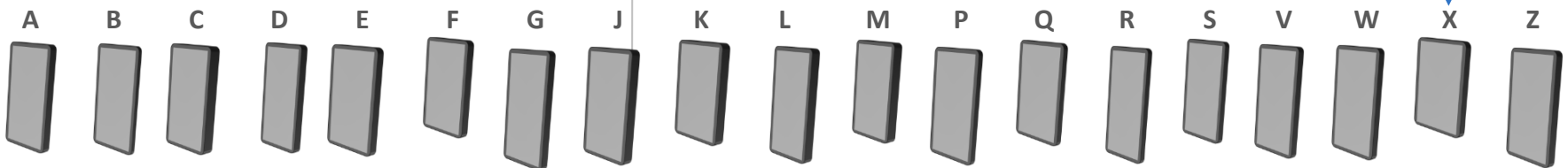
This phone has a wide screen which people liked for more content viewing with websites. Some people were drawn to the more square ratio and compactness.

- "Like the more squared off shape & a tablet shape. Good size/ratio for showing pictures"
- "I liked how you could cup it in your hand; Can reach across with thumb; Small but good size for viewing web pages. I like horizontal & vertical"
- "It has the biggest screen size for how compact it is"
- "I like the ratio of the width to height, they're closer which makes it more usable"
- "I liked the size because my hands are small, I felt like I could reach the screen with 1 hand at a point"

Least Preferred

The proportion is strange because it is so wide but short and thin as well. Hold this phone feels very awkward.

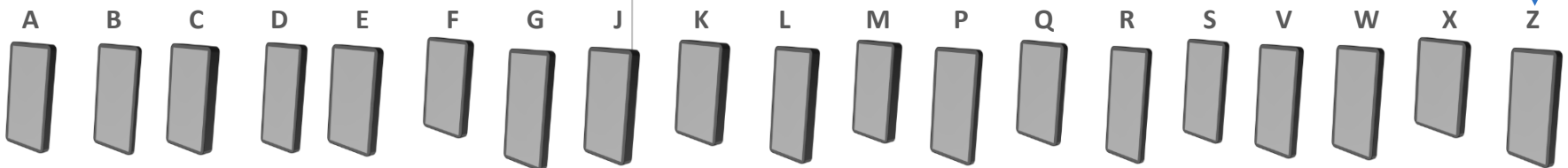
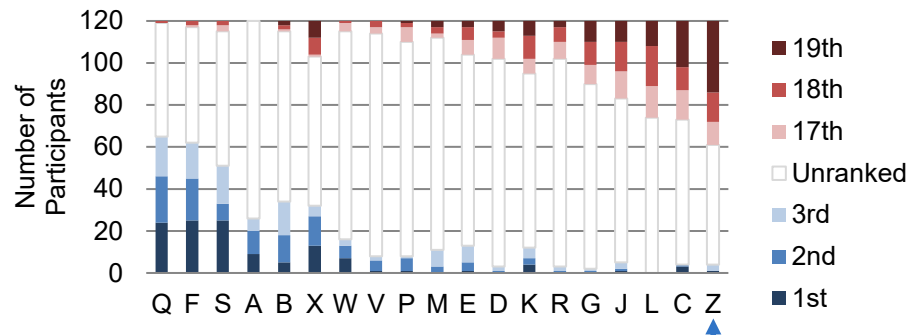
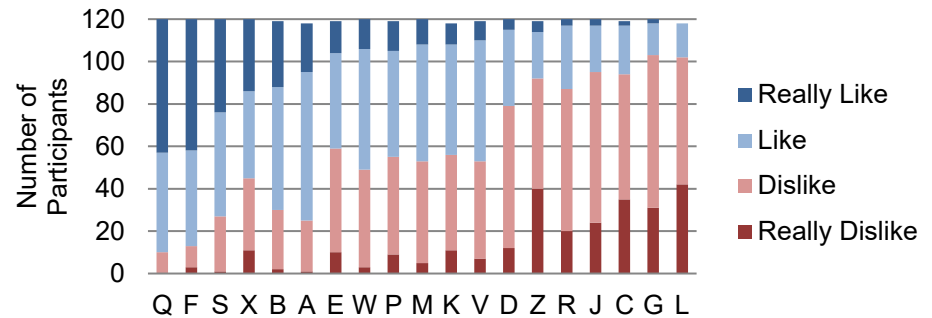
- "It was too wide, HEAVY & short for such a small phone; Can't cup your fingers around it or hold it; Keyboard would take up so much space- there wouldn't be room for the picture"
- "It was wide, short & squat; It's like a wallet, it doesn't look right"
- "You can't hold it in 1 hand; too thin"
- "I don't like the aspect ratio too square. I don't know what you would do with a size like that, you can't watch videos you can't do much of anything"



Device Z: 5.3" Screen

- Overall	Really Dislike
- Dimensions: Length	Too Tall
+ Dimensions: Width	Big But Acceptable
+ Dimensions: Thickness	Just Right
+/- Comfort: 1Hand Port	Neither
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Uncomfortable
+ Comfort: 2Hand Land.	Somewhat Comf.
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Far Too Big

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal	Weight (g)
Z	129	69	9	135mm (5.3")	133



Device Z: 5.3" Screen – Focus Group

- Overall	Really Dislike
- Dimensions: Length	Too Tall
+ Dimensions: Width	Big But Acceptable
+ Dimensions: Thickness	Just Right
+/- Comfort: 1Hand Port	Neither
+ Comfort: 2Hand Port.	Somewhat Comf.
- Comfort: 1Hand Land.	Uncomfortable
+ Comfort: 2Hand Land.	Somewhat Comf.
- Front Pocket	Too Big
- Back Pocket	Too Big
- Belt Clip	Too Big
- Shirt Pocket	Far Too Big

Most Preferred

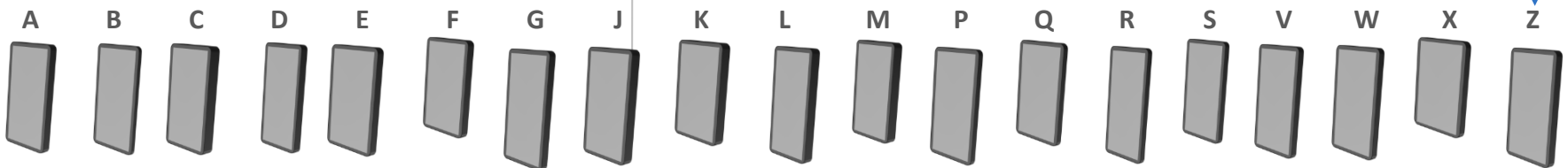
Only 2 people commented that they like big phones with big screens for better viewing.

- "I was about to get the Epic and this is similar. Big and it does a lot, but it fits my fingers"
- "The height & width were right, if it's too narrow it then websites get cut off"

Least Preferred

This phone is too big in every dimension making it too heavy and bulky for hands and pockets.

- "It hurt holding it, it was so heavy"
- "It would be a 2-handed job all the time. It's really heavy; I might drop it a lot"
- "Make it bigger and make it an iPad"
- "You couldn't slide it in any pocket in your pants or shirt; can't get your hands around it"





Appendix C: Additional Data

The following slides provide brief overviews of additional data collected during the study. These topics are outside the primary focus of the project and were not analyzed in depth.

Topics include

- Pocketability/Portability
- Comfort in the Hand
- Design Tradeoffs
- Effect of web page graphics/text (overlays)

Portability and Pocketability Summary

Chart Summary

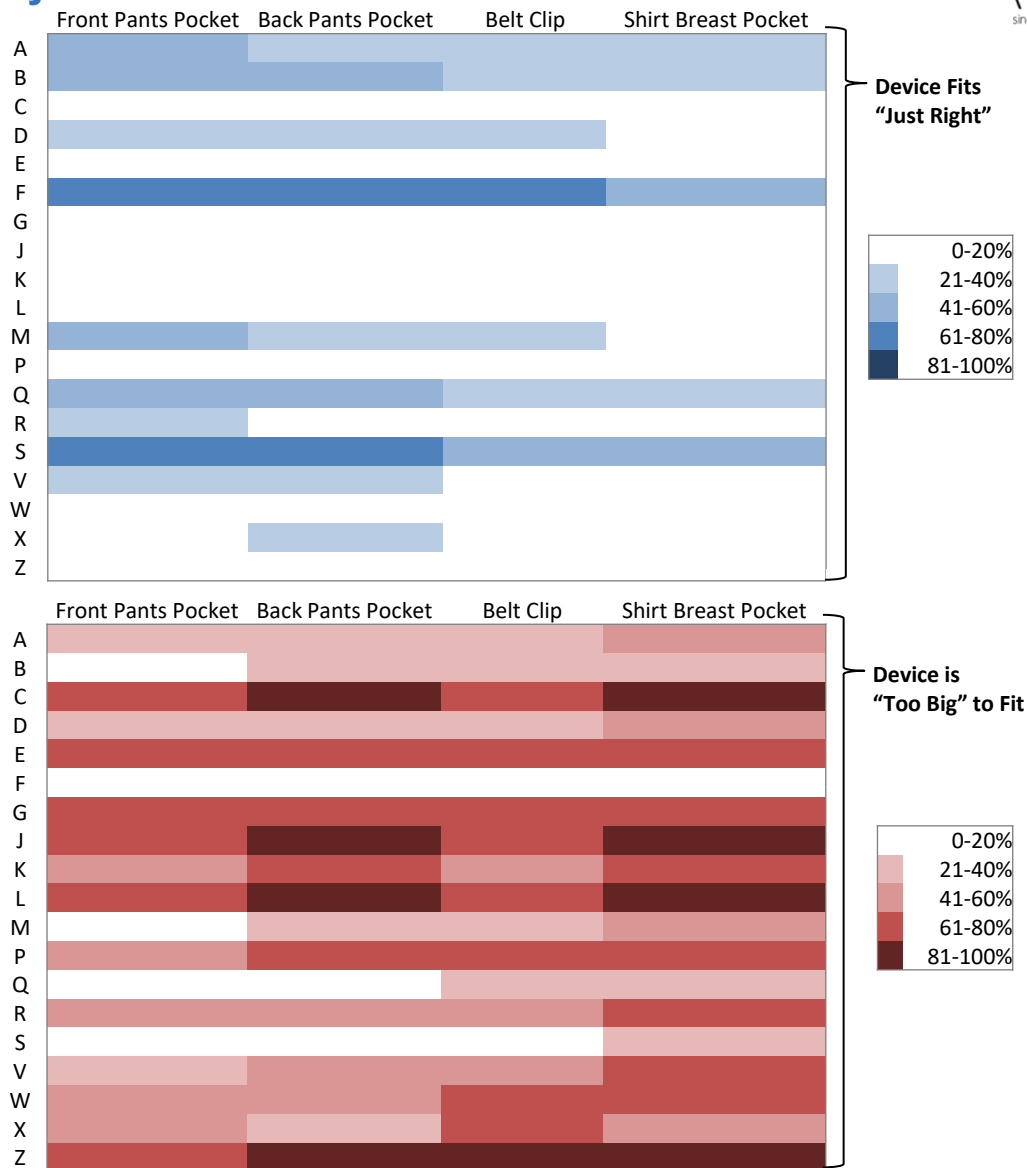
- Devices A, B, F, Q and S were the only 4 devices rated as fitting "Just Right" by at least 41-60% of people in all 4 locations
- 61-80% of people felt that devices F and S fit "Just Right" in the Front and Back Pants Pockets
- 61-80% of people felt that devices C, E, G, J, L, and Z were too big to fit in all 4 locations
- Device Z was the only device to be considered too big for all locations except the front pants pocket by more than 81% of people

Discussion

- People were often concerned about fit when phone models were:
 - Too thick/wide to fit into pockets, purses, or belt clips comfortably and without bulging. This was especially important for small women or women with tight pants.
 - Too long to fit completely into pockets while sitting or standing
- People were most interested in phones that fit into their pocket fully without bulging. Once pocketability was satisfied, they tended to express a desire to maximize screen size.
- Big/tall people have such large pockets that pocketability is not an issue

Recommendations

- Thickness – Ensure phones are thick enough to maintain positive impressions of durability but are thin enough to easily and comfortably fit in pockets
- Height - Ensure phones are not so tall that they stick out of pockets while sitting. This is difficult to satisfy completely for short people or people who wear clothing with shallow pockets.
- Consider rounding and/or tapering edges to help devices slide in/out of pockets; Some people may take a slightly thicker or larger device if it is rounded/tapered.

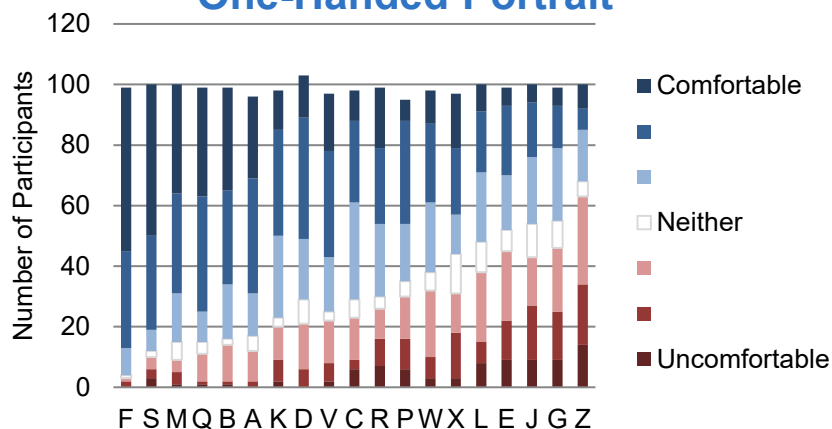


*"Too Big" to fit includes combined extreme frequency ratings (coded 3 + 4) for each location.

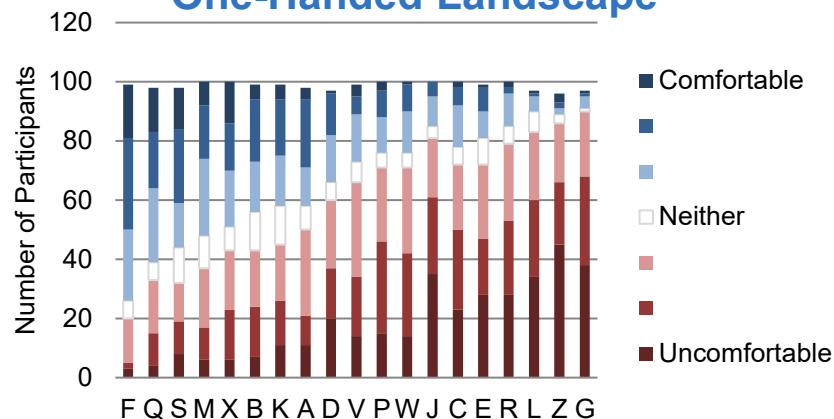
Comfort Summary: One-Handed Portrait

Based on your overall impressions of each device, rate your overall comfort from Very Uncomfortable to Very Comfortable for each orientation

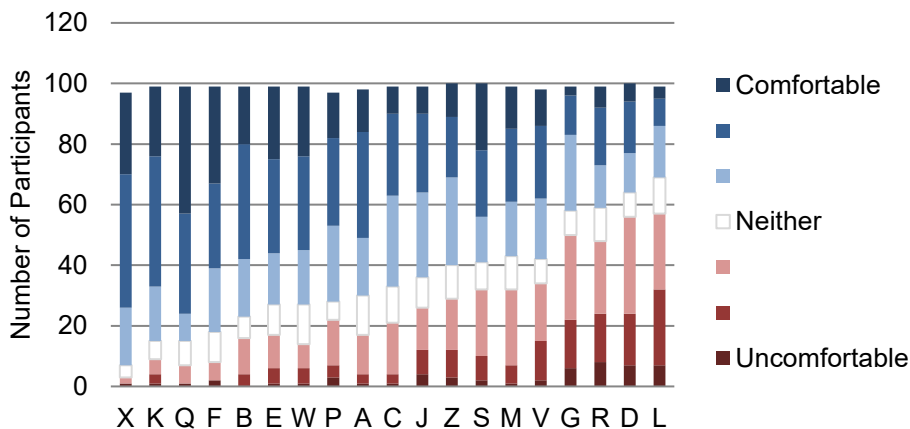
One-Handed Portrait



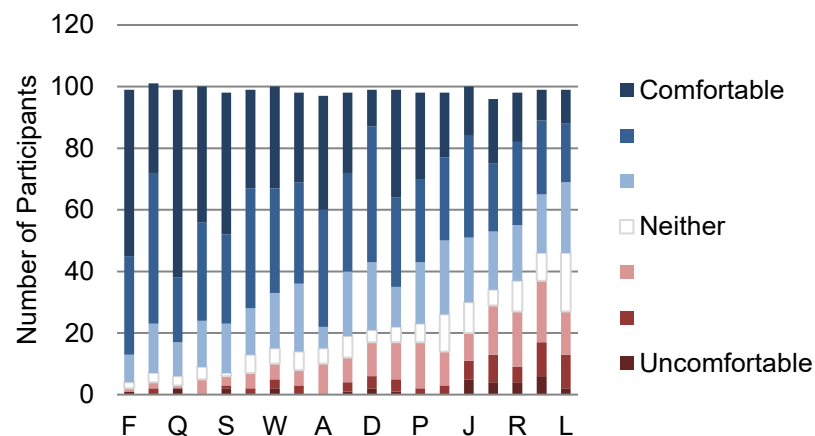
One-Handed Landscape



Two-Handed Portrait



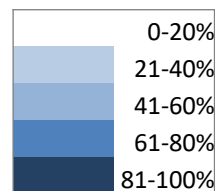
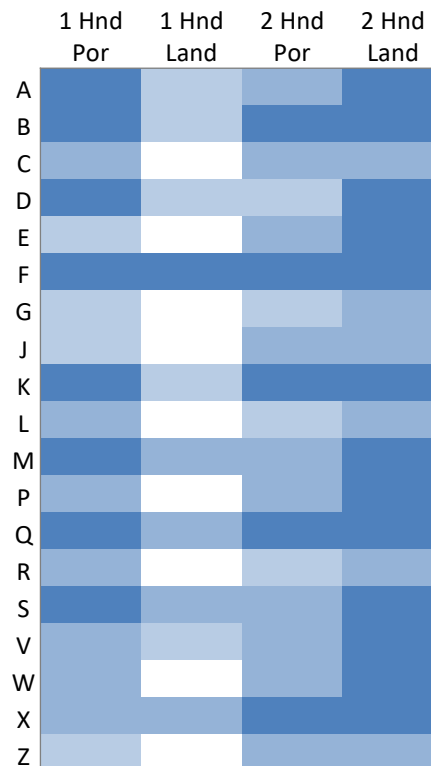
Two-Handed Landscape



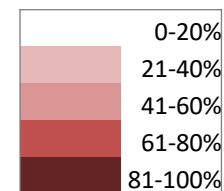
Comfort Summary: Highest and Lowest, All Orientations

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal
A	119	61	9	124mm (4.9")
J	127	66	12	132mm (5.2")
K	112	66	12	119mm (4.7")
L	127	57	12	129mm (5.1")
M	112	57	12	114mm (4.5")
P	127	66	6	132mm (5.2")
Q	112	66	6	119mm (4.7")
R	127	57	6	130mm (5.1")
S	112	57	6	114mm (4.5")
B	119	61	4	124mm (4.9")
C	119	61	14	124mm (4.9")
D	119	54	9	122mm (4.8")
E	119	69	9	127mm (5.0")
F	107	61	9	112mm (4.4")
G	132	61	9	135mm (5.3")
V	123	60	8	127mm (5.0")
W	123	66	8	127mm (5.0")
Z	129	69	9	135mm (5.3")
X	105	73	8	117mm (4.6")

Rated as "Somewhat Comfortable" or higher:



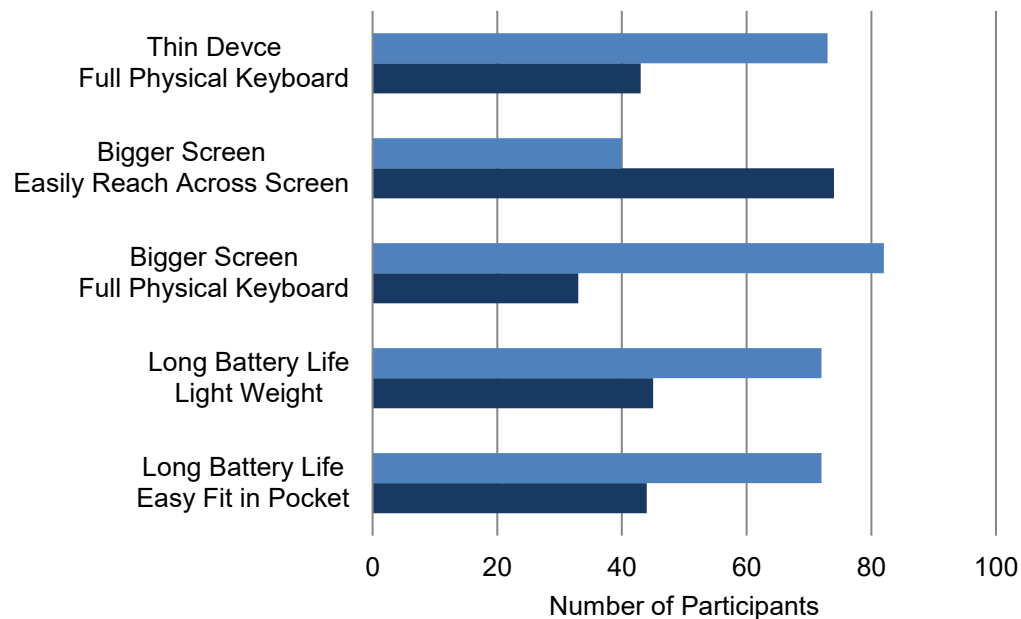
Rated as "Somewhat Uncomfortable" or lower:



Trade-offs: Battery, Pocket Fit, Screen Size, Weight, Thickness, Reach, Keyboard

Chart Summary

- People preferred a bigger screen and thin device over having a full physical keyboard (results may be impacted by touchscreen focused recruitment).
- Ability to easily reach across the screen was preferred over having a bigger screen.
- Long battery life was preferred over having a light weight phone and a phone that can easily fit into a pocket.



*Results significant at alpha = 0.05 (Chi-Square)

Screen Overlays – Effect on top rankings

Letter	Top 3 - No Overlay	Top 3 - With Overlay	Added to Top 3	Better Rank	Same Rank	Worse Rank	Dropped from Top 3
Q	65	68	4	16	33	15	1
F	62	58		18	33	7	4
S	51	44		12	29	3	7
B	34	32		4	20	8	2
X	32	36	5	1	18	12	1
A	26	25	3	4	14	4	4
W	16	17	2	3	7	5	1
E	13	15	3		7	5	1
K	12	14	3		8	3	1
M	11	8			8		3
P	8	10	2	2	3	3	
V	8	8	1	2	4	1	1
J	5	4			4		1
C	4	3		1	2		1
Z	4	6	3		3		1
D	3	3	1	1	1		1
R	3	2			2		1
G	2	2	1		1		1
L		1	1				

Changing Rank Order

People changed their rank order in order to have better text readability and image clarity. Web page viewing became the top priority and most people were willing to go slightly larger for more viewability while still maintaining decent size for hand.

- “Previously it was just on feel, now it is more visual experience.”
- “Text and images much larger and readable. Would compromise on size and weight for larger screen”
- “Text and images are a good readable size”

Most Improved (Better+New) Rank Order

A phone improved in rank order if it had a bigger screen that was easier to read/see images and if it had a wider view where they could see more content at once.

F – People felt this phone had the perfect length and width for their hands and pockets. It was compact and small but still useable.

- “Nice and small but still large enough to be useable. Fit in pockets and hand”

S – This phone was thin and light weight. The length and width were substantial and big enough for viewing with a good fit in the hand as well.

- “Slim and small but not tiny, really easy to hold”

Q – This phone felt thin and lightweight. It was easy to hold and easy to use in both orientations.

- “Out of all the phones I believe this device is the most well rounded and comfortable to use”

= better

= worse

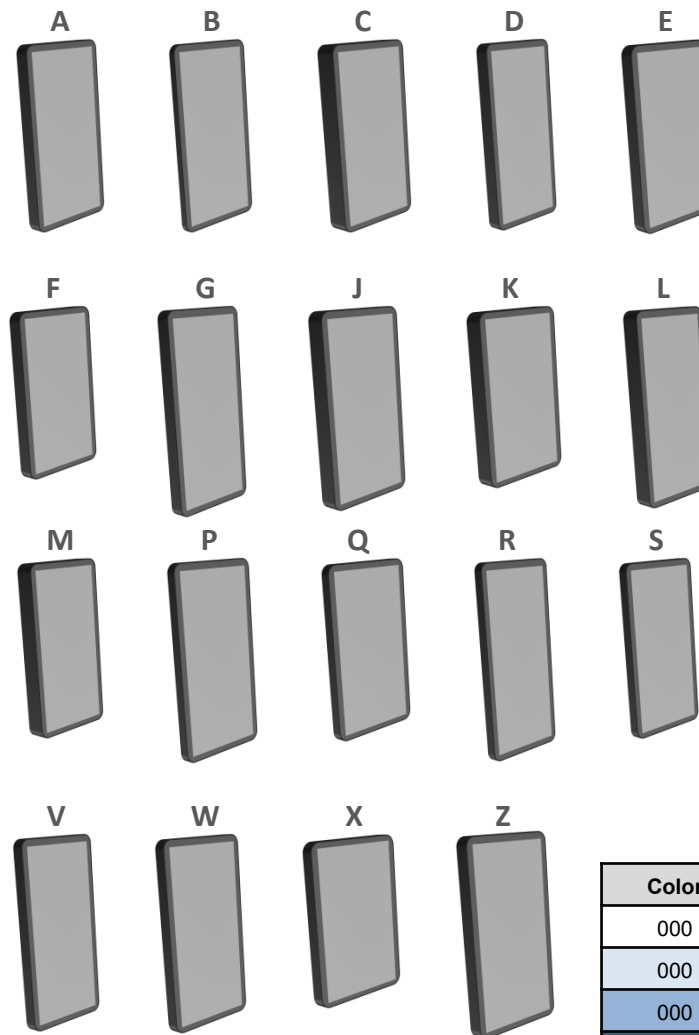


Appendix D: Device Dimensions

3D renderings of each device, to scale

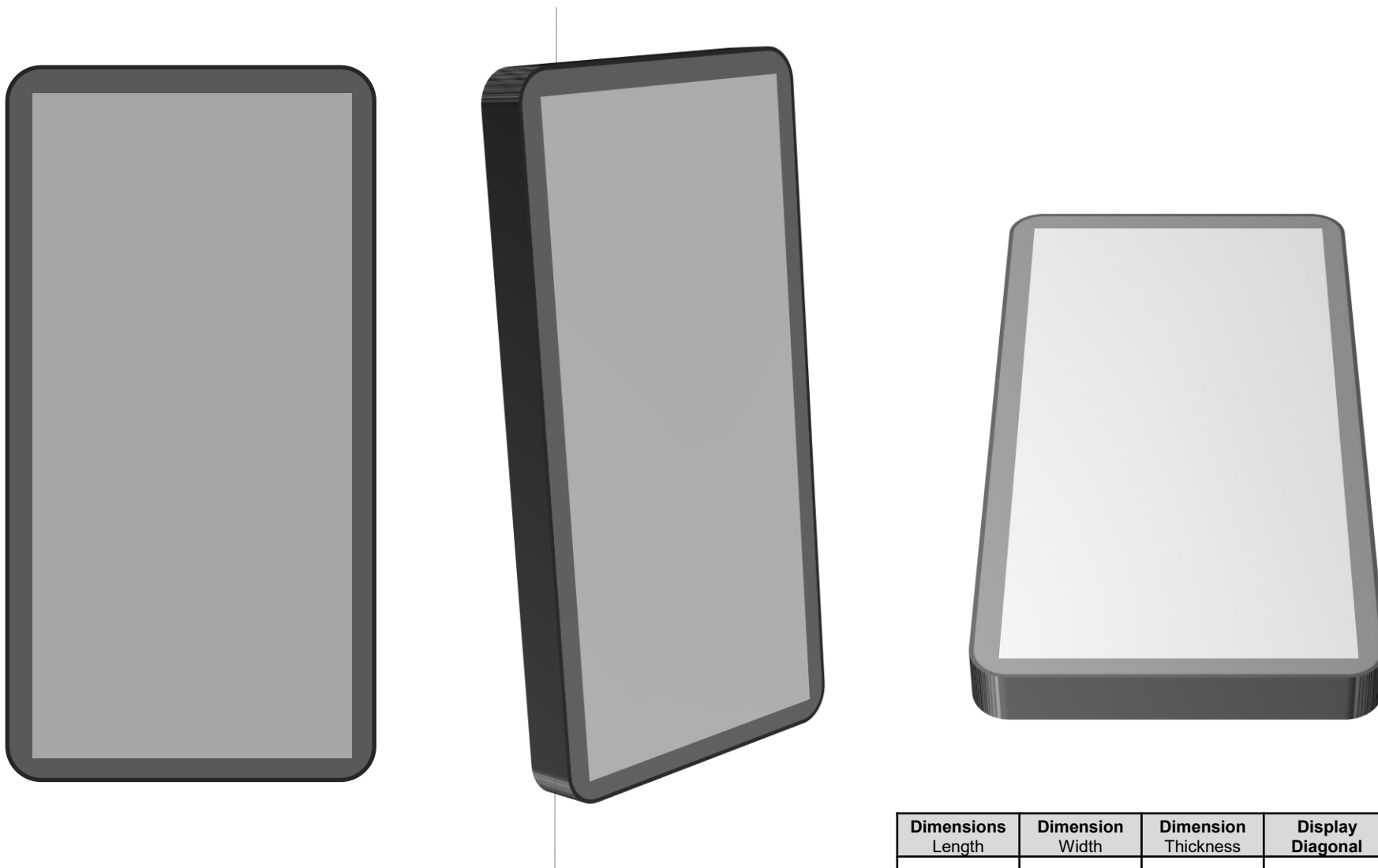
Summary of Devices

Letter	Dimensions Length (mm)	Dimension Width (mm)	Dimension Thickness (mm)	Display Diagonal
A	119	61	9	124mm (4.9")
J	127	66	12	132mm (5.2")
K	112	66	12	119mm (4.7")
L	127	57	12	129mm (5.1")
M	112	57	12	114mm (4.5")
P	127	66	6	132mm (5.2")
Q	112	66	6	119mm (4.7")
R	127	57	6	130mm (5.1")
S	112	57	6	114mm (4.5")
B	119	61	4	124mm (4.9")
C	119	61	14	124mm (4.9")
D	119	54	9	122mm (4.8")
E	119	69	9	127mm (5.0")
F	107	61	9	112mm (4.4")
G	132	61	9	135mm (5.3")
V	123	60	8	127mm (5.0")
W	123	66	8	127mm (5.0")
Z	129	69	9	135mm (5.3")
X	105	73	8	117mm (4.6")



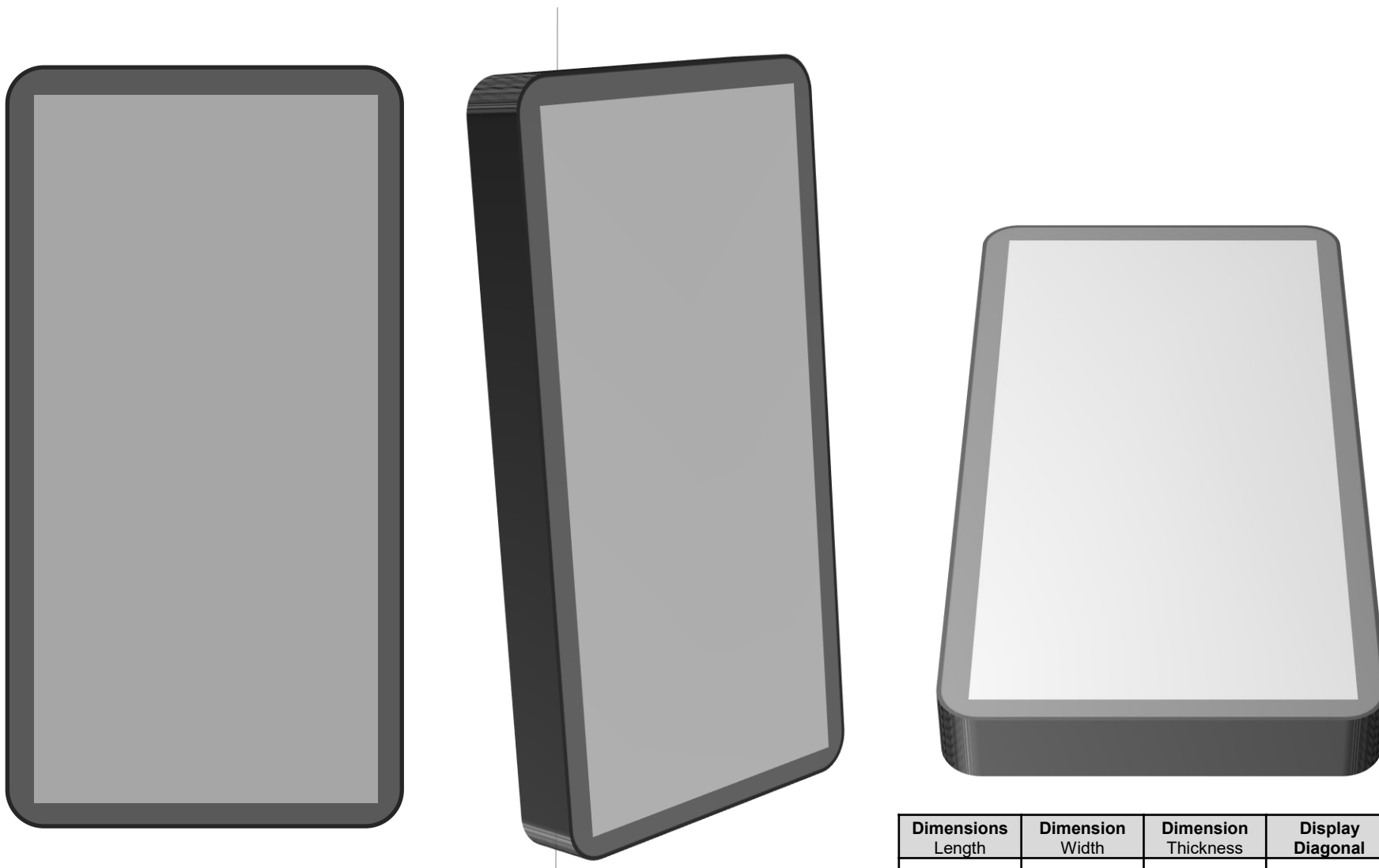
Color	Meaning
000	Minimum
000	
000	Average
000	
000	Maximum
000	Other

A



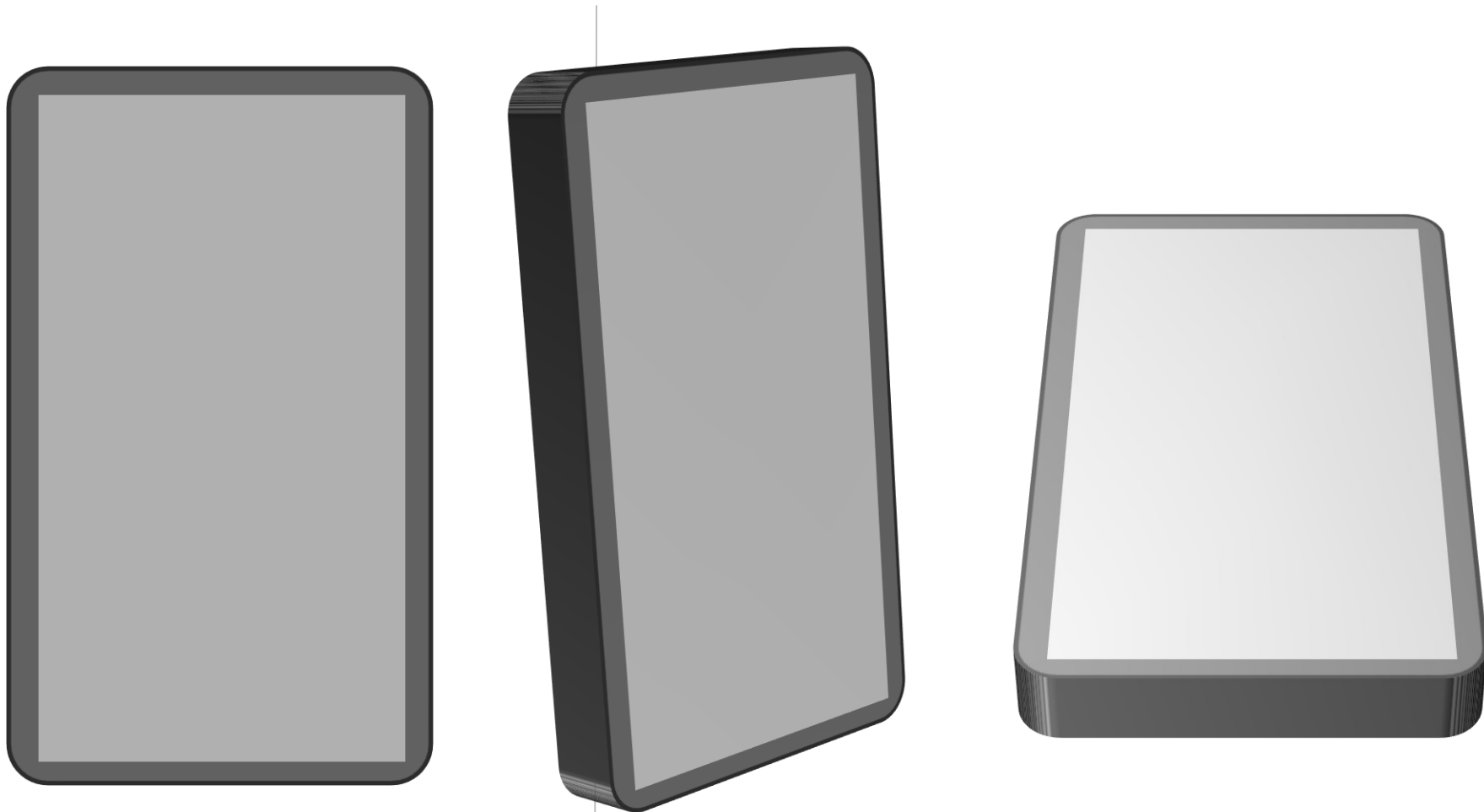
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
119mm (4.7")	61mm (2.42")	9mm (0.35")	124mm (4.9")

J



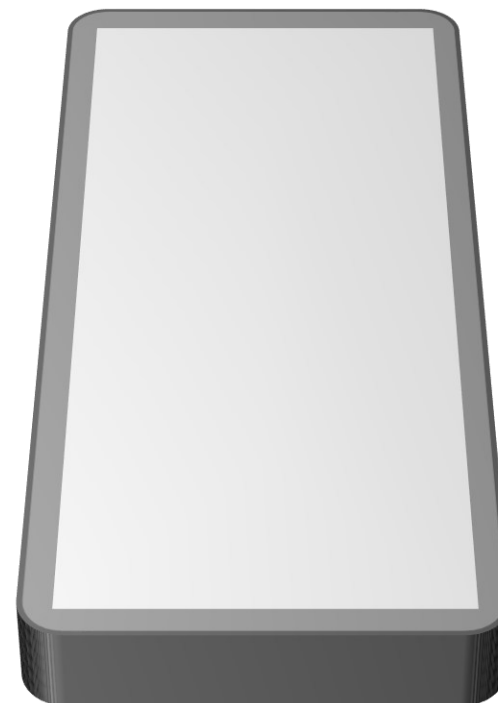
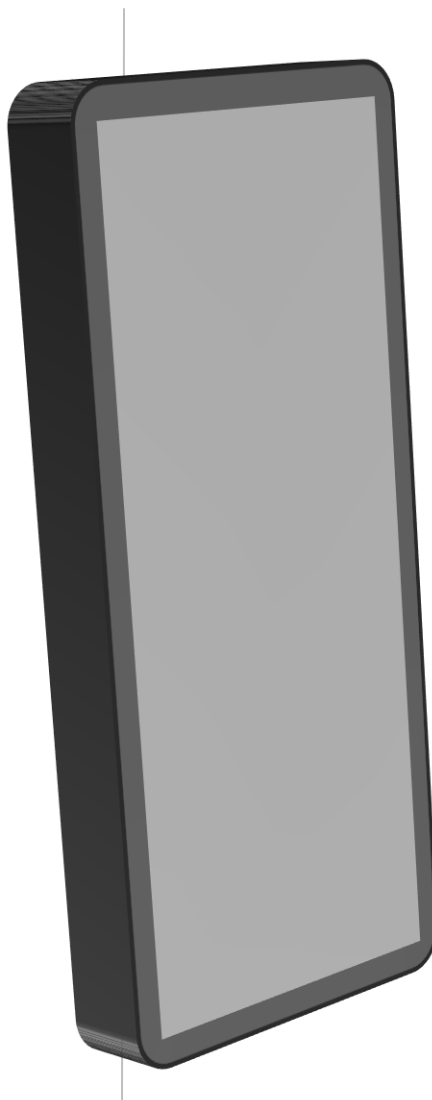
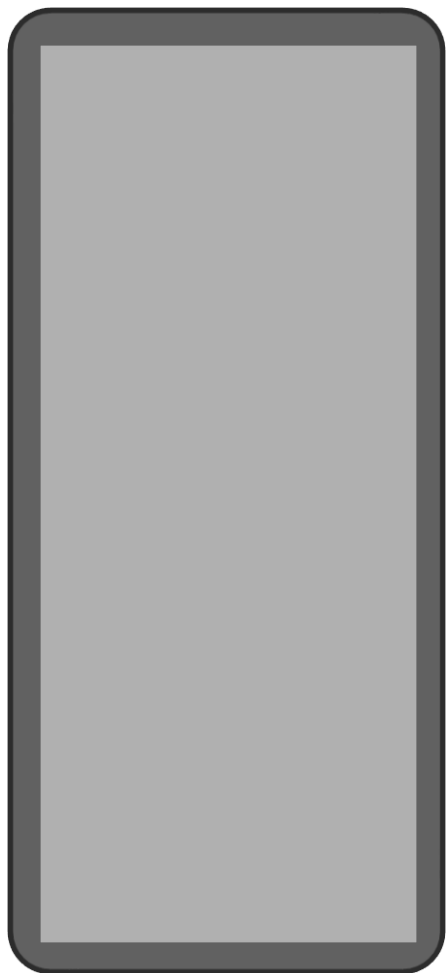
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
127mm (5")	66mm (2.6")	119mm (0.47")	132mm (5.2")

K



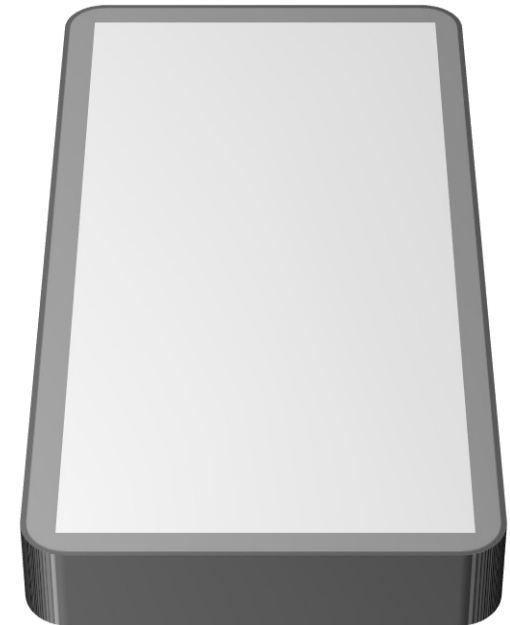
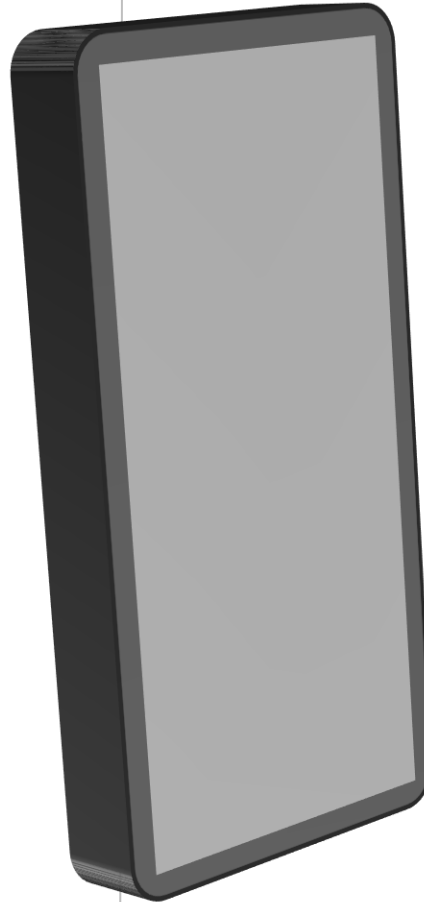
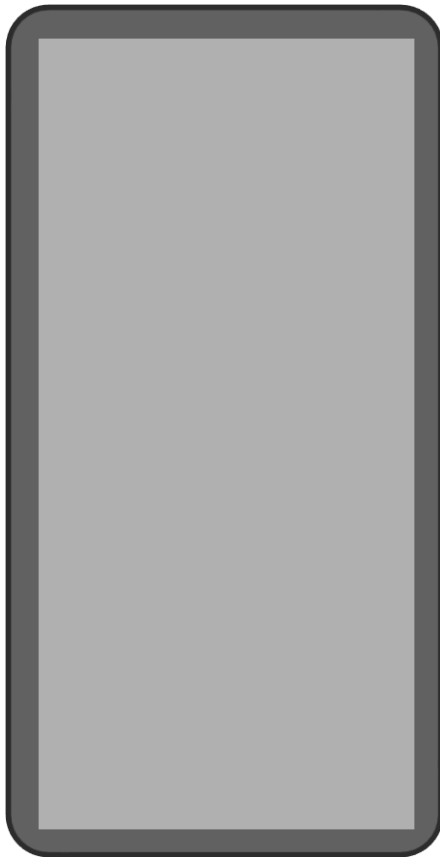
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
112mm (4.41")	66mm (2.6")	119mm (0.47")	119mm (4.7")

L



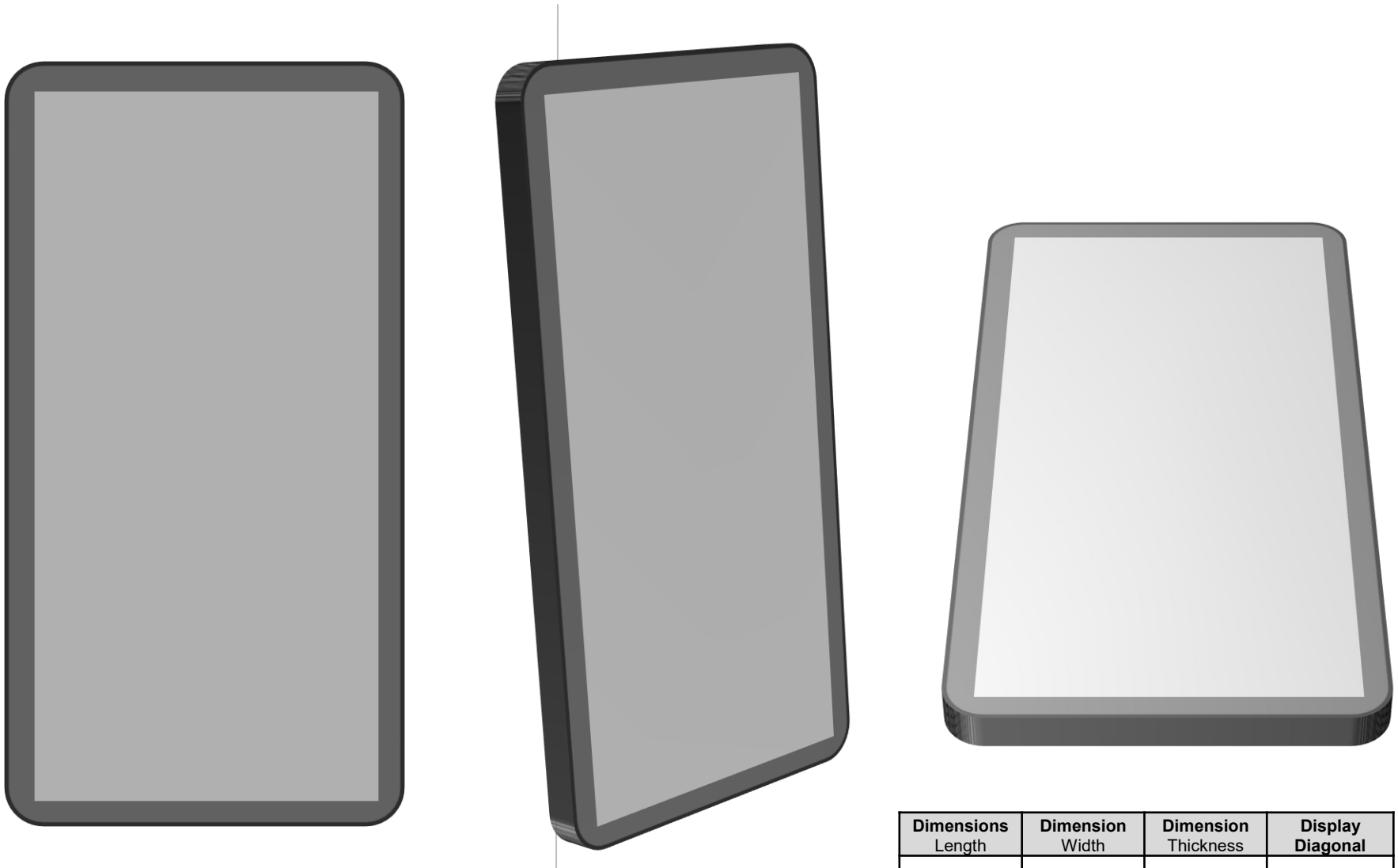
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
127mm (5")	57mm (2.24")	119mm (0.47")	129mm (5.1")

M



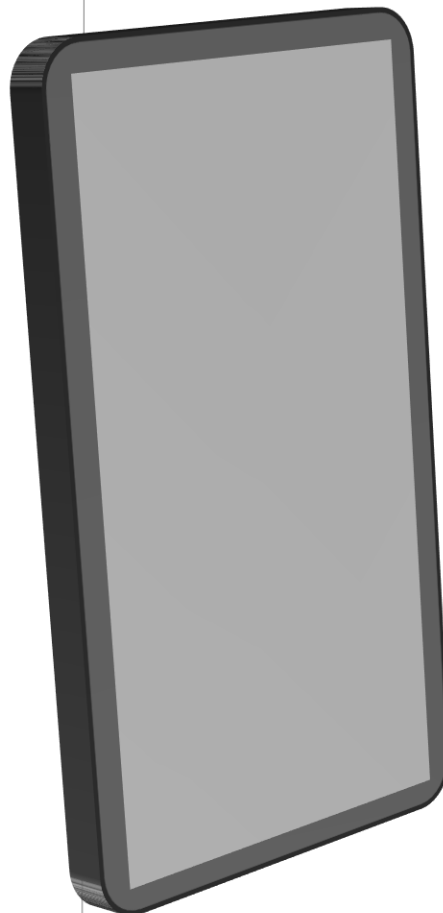
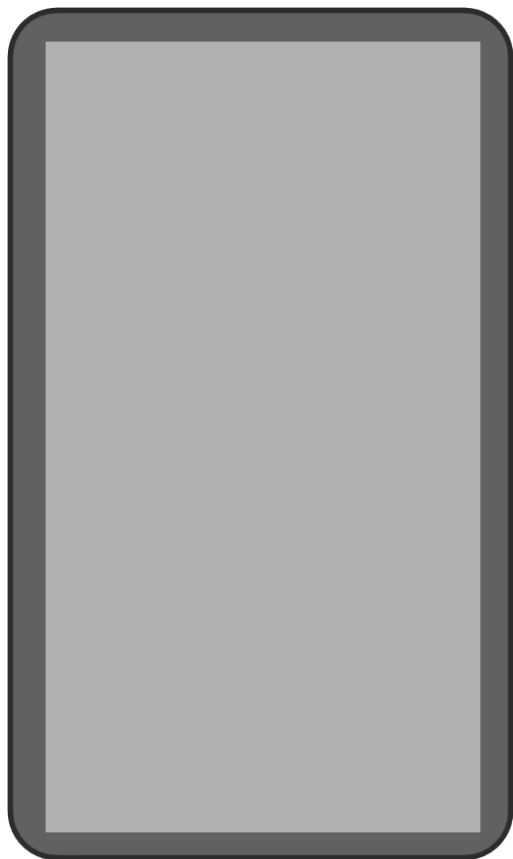
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
112mm (4.41")	57mm (2.24")	119mm (0.47")	114mm (4.5")

P



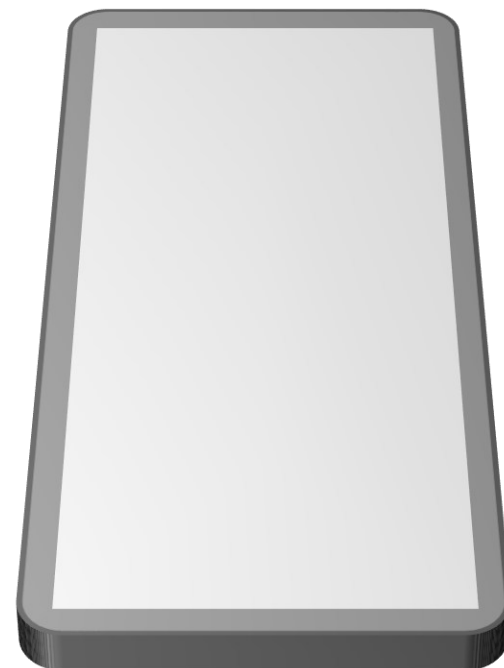
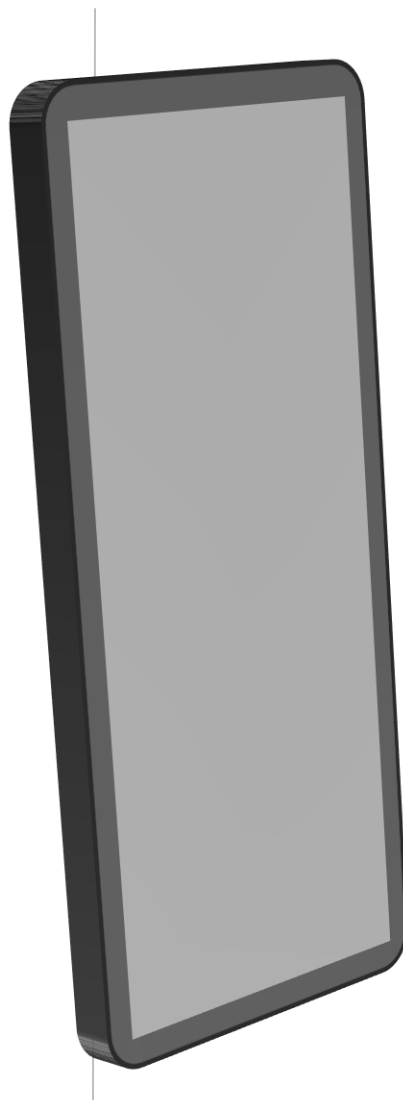
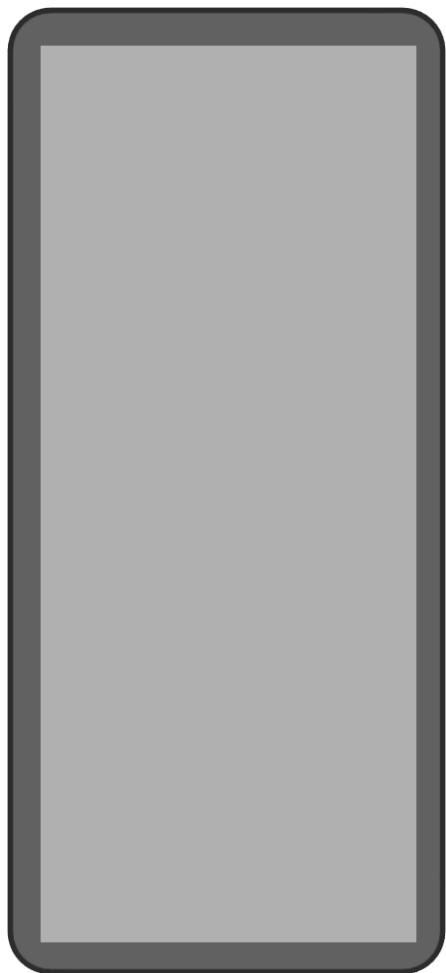
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
127mm (5")	66mm (2.6")	6mm (0.24")	132mm (5.2")

Q

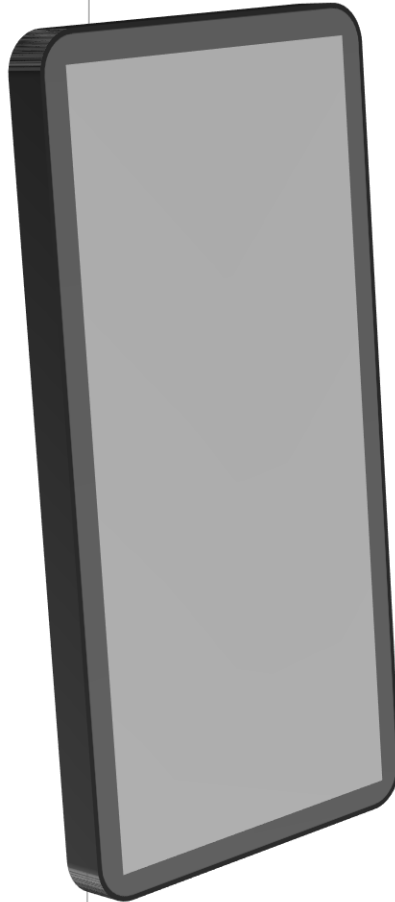
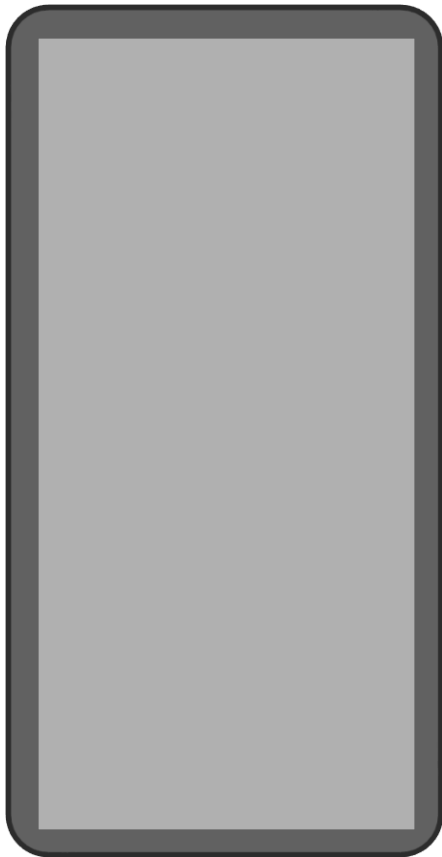


Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
112mm (4.41")	66mm (2.6")	6mm (0.24")	119mm (4.7")

R

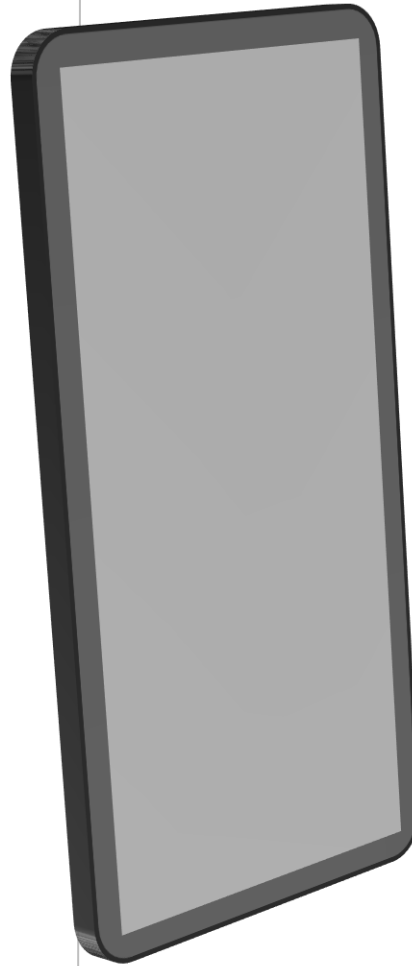
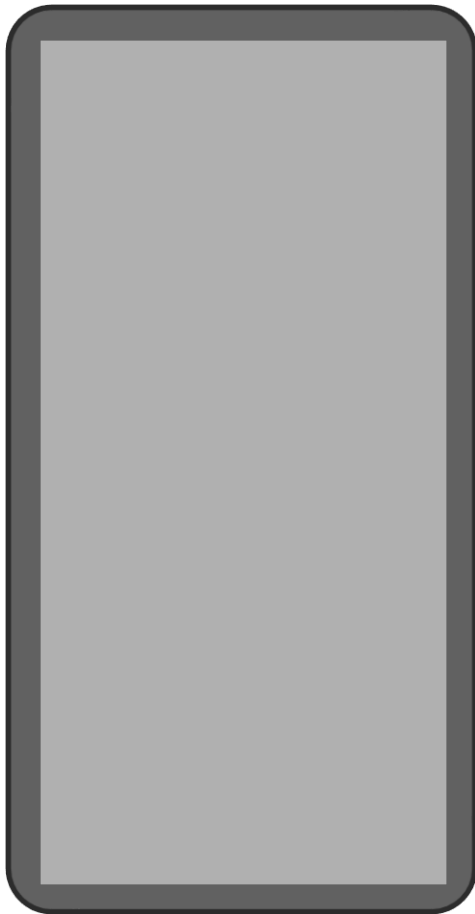


Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
127mm (5")	57mm (2.24")	6mm (0.24")	130mm (5.1")



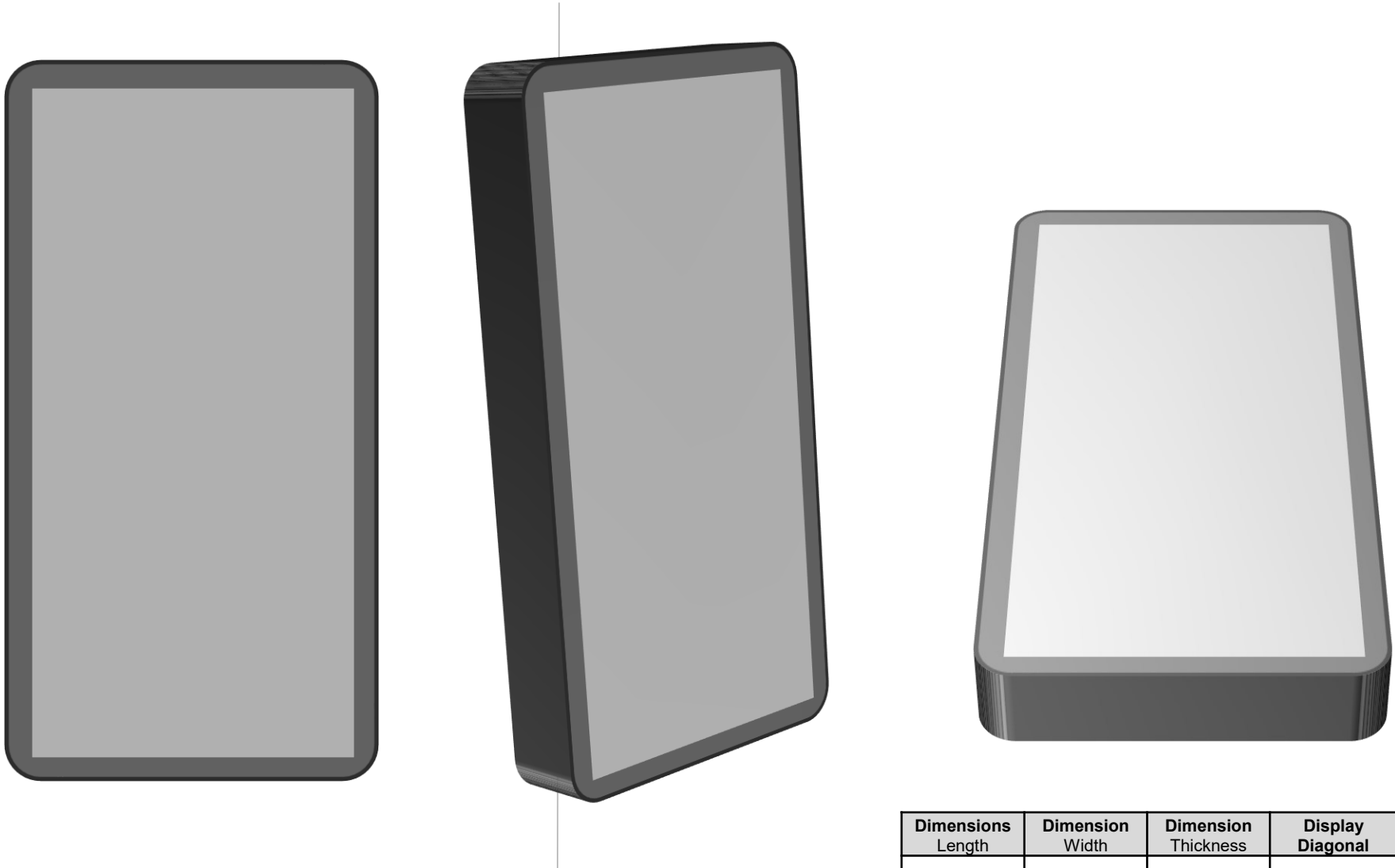
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
112mm (4.41")	57mm (2.24")	6mm (0.24")	114mm (4.5")

B



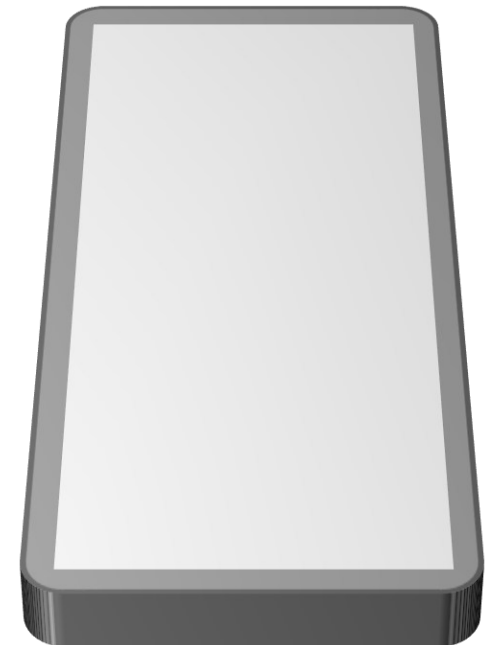
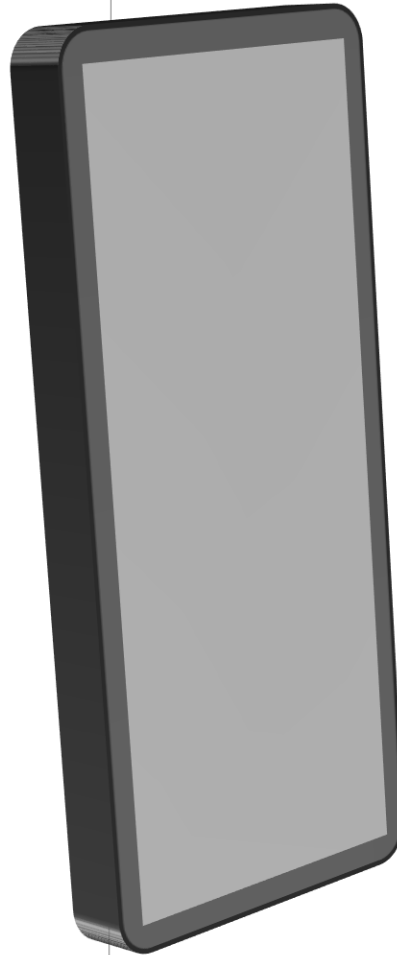
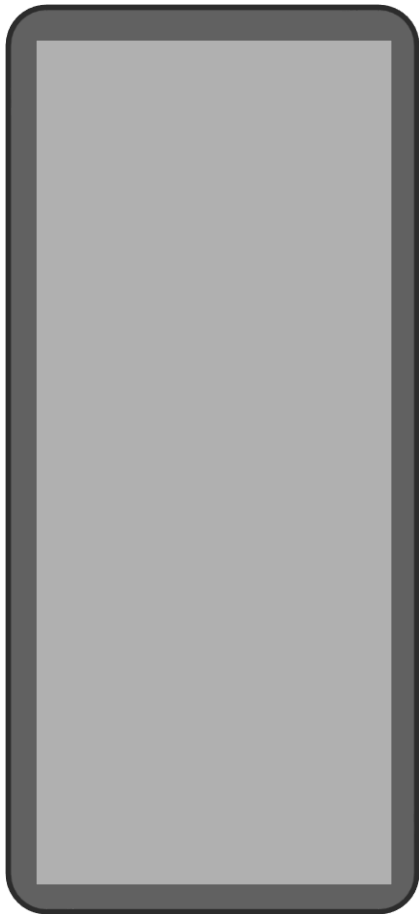
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
119mm (4.7")	61mm (2.42")	4mm (0.16")	124mm (4.9")

C



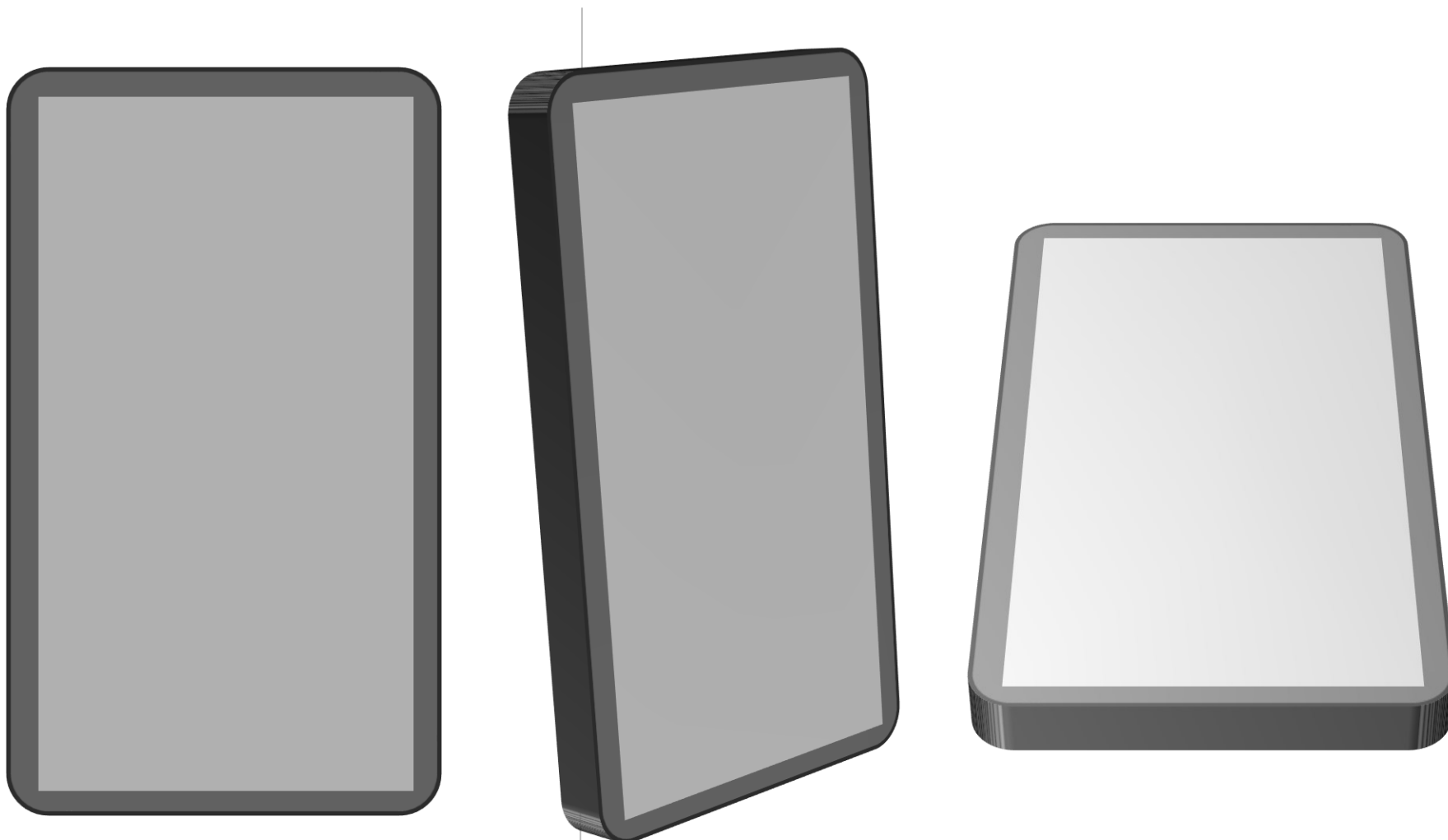
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
119mm (4.7")	61mm (2.42")	14mm (0.55")	124mm (4.9")

D



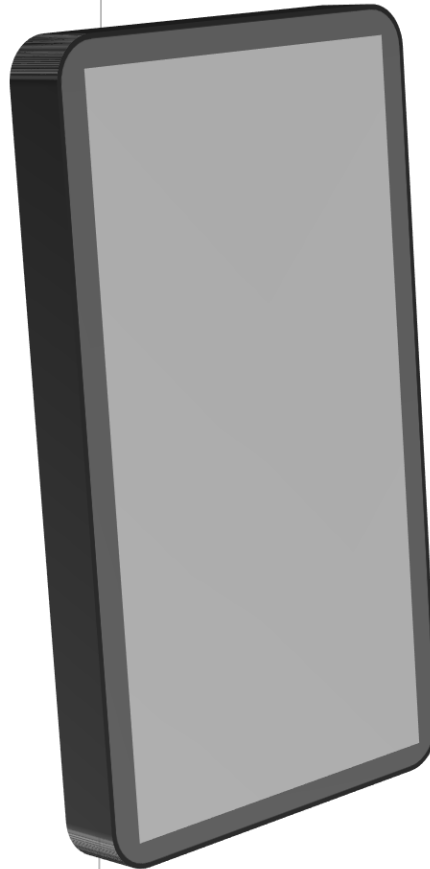
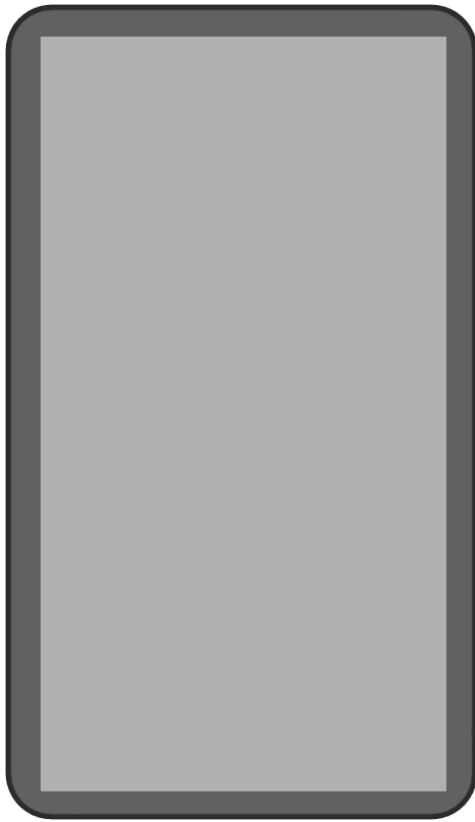
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
119mm (4.7")	54mm (2.12")	9mm (0.35")	122mm (4.8")

E



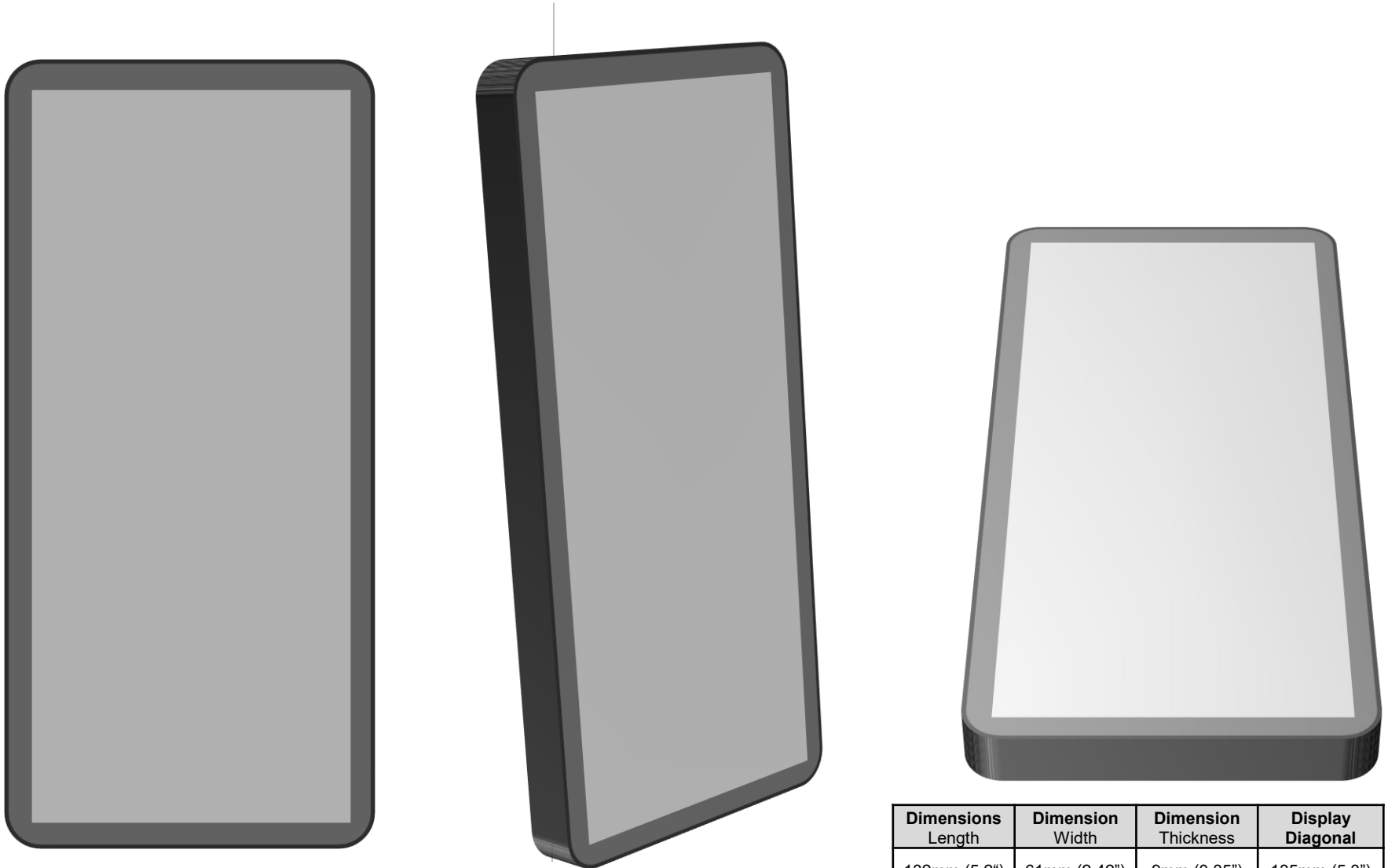
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
119mm (4.7")	69mm (2.72")	9mm (0.35")	127mm (5.0")

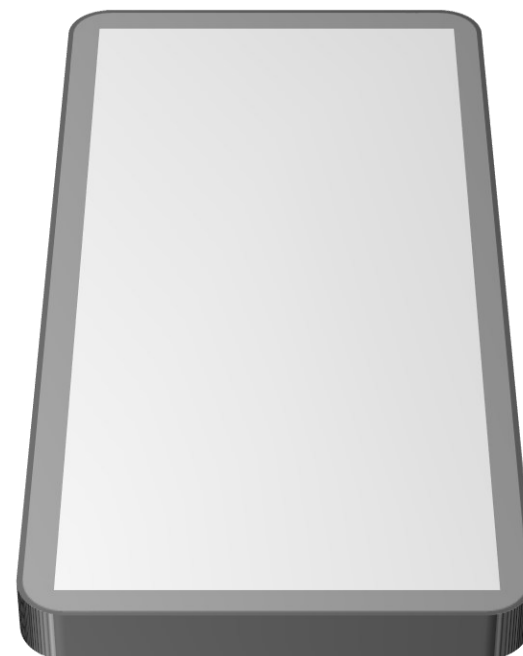
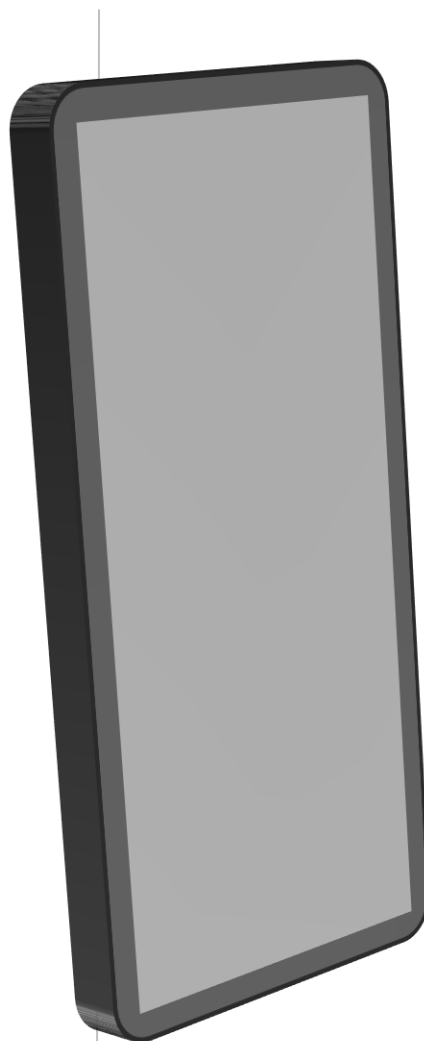
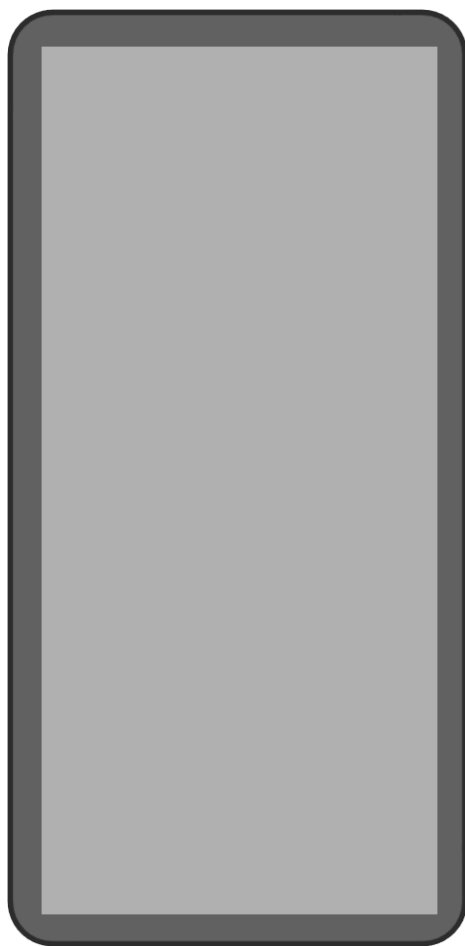
F



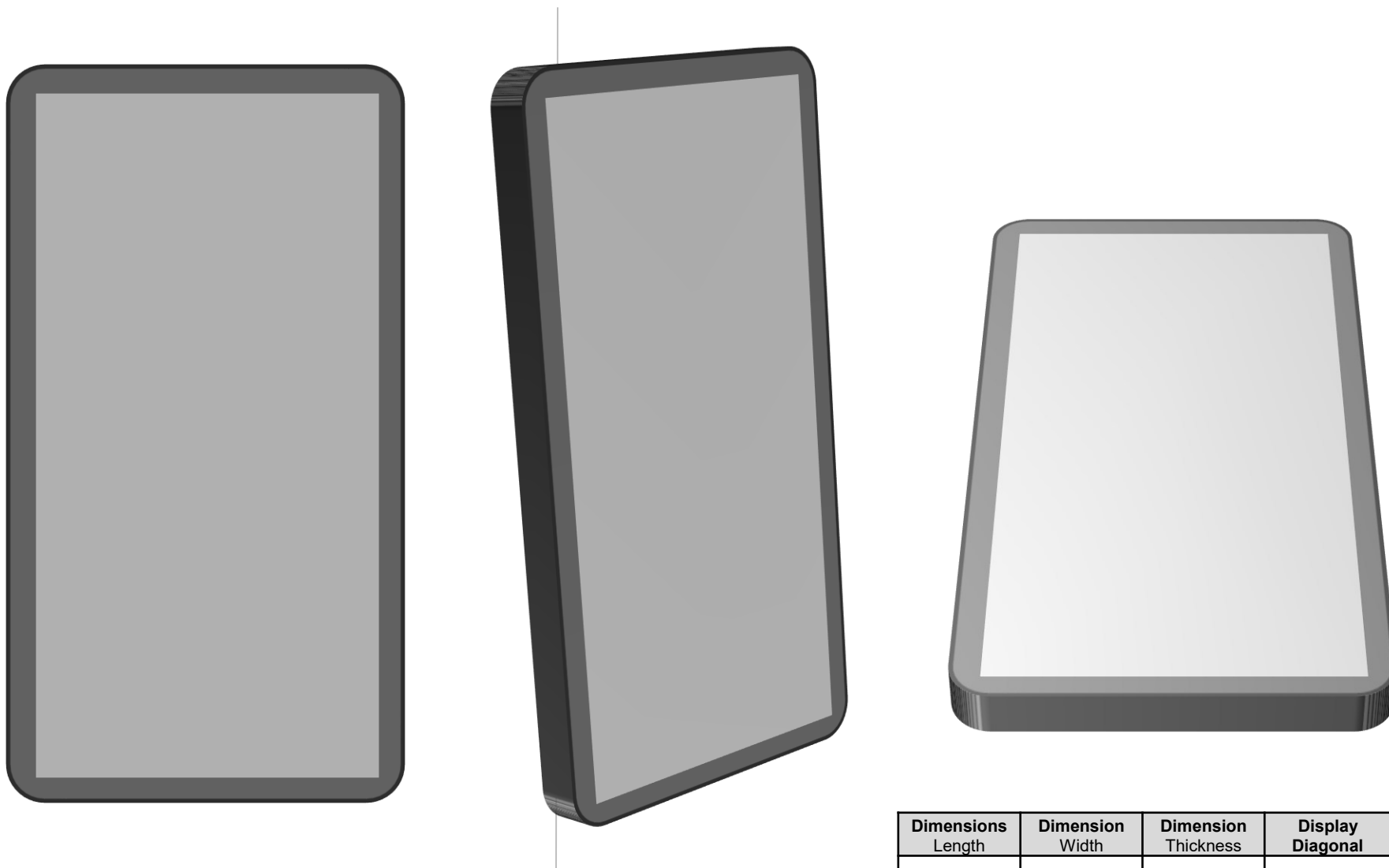
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
107mm (4.21")	61mm (2.42")	9mm (0.35")	112mm (4.4")

G



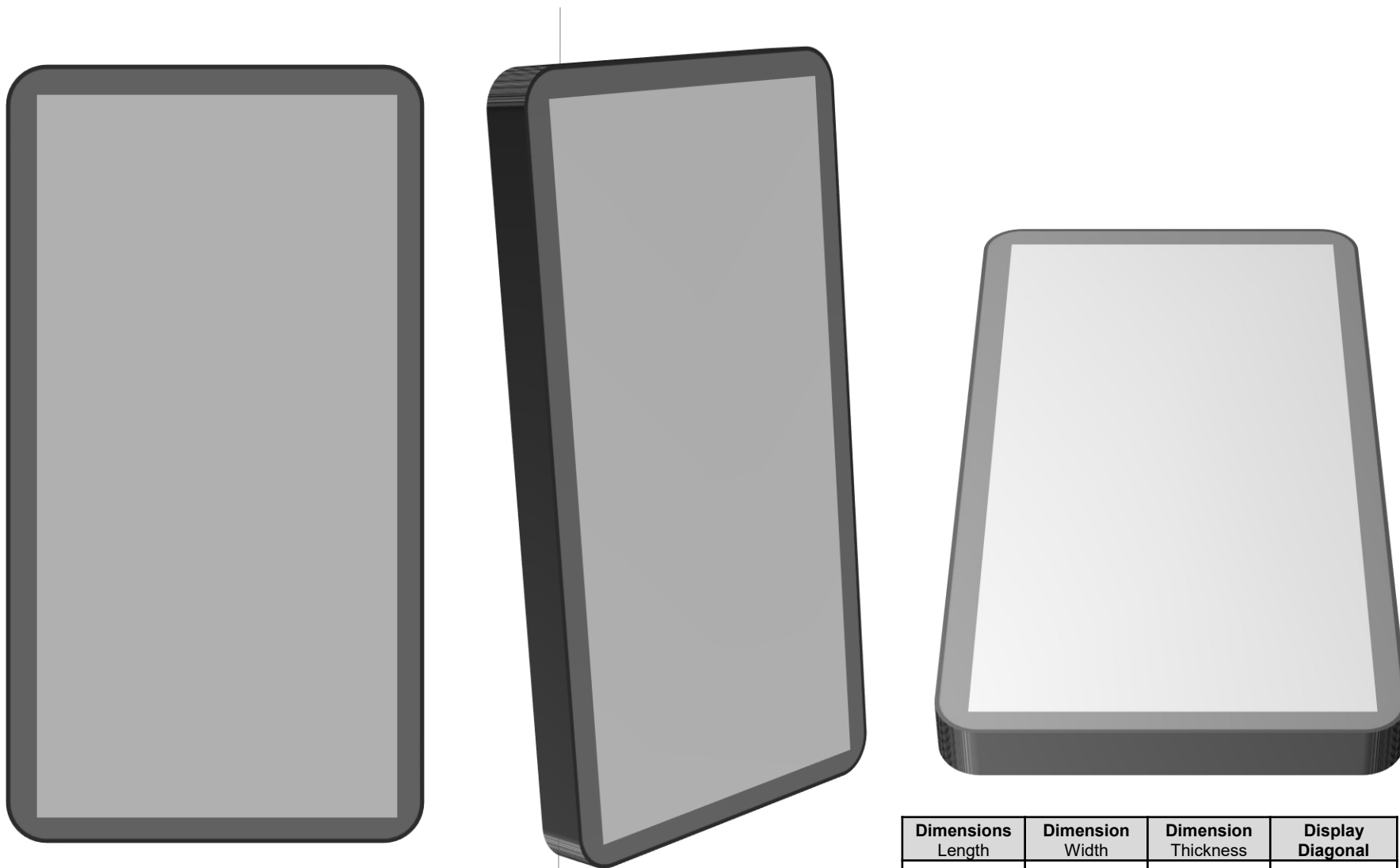


Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
123mm (4.84")	60mm (2.36")	8mm (0.3")	127mm (5.0")



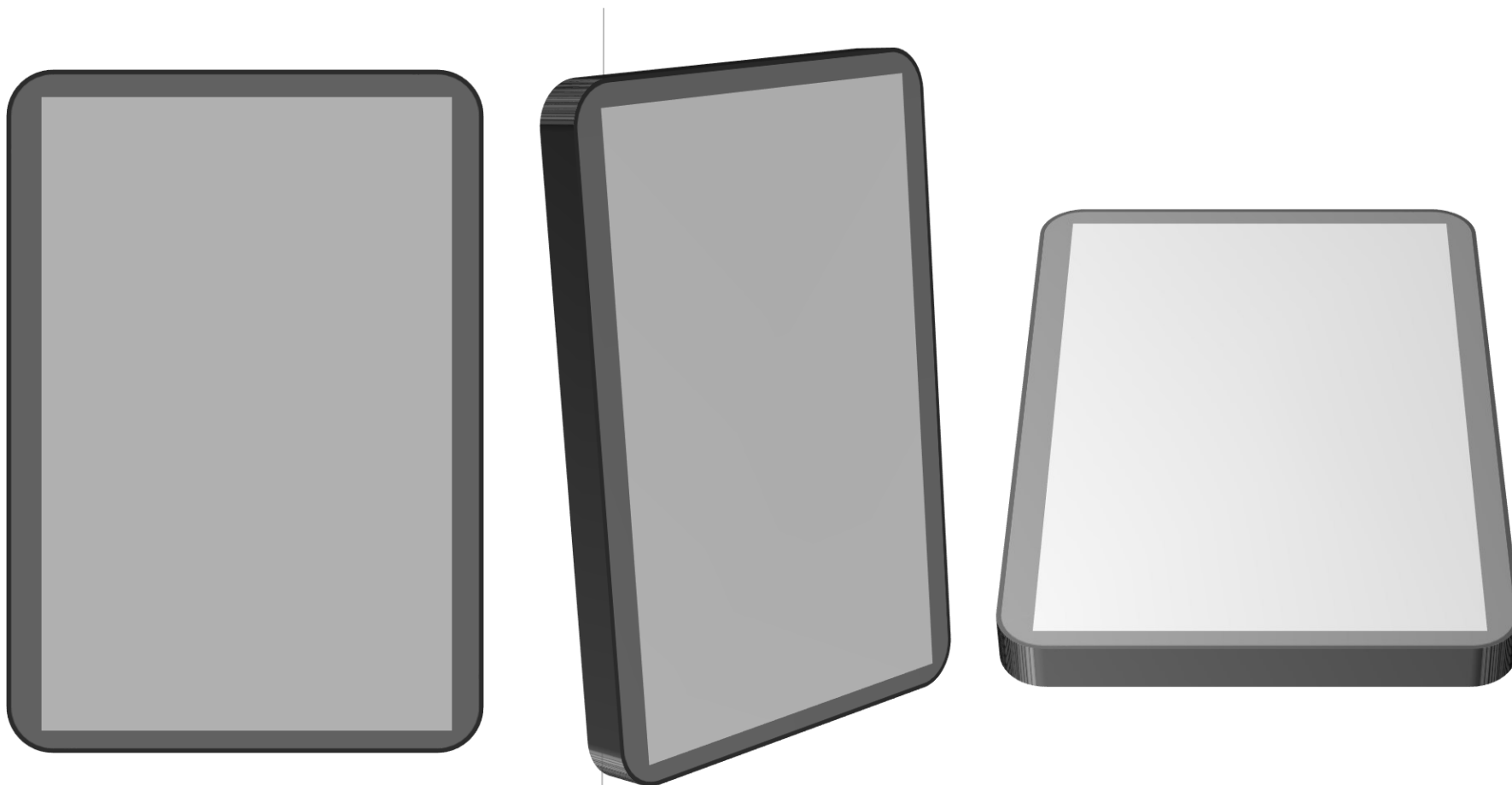
Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
123mm (4.84")	66mm (2.6")	8mm (0.3")	127mm (5.0")

Z



Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
129mm (5.08")	69mm (2.72")	9mm (0.35")	135mm (5.3")

X



Dimensions Length	Dimension Width	Dimension Thickness	Display Diagonal
105mm (4.13")	73mm (2.87")	8mm (0.3")	117mm (4.6")

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Thank You!